Supplementary Information

Poly(pentafulvalene)s: One-Dimensional Conjugated Hydrocarbons toward Stable and Narrow-Bandgap Polyenes

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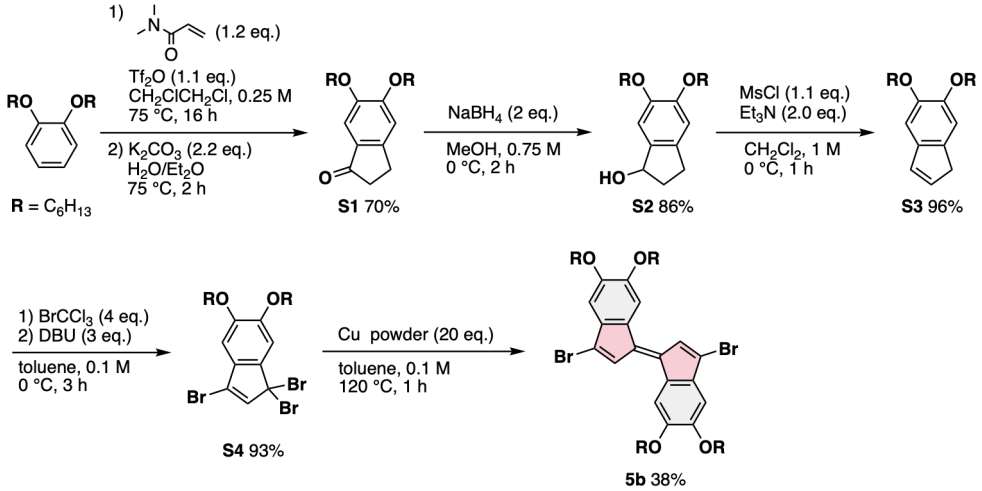
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1. Experimental Details of the Synthesis

General. The names of the molecules in the main manuscript and supporting information do not necessarily follow IUPAC recommendations to guide the reader. Melting points (mp) or decomposition temperatures were determined with a Yanaco MP-S3 instrument. ^{1}H, ^{13}C{^{1}H}, and ^{119}Sn{^{1}H} NMR spectra were recorded with a Bruker AVANCE III 500 MHz spectrometer (500 MHz for ^{1}H, 126 MHz for ^{13}C, and 187 MHz for ^{119}Sn). The chemical shifts in ^{1}H NMR spectra are reported in δ ppm using the residual protons of the solvents, $CHCl_3$ 7.26 ppm, CH_2Cl_2 5.32 ppm, 1,1,2,2-tetrachloroethane-d_2 (TCE-d_2) 5.97 ppm, acetone-d_6 2.05 ppm, and DMSO 2.50 ppm as an internal standard. The chemical shifts in ^{13}C NMR spectra are reported using the solvent signals of $CDCl_3$ 77.16 ppm, CD_2Cl_2 53.84 ppm, TCE-d_2 73.78 ppm, acetone-d_6 29.84 ppm, and DMSO-d_6 39.52 ppm as an internal standard. The chemical shifts of ^{119}Sn{^{1}H} NMR spectra are reported in δ (ppm) relative to external tetramethyltin ($SnMe_4$, δ = 0.00 ppm) measured in the corresponding deuterated solvent. High-resolution mass spectra (HRMS) were measured using a Bruker timsTOF (IMS-QTOF) system with APCI and ESI ionization, or a Bruker solarix (FT-ICR) system with MALDI ionization. Thin layer chromatography (TLC) was performed on glass plates coated with 0.25 mm thickness of silica gel $60F_{254}$ (Merck) or glass plates coated with 0.50 mm thickness of NH_2 silica gel $60F_{254}$ (FUJIFILM Wako Pure Chemical Corporation). Flash chromatography was performed with a Biotage Isolera One instrument or Biotage Isolera Selekt instrument using normal-phase silica gel of Wakosil® HC-N (FUJIFILM Wako Pure Chemical Corporation) or Wakosil® $50NH_2$(HC) (FUJIFILM Wako Pure Chemical Corporation). Commercially available solvents and reagents were used without further purification unless otherwise mentioned. Anhydrous tetrahydrofuran (THF), Et_2O, CH_2Cl_2, and toluene were purchased from Wako Pure Chemical Corporation and further purified by Glass Contour Solvent Systems. Anhydrous MeOH, anhydrous 1,2-dichloroethane, *N,N*-dimethylacrylamide, potassium carbonate (K_2CO_3), methanesulfonyl chloride (MsCl), bromotrichloromethane ($BrCCl_3$), copper powder, and hexamethyldistannane (Me_6Sn_2) were purchased from Wako Pure Chemical Corporation. Trifluoromethanesulfonic anhydride (Tf_2O), tri-*tert*-butylphosphine (*t*-Bu_3P), tetrakis(triphenylphosphine)palladium(0) ($Pd(PPh_3)_4$), and 2,4,6-trimethylphenylboronic acid were purchased from Tokyo Chemical Industry (TCI). Sodium borohydride ($NaBH_4$), triethylamine (Et_3N), and 1,8-diazabicyclo[5.4.0]-7-undecene (DBU) were purchased from Nacalai Tesque. Palladium(II) acetate ($Pd(OAc)_2$) was purchased from BLD Pharmatech. The literature-known compounds, *i.e.,* 1,2-bis(hexyloxy)benzene,[S1] and (*E*)-3,3′-Dibromo-1,1′-biindenylidene **5a**,[S2] [1(1′)E]-3-bromo-3′-mesityl-1,1′-biindenylidene **9**,[S3] and [1(1′)E,1″(1‴)E]-3,3‴-dimesityl-[1,1′:3′,3″:1″,1‴-quaterindene]-1(1′),1″(1‴)-diene **4b**[S3] were prepared according to the literature method. All reactions were performed under a nitrogen atmosphere in flame-dried glassware using anhydrous solvents unless stated otherwise.

Scheme S1



5,6-Bis(hexyloxy)-2,3-dihydro-1*H*-inden-1-one (S1). To a solution of *N,N*-dimethylacrylamide (12.4 mL, 120 mmol) in 1,2-dichloroethane (300 mL) was added Tf_2O (20.2 mL, 120 mmol) dropwise over 20 min at 0 °C. After stirring for 40 min, a solution of 1,2-bis(hexyloxy)benzene (29.8 g, 107 mmol) in 1,2-dichloroethane (100 mL) was added dropwise over 20 min. After stirring for 16 h at 75 °C, a solution of K_2CO_3 (33 g, 240 mmol) in H_2O/Et_2O (2/1, 300 mL) was added. After stirring for 2 h at 75 °C, all volatiles were removed in vacuo. The resulting mixture was diluted with ethyl acetate, followed by addition of H_2O, and separated into two layers. The aqueous layer was extracted with ethyl acetate (2 × 200 mL), and the combined organic layer was washed with brine and dried over Na_2SO_4. After filtration and removal of all volatiles in vacuo, the crude product was purified by silica gel column chromatography (Wakosil® HC-N, hexane to hexane/ethyl acetate 85/15 as an eluent, R_f = 0.10 (hexane/ethyl acetate 9/1)) to afford 24.9 g of **S1** as orange solids (74.7 mmol, 70% yield). Mp: 44.8–45.7 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.17 (s, 1H), 6.86 (s, 1H), 4.07 (t, J = 6.6 Hz, 2H), 4.01 (t, J = 6.7 Hz, 2H), 3.01–3.05 (m, 2H), 2.64–2.68 (m, 2H), 1.79–1.90 (m, 4H), 1.42–1.53 (m, 4H), 1.27–1.40 (m, 8H), 0.84–0.97 (m, 6H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$): δ 206.0, 155.7, 150.4, 149.3, 129.8, 108.7, 105.9, 69.27, 69.25, 36.7, 31.69, 31.67, 29.11, 29.06, 25.80, 25.78, 25.7, 22.8, 22.7, 14.17, 14.16. HRMS (APCI): m/z Calcd. for $C_{21}H_{33}O_3$: 333.2424 ([M+H]$^+$). Obsd. 333.2424.

5,6-Bis(hexyloxy)-2,3-dihydro-1*H*-inden-1-ol (S2). To a solution of **S1** (24.9 g, 74.7 mmol) in MeOH (100 mL) was added $NaBH_4$ (5.78 g, 153 mmol) at 0 °C. After stirring for 2 h, the resulting mixture was quenched with H_2O, diluted with ethyl acetate, and separated into two layers. The aqueous layer was extracted with ethyl acetate (3 × 100 mL), and the combined organic layer was washed with brine and dried over Na_2SO_4. Filtration and removal of all volatiles in vacuo afforded 21.7 g of **S2** as orange solids (64.7 mmol, 86% yield). R_f = 0.06 (hexane/ethyl acetate 9/1). Mp: 32.7–33.6 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 6.94 (s, 1H), 6.76 (s, 1H), 5.18 (ddd, J = 7.4, 6.9, 4.7 Hz, 1H), 3.94–4.01 (m, 4H), 2.98 (ddd, J = 15.6, 8.5, 5.0 Hz, 1H), 2.74 (ddd, J = 15.6, 8.4, 6.0 Hz, 1H), 2.49 (dddd, J = 13.4, 8.4, 6.9, 5.0 Hz, 1H), 1.94 (dddd, J = 13.4, 8.5, 6.0, 4.7 Hz, 1H), 1.76–1.85 (m, 4H), 1.65 (d, J = 7.4 Hz, 1H), 1.42–1.51 (m, 4H), 1.30–1.38 (m, 8H), 0.87–0.93 (m, 6H). ^{13}C{^{1}H} NMR (126 MHz, CD_2Cl_2): δ 150.3, 149.0, 137.6, 136.0, 110.6, 110.3, 77.0, 70.0, 69.9, 36.8, 32.07, 32.06, 30.0, 29.91, 29.86, 29.8, 26.18, 26.17, 23.1, 14.2. A signal

attributable to the carbon atoms in the aliphatic region was not observed due to overlap with the other signals. HRMS (ESI): *m/z* Calcd. for $C_{21}H_{34}O_3Na$: 357.2400 ([*M*+Na]$^+$). Obsd. 357.2399.

5,6-Bis(hexyloxy)-1*H*-indene (S3). To a solution of **S2** (21.7 g, 64.7 mmol) and triethylamine (19 mL, 130 mmol) in CH_2Cl_2 (65 mL) was added MsCl (5.9 mL, 72 mmol) at 0 °C. After stirring for 1 h, the resulting mixture was quenched by H_2O, diluted with CH_2Cl_2, and separated into two layers. The aqueous layer was extracted with CH_2Cl_2 (3 × 50 mL), and the combined organic layer was washed with brine and dried over Na_2SO_4. After filtration and removal of all volatiles in vacuo, the crude product was purified by silica gel column chromatography (Wakosil® HC-N, hexane to hexane/ethyl acetate 9/1 as an eluent, R_f = 0.55 (hexane/ethyl acetate 9/1)) to afford 19.6 g of **S3** as yellow liquid (61.9 mmol, 96% yield). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.08 (s, 1H), 6.97 (s, 1H), 6.77 (dt, *J* = 5.4, 1.8 Hz, 1H), 6.44 (dt, *J* = 5.4, 1.9 Hz, 1H), 4.01 (t, *J* = 6.7 Hz, 2H), 4.00 (t, *J* = 6.7 Hz, 2H), 3.32 (dd, *J* = 1.9, 1.8 Hz, 2H), 1.77–1.86 (m, 4H), 1.44–1.51 (m, 4H), 1.28–1.39 (m, 8H), 0.86–0.95 (m, 6H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$): δ 148.7, 147.6, 138.2, 136.6, 132.9, 131.9, 111.4, 107.7, 70.3, 70.0, 39.1, 31.8, 29.66, 29.60, 25.9, 22.8, 14.2. Four signals attributable to the carbon atoms in the aliphatic region were not observed due to overlap with the other signals. HRMS (APCI): *m/z* Calcd. for $C_{21}H_{33}O_2$: 317.2475 ([*M*+H]$^+$). Obsd. 317.2474.

1,1,3-Tribromo-5,6-bis(hexyloxy)-1*H*-indene (S4). To a solution of **S3** (19.6 g, 61.9 mmol) in toluene (400 mL) was added a solution of $BrCCl_3$ (24.0 mL, 243 mmol) in toluene (110 mL) dropwise over 20 min at 0 °C. After stirring for 40 min, a solution of DBU (28.0 mL, 187 mmol) in toluene (110 mL) was added dropwise over 20 min. After stirring for 3 h, the resulting mixture was quenched by aqueous HCl. After removal of all volatiles in vacuo, the mixture was diluted with ethyl acetate. After addition of H_2O, the mixture separated into two layers. The aqueous layer was extracted with ethyl acetate (3 × 100 mL), and the combined organic layer was washed with brine and dried over Na_2SO_4. After filtration and removal of all volatiles in vacuo, the crude product was purified by silica gel column chromatography (Wakosil® HC-N, hexane as an eluent, R_f = 0.67 (hexane/ethyl acetate 9/1)) to afford 31.8 g of **S4** as yellow liquid (57.5 mmol, 93% yield). ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.22 (s, 1H), 6.74 (s, 1H), 6.72 (s, 1H), 4.03–4.10 (m, 4H), 1.80–1.90 (m, 4H), 1.45–1.53 (m, 4H), 1.31–1.40 (m, 8H), 0.88–0.95 (m, 6H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$): δ 150.8, 150.2, 139.4, 138.9, 128.2, 122.2, 109.7, 106.1, 69.9, 69.7, 51.2, 31.72, 31.68, 29.29, 29.26, 25.83, 25.82, 22.8, 14.2. Two signals attributable to the carbon atoms in the aliphatic region were not observed due to overlap with the other signals. HRMS (APCI): *m/z* Calcd. for $C_{21}H_{30}{}^{79}Br_3O_2$: 550.9790 ([*M*+H]$^+$). Obsd. 550.9792.

[1(1′)*E*]-3,3′-Dibromo-5,5′,6,6′-tetrakis(hexyloxy)-1,1′-biindenylidene (5b). A mixture of **S4** (31.8 g, 57.5 mmol) and copper powder (74.2 g, 1.17 mol) in toluene (570 mL) was stirred for 1 h at 120 °C. After Celite® filtration and removal of all volatiles in vacuo, the crude product was washed with hexane to afford 8.52 g of **5b** as red solids (10.8 mmol, 38% yield). R_f = 0.37 (hexane/CH_2Cl_2 9/1). Mp: 150.1–150.9 °C. ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.41 (s, 2H), 7.25 (s, 2H), 6.88 (s, 2H), 4.03–4.15 (m, 8H), 1.80–1.91 (m, 8H), 1.47–1.55 (m, 8H), 1.31–1.43 (m, 16H), 0.88–0.97 (m, 12H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$): δ 150.9, 148.4, 137.3, 135.7, 128.6, 128.0,

125.7, 113.4, 106.8, 71.1, 69.5, 31.83, 31.75, 29.7, 29.4, 25.9, 22.81, 22.78, 14.21, 14.18. One signal attributable to the carbon atom in the aliphatic region was not observed due to overlap with another signal. HRMS (ESI): *m/z* Calcd. for $C_{42}H_{59}{}^{79}Br_2O_4$: 785.2775 ([*M*+H]$^+$). Obsd. 785.2778.

[1(1′)*E*]-5,5′,6,6′-Tetrakis(hexyloxy)-3,3′-bis(trimethylstannyl)-1,1′-biindenylidene (7b). To a solution of **5b** (1.20 g, 1.52 mmol) and $Pd(PPh_3)_4$ (87.5 mg, 75.7 μmol) in THF (15 mL) was added Me_6Sn_2 (5 g, 15 mmol). After stirring for 21 h at 75 °C, the resulting mixture was filtered through a plug of Wakosil® 50NH$_2$(HC), rinsed with CH_2Cl_2. After the removal of all volatiles in vacuo from the resulting filtrate, the resulting crude product was purified by silica gel column chromatography (Wakosil® 50NH$_2$(HC), hexane to hexane/CH_2Cl_2 96/4 as an eluent, R_f = 0.61 (hexane/CH_2Cl_2 9/1)) to afford 933 mg of **7b** as red solids (0.976 mmol, 64% yield). Mp: 90.9–91.7 °C. ^{1}H NMR (500 MHz, CD_2Cl_2): δ 7.54 (s, 2H), 7.33 (s, ^3J(H–Sn) = 33.3 Hz, 2H), 6.81 (s, 2H), 4.05 (t, J = 6.6 Hz, 4H), 4.04 (t, J = 6.5 Hz, 4H), 1.78–1.86 (m, 8H), 1.45–1.55 (m, 8H), 1.33–1.43 (m, 16H), 0.88–0.96 (m, 12H), 0.39 (s, ^2J(H–^{119}Sn) = 56.6 Hz, ^2J(H–^{117}Sn) = 54.3 Hz, 18H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$): δ 152.3, 150.2, 147.0, 143.2, 139.2, 136.1, 130.1, 114.3, 109.6, 70.9, 69.6, 31.81, 31.79, 29.7, 29.5, 25.9, 22.83, 22.80, 14.23, 14.21, −9.1 (^1J(^{119}Sn–^{13}C) = 360.4 Hz, ^1J(^{117}Sn–^{13}C) = 345.2 Hz). One signal attributable to the carbon atom in the aliphatic region was not observed due to overlap with the other signals. ^{119}Sn{^{1}H} NMR (187 MHz, CD_2Cl_2): δ –47.24. HRMS (MALDI): *m/z* Calcd. for $C_{48}H_{76}O_4{}^{120}Sn_2$: 956.3782 ([*M*]$^+$). Obsd. 956.3787.

Poly[5,5′,6,6′-tetrakis(hexyloxy)-1,1′-biindenylidene-3,3′-diyl] (3b). A solution of $Pd(OAc)_2$ (6.0 mg, 26.7 μmol) and *t*-Bu_3P (10.8 mg, 53.4 μmol) in toluene (5.4 mL) was stirred at an ambient temperature for 30 min. An aliquot of the resulting solution (0.50 mL), corresponding to $Pd(OAc)_2$ (2.47 μmol) and *t*-Bu_3P (4.94 μmol), was added to a mixture of **5b** (38.9 mg, 49.4 μmol) and **7b** (47.3 mg, 49.6 μmol). The resulting mixture was heated at 200 °C for 5 min under microwave irradiation using a Biotage Initiator+ microwave reactor. The reaction mixture was diluted with toluene (5 mL) and added dropwise to CH_3CN (100 mL). The resulting precipitates were collected by filtration, washed with CH_3CN, and dried under reduced pressure to afford 52.0 mg of the crude polymer. The crude polymer was purified by recycling preparative GPC using a LaboACE LC-5060 Plus II (Japan Analytical Industry) equipped with polystyrene gel columns (JAIGEL-3HR-Plus and JAIGEL-4HR-Plus, Japan Analytical Industry) using $CHCl_3$ as eluent at a flow rate of 10 mL min^{-1}. The fractions eluted from 40.8 to 50.0 min during the second cycle and from 99.6 to 109.9 min during the fourth cycle were collected and combined. The combined fractions were concentrated under reduced pressure and dried under reduced pressure to afford **3b** as dark purple solids (21.8 mg, 35% yield). GPC analysis was performed using an EcoSEC Elite HLC-8420GPC system (Tosoh) equipped with TSKgel SuperHZM-M and TSKgel SuperH-RC columns using THF as the eluent at a flow rate of 0.35 mL min^{-1} at 40 °C with UV detection at 254 nm. The molecular weights were determined relative to polystyrene standards (PStQuick® Kit-M): M_n = 9.70 × 10^3, M_w = 1.68 × 10^4, $Đ$ = 1.73, DP_n = 15.5, and DP_w = 26.8. ^{1}H NMR (500 MHz, TCE-d_2): δ 8.10–6.76 (6H), 4.30–3.79 (8H), 1.98–1.69 (8H), 1.54–1.21 (24H), 0.98–0.78 (12H).

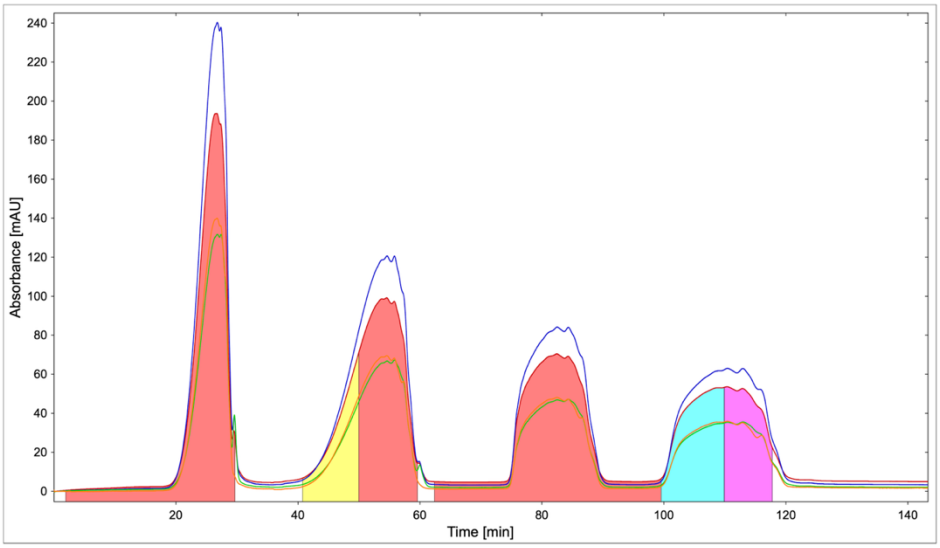


Figure S1. Recycling preparative GPC chromatogram of the crude polymer. The collected fractions are indicated by yellow and light blue.

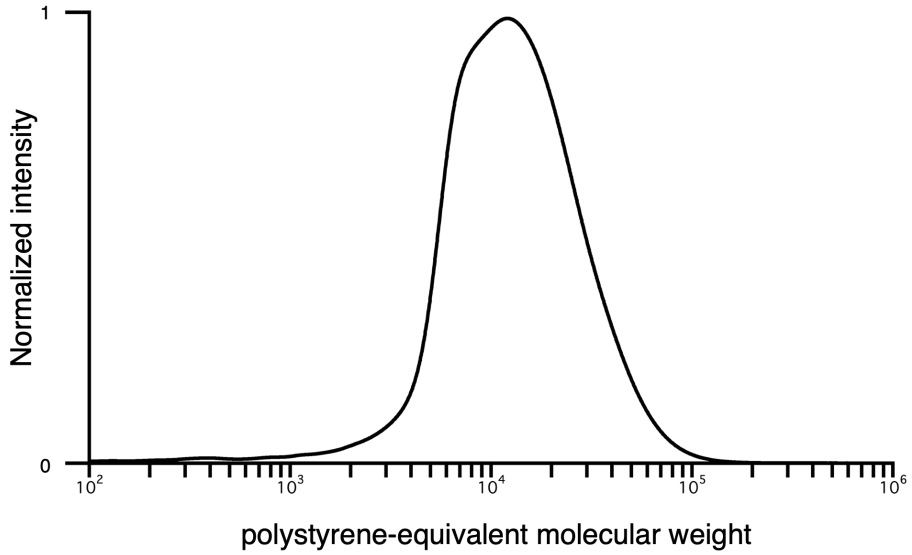
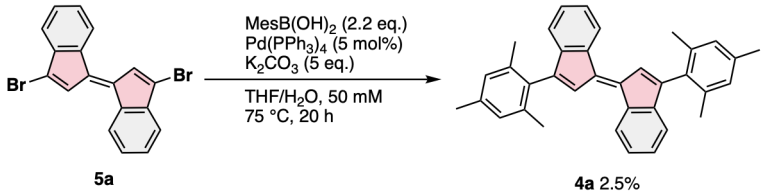


Figure S2. Analytical GPC chromatogram of **3b**.

[1(1′)*E*]-3,3′-Bis(trimethylstannyl)-1,1′-biindenylidene (7a). A solution of **5a** (387 mg, 1.00 mmol), Me_6Sn_2 (944 mg, 2.88 mmol), and $Pd(PPh_3)_4$ (59.6 mg, 51.6 µmol) in anhydrous THF (5 mL) was stirred at 75 °C for 24 h. The resulting mixture was passed through a plug of silica gel (Wakosil® $50NH_2$(HC)) by eluting with CH_2Cl_2. After filtration and removal of all volatiles under reduced pressure, the crude product was purified by silica gel column chromatography (Wakosil® $50NH_2$(HC), hexane to hexane/CH_2Cl_2 9/1 as eluent, R_f = 0.40 (hexane)) followed by recrystallization with CH_2Cl_2/acetonitrile 1/1 to afford 362 mg of **7a** as red solids (0.653 mmol, 65% yield). Mp: 169.8–170.5 °C. ^{1}H NMR (500 MHz, acetone-d_6): δ 8.08 (dd, J = 7.5, 1.3 Hz, 2H), 7.69 (s, ^3J(Sn–H) = 33.4 Hz, 2H), 7.35 (dd, J = 7.1, 1.1 Hz, 2H), 7.28 (app td, J = 7.4, 1.1 Hz, 2H), 7.23 (app td, J = 7.4, 1.3 Hz, 2H), 0.42 (s, $^2J(^{119}$Sn–H) = 57.6 Hz, $^2J(^{117}$Sn–H) = 55.1 Hz,

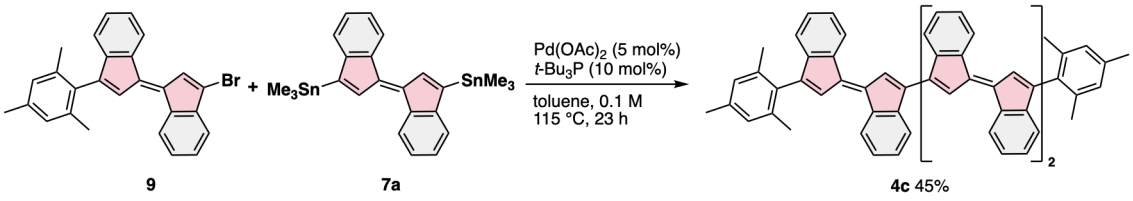
18H). $^{13}C\{^{1}H\}$ NMR (126 MHz, acetone-d_6): δ 154.6, 149.6, 140.7, 138.2, 137.6 (^{2}J(Sn–^{13}C) = 32.7 Hz), 129.4, 126.5, 126.0, 123.9, −9.4 (^{1}J(^{119}Sn–^{13}C) = 365.5 Hz, ^{1}J(^{117}Sn–^{13}C) = 349.6 Hz). $^{119}Sn\{^{1}H\}$ NMR (187 MHz, acetone-d_6) δ –43.3. HRMS (APCI): *m/z* Calcd. for $C_{24}H_{28}{}^{120}Sn_2$: 556.0241 ([*M*]$^-$). Obsd. 556.0245.

Scheme S2



[1(1′)*E*]-3,3′-Dimesityl-1,1′-biindenylidene (4a). A solution of **5a** (386 mg, 1.00 mmol), 2,4,6-trimethylphenylboronic acid (361 mg, 2.20 mmol), K_2CO_3 (694 mg, 5.02 mmol) and $Pd(PPh_3)_4$ (57.8 mg, 50.0 μmol) in THF (20 mL) and H_2O (4 mL) was stirred for 20 h at 75 °C. The resulting mixture was quenched by H_2O, diluted by CH_2Cl_2, and separated into two layers. The aqueous layer was extracted with CH_2Cl_2 (3 × 50 mL). The combined organic layer was dried over Na_2SO_4. After filtration and removal of all volatiles under reduced pressure, the crude product was purified by silica gel column chromatography (Wakosil® HC-N, hexane to hexane/CH_2Cl_2 4/1 as eluent, R_f = 0.50 (hexane/CH_2Cl_2 4/1)), to afford 11.8 mg of **4a** as orange solids (25.4 μmol, 2.5% yield). Mp: > 300 °C. ^{1}H NMR (500 MHz, CD_2Cl_2): δ 8.00–8.05 (m, 2H), 7.36 (s, 2H), 7.17–7.24 (m, 4H), 7.00 (s, 4H), 6.84–6.87 (m, 2H), 2.35 (s, 6H), 2.18 (s, 12H). $^{13}C\{^{1}H\}$ NMR (126 MHz, CD_2Cl_2): δ 149.4, 144.0, 140.3, 137.8, 137.6, 136.9, 132.0, 128.53, 128.49, 126.3, 126.1, 125.4, 121.1, 21.3, 20.5. HRMS (ESI): *m/z* Calcd. for $C_{36}H_{33}$: 465.2577 ([*M*+H]$^+$). Obsd. 465.2577.

Scheme S3



[1(1′)*E*,1″(1‴)*E*,1⁗(1⁗′)*E*]-3,3‴⁗-Dimesityl-[1,1′:3′,3″:1″,1‴:3‴,3⁗:1⁗,1⁗′-sexiindene]-1(1′),1″(1‴),1⁗(1⁗′)-triene (4c). A solution of **9** (501 mg, 1.18 mmol), **7a** (283 mg, 0.51 mmol), $Pd(OAc)_2$ (11.2 mg, 50 μmol), and *t*-Bu_3P (20.2 mg, 0.10 mmol) in toluene (10 mL) was stirred at 115 °C for 23 h. The resulting mixture was passed through a plug of silica gel (Wakosil® 50NH_2(HC)) by eluting with CH_2Cl_2. After filtration and removal of all volatiles under reduced pressure, the crude product was purified by silica gel column chromatography (Wakosil® 50NH_2(HC), hexane to hexane/CH_2Cl_2 1/1, R_f = 0.10 (hexane/CH_2Cl_2 9/1)) to afford 210 mg of **4c** as dark green solids (0.228 mmol, 45% yield). Mp: > 300 °C. ^{1}H NMR (500 MHz, TCE-d_2): δ 8.14 (d, *J* = 7.5 Hz, 2H), 8.04–8.09 (m, 4H), 8.01 (s, 2H), 7.96 (s, 2H), 7.84 (d, *J* = 7.1 Hz, 2H), 7.83 (d, *J* = 7.6 Hz, 2H), 7.41–7.50 (m, 6H), 7.34–7.39 (m, 4H), 7.29 (app t, *J* = 7.4 Hz, 2H), 7.23 (app t, *J* = 7.3 Hz, 2H), 7.00 (s, 4H), 6.92 (d, *J* = 7.3 Hz, 2H), 2.36 (s, 6H), 2.20 (s, 12H). $^{13}C\{^{1}H\}$

NMR (126 MHz, TCE-d_2): δ 150.1, 143.4, 142.02, 141.96, 141.81, 141.75, 141.13, 141.08, 139.4, 137.87, 137.84, 137.3, 136.9, 136.4, 131.3, 128.50, 128.46, 128.3, 128.2, 126.8, 126.5, 126.3, 126.1, 125.9, 125.7, 125.3, 125.2, 124.9, 121.8, 121.5, 121.1, 21.2, 20.4. HRMS (MALDI): *m/z* Calcd. for $C_{72}H_{52}$: 916.4075 ([*M*]$^-$). Obsd. 916.4073.

[1(1′)*E*]-3-Trimethylstannyl-3′-mesityl-1,1′-biindenylidene (10). A solution of **9** (426 mg, 1.00 mmol), Me_6Sn_2 (969 mg, 2.96 mmol), and $Pd(PPh_3)_4$ (57 mg, 49 μmol) in anhydrous THF (10 mL) was stirred at 75 °C for 12 h. The resulting mixture was passed through a plug of silica gel (Wakosil® 50NH_2(HC)) by eluting with CH_2Cl_2. After filtration and removal of all volatiles under reduced pressure, the crude product was purified by silica gel column chromatography (Wakosil® 50NH_2(HC), hexane to hexane/CH_2Cl_2 1/1 as eluent, R_f = 0.30 (hexane)) to afford 509 mg of **10** as red solids (1.00 mmol, > 99% yield). Mp: 77.8–78.6 °C. ^{1}H NMR (500 MHz, CD_2Cl_2): δ 8.08 (d, *J* = 7.5 Hz, 1H), 7.93 (d, *J* = 7.6 Hz, 1H), 7.66 (s, ^3J(Sn–H) = 33.1 Hz, 1H), 7.20–7.33 (m, 5H), 7.16 (app td, *J* = 7.5, 1.0 Hz, 1H), 7.00 (s, 2H), 6.85 (d, *J* = 7.0 Hz, 1H), 2.35 (s, 3H), 2.16 (s, 6H), 0.43 (s, ^2J(^{119}Sn–H) = 56.8 Hz, ^2J(^{117}Sn–H) = 54.5 Hz, 9H). ^{13}C{^{1}H} NMR (126 MHz, CD_2Cl_2): δ 153.4, 149.9, 149.1, 144.1, 141.2, 139.4, 137.8, 137.70, 137.63, 137.0, 136.8, 132.0, 128.7, 128.6, 128.5, 126.4, 125.8, 125.6, 125.3, 123.4, 121.1, 21.3, 20.4, −9.2 (^1J(^{119}Sn–^{13}C) = 363.4 Hz, ^1J(^{117}Sn–^{13}C) = 347.2 Hz). ^{119}Sn{^{1}H} NMR (187 MHz, CD_2Cl_2) δ –46.02. HRMS (APCI): *m/z* Calcd. for $C_{30}H_{31}{}^{120}$Sn: 511.1442 ([*M*+H]$^+$). Obsd. 511.1445.

[1(1′)*E*,1″(1‴)*E*]-3-Bromo-3‴-mesityl-[1,1′:3′,3″:1″,1‴-quaterindene]-1(1′),1″(1‴)-diene (11). A solution of **10** (255 mg, 0.50 mmol), **5a** (579 mg, 1.50 mmol), $Pd(OAc)_2$ (5.6 mg, 25 μmol), and *t*-Bu_3P (10.1 mg, 50 μmol) in toluene (20 mL) was stirred for 12 h at 115 °C. The resulting mixture was passed through a plug of silica gel (Wakosil® 50NH_2(HC)) by eluting with CH_2Cl_2. After filtration and removal of all volatiles under reduced pressure, the crude product was purified by silica gel column chromatography (Wakosil® 50NH_2(HC), hexane/CH_2Cl_2 19/1 to 1/1 as eluent, R_f = 0.23 (hexane/CH_2Cl_2 9/1)) followed by recrystallization with CH_2Cl_2/acetonitrile 1/1 to afford 58 mg of **11** as dark purple solids (89 μmol, 18% yield). Mp: > 300 °C. ^{1}H NMR (500 MHz, TCE-d_2): δ 8.01–8.10 (m, 3H), 7.94–8.00 (m, 2H), 7.81 (s, 1H), 7.78 (d, *J* = 7.4 Hz, 1H), 7.76 (d, *J* = 7.6 Hz, 1H), 7.66 (s, 1H), 7.31–7.49 (m, 8H), 7.28 (app td, *J* = 7.5, 1.0 Hz, 1H), 7.23 (app td, *J* = 7.4, 0.6 Hz, 1H), 6.99 (s, 2H), 6.91 (d, *J* = 7.1 Hz, 1H), 2.35 (s, 3H), 2.19 (s, 6H). ^{13}C{^{1}H} NMR (126 MHz, TCE-d_2): δ 150.1, 143.4, 142.5, 142.0, 141.8, 141.7, 140.9, 140.8, 140.1, 139.3, 138.7, 137.8, 137.4, 137.3, 136.9, 136.4, 135.5, 131.3, 128.9, 128.8, 128.6, 128.5, 128.3, 128.2, 127.2, 127.1, 126.8, 126.5, 126.3, 126.2, 125.7, 125.6, 125.3, 124.9, 124.4, 121.7, 121.4, 121.1, 121.0, 21.2, 20.4. One signal attributable to the carbon atom in the aromatic region was not observed due to overlap with the other signal. HRMS (APCI): *m/z* Calcd. for $C_{45}H_{32}{}^{79}$Br: 651.1682 ([*M*+H]$^+$). Obsd. 651.1679.

[1(1′)*E*,1″(1‴)*E*,1⁗(1⁗⁗)*E*,1⁗⁗⁗(1⁗⁗⁗⁗)*E*]-3,3⁗⁗⁗⁗-Dimesityl-[1,1′:3′,3″:1″,1‴:3‴,3⁗:1⁗,1⁗⁗⁗:3⁗⁗⁗,3⁗⁗⁗⁗:1⁗⁗⁗⁗,1⁗⁗⁗⁗⁗-octiindene]-1(1′),1″(1‴),1⁗⁗⁗(1⁗⁗⁗⁗),1⁗⁗⁗⁗⁗(1⁗⁗⁗⁗⁗)-tetraene (4d). A solution of **11** (65 mg, 0.10 mmol), Me_6Sn_2 (17 mg, 50 μmol), $Pd(OAc)_2$ (1.2 mg, 5.2 μmol), and *t*-Bu_3P (2.1 mg, 10 μmol) in anhydrous toluene (2 mL) was stirred for 12 h at 115 °C. After cooling the reaction mixture to

ambient temperature, the resulting precipitate was collected by vacuum filtration, washed with hexane, and recrystallized from $CHCl_2CHCl_2$/acetonitrile 1/1 to afford 25 mg of **4d** as dark green solids (22 μmol, 44% yield). Mp: > 300 °C. ^{1}H NMR (500 MHz, TCE-d_2): δ 8.11–8.19 (m, 4H), 8.04–8.10 (m, 4H), 8.01 (s, 2H), 8.00 (s, 2H), 7.96 (s, 2H), 7.81–7.88 (m, 6H), 7.41–7.54 (m, 10H), 7.34–7.39 (m, 4H), 7.29 (app t, J = 7.4 Hz, 2H), 7.23 (app t, J = 7.4 Hz, 2H), 7.00 (s, 4H), 6.92 (d, J = 7.3 Hz, 2H), 2.36 (s, 6H), 2.20 (s, 12H). ^{13}C{^{1}H} NMR (126 MHz, TCE-d_2): δ 150.1, 143.5, 142.2, 142.0, 141.9, 141.85, 141.81, 141.4, 141.09, 141.08, 139.4, 138.6, 137.90, 137.89, 137.3, 136.9, 136.4, 131.3, 128.55, 128.47, 128.3, 128.2, 126.90, 126.85, 126.8, 126.7, 126.58, 126.56, 126.4, 126.3, 126.1, 125.7, 125.3, 125.2, 121.9, 121.8, 120.3, 120.2, 21.2, 20.3. Two signals attributable to the carbon atoms in the aromatic region were not observed due to overlap with the other signals. HRMS (MALDI): *m/z* Calcd. for $C_{90}H_{62}$: 1142.4857 ([M]$^-$). Obsd. 1142.4839.

[1(1′)*E*,1″(1‴)*E*,1⁗(1⁗′)*E*,1⁗′(1⁗′′)*E*,1⁗′′(1⁗′′′)*E*,1⁗′′′(1⁗′′′′)*E*]-3,3⁗′′′′-Dimesityl-[1,1′:3′,3″:1″,1‴:3‴,3⁗:1⁗,1⁗′:3⁗′,3⁗′′:1⁗′,1⁗′′:3⁗′′,3⁗′′′:1⁗′′′,1⁗′′′′:3⁗′′′′,3⁗′′′′′:1⁗′′′′′,1⁗′′′′′′:3⁗′′′′′′,3⁗′′′′′′′:1⁗′′′′′′′,1⁗′′′′′′′′-deciindene]-1(1′),1″(1‴),1⁗(1⁗′),1⁗′(1⁗′′),1⁗′′(1⁗′′′),1⁗′′′(1⁗′′′′)-pentaene (4e). A solution of **11** (65 mg, 0.10 mmol), **7a** (28 mg, 50 μmol), $Pd(OAc)_2$ (1.2 mg, 5.2 μmol), and t-Bu_3P (2.1 mg, 10 μmol) in anhydrous toluene (2 mL) was stirred for 12 h at 115 °C. After cooling the reaction mixture to an ambient temperature, the resulting precipitate was collected by vacuum filtration and washed with hexane, and recrystallized from $CHCl_2CHCl_2$/acetonitrile 1/1 to afford 25 mg of **4e** as dark green solids (18 μmol, 37% yield). Mp: > 300 °C. ^{1}H NMR (500 MHz, TCE-d_2): δ 8.10–8.21 (m, 6H), 8.04–8.10 (m, 4H), 7.96–8.04 (m, 6H), 7.95 (s, 2H), 7.80–7.90 (m, 8H), 7.42–7.54 (m, 14H), 7.34–7.39 (m, 4H), 7.29 (app t, J = 7.4 Hz, 2H), 7.23 (app t, J = 7.3 Hz, 2H), 7.00 (s, 4H), 6.92 (d, J = 7.2 Hz, 2H), 2.36 (s, 6H), 2.20 (s, 12H). ^{13}C{^{1}H} NMR (126 MHz, TCE-d_2): δ 150.1, 143.5, 142.2, 142.1, 142.0, 141.9, 141.84, 141.81, 141.5, 141.4, 141.3, 141.1, 139.4, 137.91, 137.89, 137.85, 137.4, 137.0, 136.9, 136.6, 136.4, 130.4, 128.59, 128.55, 128.50, 128.48, 128.3, 128.2, 126.87, 126.85, 126.83, 126.80, 126.5, 126.42, 126.40, 126.3, 126.2, 126.1, 125.9, 125.7, 125.33, 125.30, 125.25, 123.4, 121.8, 120.2, 116.7, 116.4, 21.2, 20.4. One signal attributable to the carbon atom in the aromatic region was not observed due to overlap with the other signal. HRMS (MALDI): *m/z* Calcd. for $C_{108}H_{72}$: 1368.5640 ([M]$^-$). Obsd. 1368.5603.

2. Single-Crystal X-ray Diffraction

X-ray crystallographic analysis of 7a. Red block single crystals were grown by slow diffusion of acetonitrile into a solution of **7a** in CH_2Cl_2. Intensity data were collected at 100 K on synchrotron radiation (λ = 0.8124 Å) at the BL40XU beamline in SPring-8 (JASRI). A total of 12146 reflections were measured with the maximum 2θ angle of 61.0°, of which 2285 were independent reflections (R_{int} = 0.1022). The structure was solved by direct methods (SHELXT 2018/2) [S4] and refined by the full-matrix least-squares on F^2 (SHELXL-2019/2) [S5]. All non-hydrogen atoms were refined anisotropically, and all hydrogen atoms were placed using AFIX instructions. The crystal data are as follows: $C_{24}H_{28}Sn_2$; FW = 553.84, crystal size 0.1 × 0.1 × 0.1 mm^3, Monoclinic, $P2_1/c$ (#14), a = 8.95170(10) Å, b = 14.1982(2) Å, c = 9.6204(2) Å, β = 113.481(2)°, V = 1121.48(3) Å^3, Z = 2, D_c = 1.640 g cm^{-3}, μ = 3.156 mm^{-1}, R_1 = 0.0605 (I >

$2\sigma(I)$), wR_2 = 0.1397 (all data), GOF = 1.296. CCDC 2598311 contains the supplementary crystallographic data for this compound. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) at www.ccdc.cam.ac.uk/data_request/cif.

X-ray crystallographic analysis of 4a. Orange plate single crystals were grown by slow diffusion of ethanol into a solution of **4a** in CH_2Cl_2. Intensity data were collected at 100 K on MoKα (λ = 0.71073 Å) using rotating-anode X-ray tube. A total of 10706 reflections were measured with the maximum 2θ angle of 60.9°, of which 3432 were independent reflections (R_{int} = 0.0232). The structure was solved by direct methods (SHELXT 2018/2) [S4] and refined by the full-matrix least-squares on F^2 (SHELXL-2019/2) [S5]. All non-hydrogen atoms were refined anisotropically, and all hydrogen atoms were placed using AFIX instructions. The crystal data are as follows: $C_{36}H_{32}$; FW = 464.61, crystal size 0.2 × 0.2 × 0.05 mm^3, Monoclinic, $P2_1/c$ (#14), a = 15.0290(7) Å, b = 6.9537(3) Å, c = 12.7887(6) Å, β = 104.812(5)°, V = 1292.10(11) Å^3, Z = 2, D_c = 1.194 g cm^{-3}, μ = 0.067 mm^{-1}, R_1 = 0.0454 ($I > 2\sigma(I)$), wR_2 = 0.1299 (all data), GOF = 1.080. CCDC 2598310 contains the supplementary crystallographic data for this compound. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) at www.ccdc.cam.ac.uk/data_request/cif.

X-ray crystallographic analysis of 11. Purple plate single crystals were grown by slow diffusion of ethanol into a solution of **11** in CH_2Cl_2. Intensity data were collected at 100 K on synchrotron radiation (λ = 0.4133 Å) at the BL02B1 beamline in SPring-8 (JASRI). A total of 86655 reflections were measured with the maximum 2θ angle of 31.1°, of which 7041 were independent reflections (R_{int} = 0.0466). The structure was solved by direct methods (SHELXT 2018/2) [S4] and refined by the full-matrix least-squares on F^2 (SHELXL-2019/2) [S5]. All non-hydrogen atoms were refined anisotropically, and all hydrogen atoms were placed using AFIX instructions. The crystal data are as follows: $C_{45}H_{31}Br$; FW = 651.61, crystal size 0.15 × 0.05 × 0.02 mm^3, Monoclinic, $P2_1/c$ (#14), a = 29.4863(6) Å, b = 5.19410(10) Å, c = 20.7676(4) Å, β = 104.993(2)°, V = 3072.38(11) Å^3, Z = 4, D_c = 1.409 g cm^{-3}, μ = 0.335 mm^{-1}, R_1 = 0.0455 ($I > 2\sigma(I)$), wR_2 = 0.1102 (all data), GOF = 1.146. CCDC 2598312 contains the supplementary crystallographic data for this compound. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) at www.ccdc.cam.ac.uk/data_request/cif.

3. Optical and Electronic Properties

Measurement conditions. UV–vis–NIR absorption spectra were recorded on a Shimadzu UV-3600i Plus spectrophotometer. Solution spectra were measured using a quartz cuvette with a 1 cm optical path length, and thin-film spectra were measured on quartz substrates. All spectra were recorded at room temperature.

Stability test of the polymer thin film. A solution of **3b** (1.8 mg) in spectroscopic-grade $CHCl_3$ (360 µL) was stirred for 30 min. An aliquot of the resulting solution (50 µL) was spin-coated onto a quartz substrate at 700 rpm for 30 s to afford a thin film. Three thin-film samples were separately placed in glass vials and stored under the following conditions: (i) under ambient air and

laboratory illumination, (ii) under an argon atmosphere and laboratory illumination, and (iii) under ambient air in the dark. For the dark condition, the vial was completely wrapped with aluminum foil. UV–vis–NIR absorption spectra were recorded for the as-prepared films and after storage for 24, 72, and 168 h at a data interval of 0.2 nm. It should be noted that the glass vials partially attenuated the UV component of the incident laboratory light during storage. The results are shown in Figures 2a–c.

Photoelectron yield spectroscopy (PYS) measurement. A thin film of **3b** was prepared by drop-casting a $CHCl_3$ solution onto a quartz substrate. Photoelectron yield spectroscopy measurements were performed using a Bunko Keiki BIP-KV202GD. The ionization energy was determined from the photoelectron yield spectrum by linear extrapolation of the threshold region. Specifically, the cube root of the photoelectron yield intensity was plotted against the excitation energy, and the threshold energy was determined from the intersection of linear fits of the background and rising regions. Two independent measurements gave ionization energies of 5.64 and 5.65 eV, confirming the reproducibility of the measurement (Figure 3a and Figure S3).

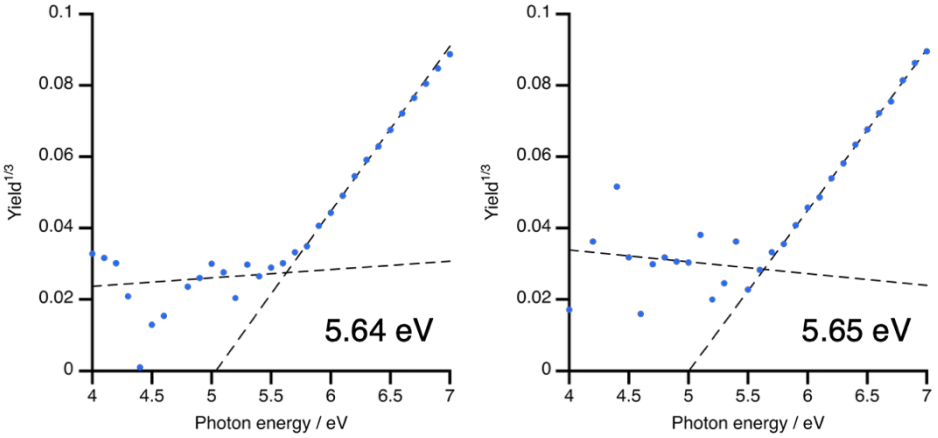


Figure S3. Photoelectron yield spectra of **3b** in the film.

Optical band gap of the polymer. The absorption spectrum of **3b** in $CHCl_3$ and that of the as-prepared thin film used for the stability test were used to estimate the optical band gaps. Absorption spectra of the polymer were measured in spectroscopic-grade $CHCl_3$ at a data interval of 0.2 nm. The absorption spectra were normalized prior to the analysis. The absorption onset was determined from the intersection of linear fits to the absorption-edge and background regions. For the solution spectrum, the regions of 750–830 and 960–1100 nm were used for the absorption edge and background, respectively. For the thin-film spectrum, the corresponding regions were 840–940 and 1017–1150 nm, respectively. The optical band gap ($E_{g,opt}$) was calculated from the absorption onset wavelength (λ_{onset}) using $E_{g,opt} = 1240/\lambda_{onset}$. The results are shown in Figure 3b.

Absorption spectra of oligomers. Absorption spectra of **4a–e** were measured in 1,2-dichlorobenzene (ODCB; anhydrous, 99%, Sigma-Aldrich) at a data interval of 0.1 nm. The results are shown in Figure 4a.

Extrapolation of the longest-wavelength absorption maxima of π-conjugated oligomers. The longest-wavelength absorption bands of oligomers **4a–e** were analyzed by Gaussian peak fitting using Origin® to determine the absorption maxima. The absorption maxima of oligoenes[S6] and oligo(EDOT)s[S7] were obtained from the reported literature values. The obtained absorption maxima were summarized in Table S1. The absorption maxima were plotted against the number of repeating units (n) and fitted using the Meier equation using KaleidaGraph. The fitted curves and parameters are shown in Figure 6b, Figure S4, and Table S2, respectively.

Table S1. Longest-wavelength absorption maxima (λ_n, nm) of oligo(DBF)s **4a–e**, oligoenes **12**[S6], and oligo(EDOT)s **13**[S7]

Number of repeating units (n)	1	2	3	4	5	6	7	8	9
Oligo(DBF)s **4**	475	596	665	702	728	–	–	–	–
Oligoenes **12**	–	–	308	342	370	398	420	442	460
Oligo(EDOT)s **13**	–	350	412	458	498	–	–	–	–

Table S2. Extrapolation of the longest-wavelength absorption maxima of π-conjugated oligomers using the Meier equation

	n_0[a]	λ_{n_0} / nm[b]	λ_∞ / nm[c]	ΔE_{opt} / eV[d]	a	R^2[e]
Oligo(DBF)s **4**[f]	1	475	757	1.64	0.557	0.99984
Oligoenes **12**[g]	3	308	610	2.03	0.117	0.99986
Oligo(EDOT)s **13**[h]	2	350	644	1.92	0.232	0.99976

[a] The number of repeating units in the shortest member examined, [b] longest absorption wavelength of the shortest oligomer, [c] longest absorption wavelength of the ideal polymer. [d] Optical band gap calculated from λ_∞. [e] Coefficient of determination for the fitting based on eq. (1). [f] Extrapolated by using λ_n data determined for **4a–e** as model compounds. [g] Extrapolated by using λ_n data determined for **12** (n = 3–9) as model compounds. [h] Extrapolated by using λ_n data determined for **13** (n = 2–5) as model compounds.

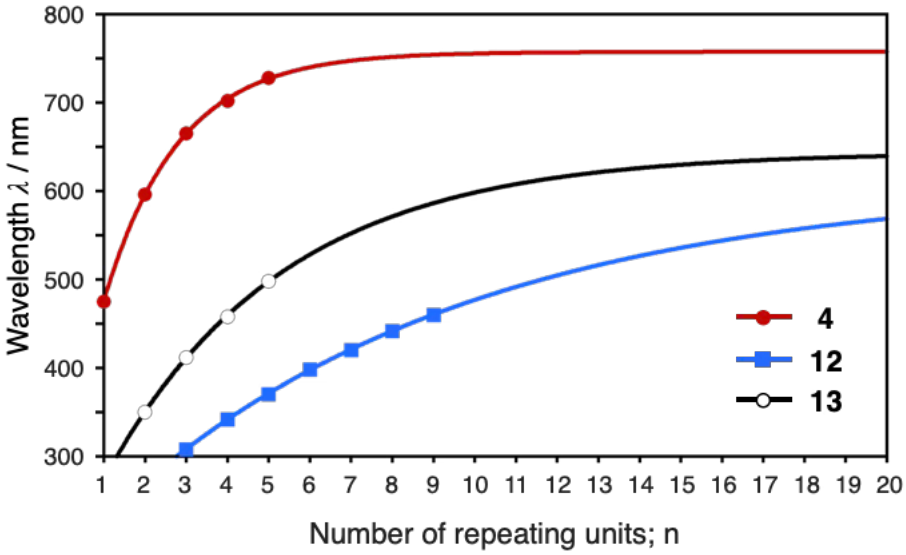


Figure S4. Plots of the longest-wavelength absorption maximum and fitted curves of **4**, **12**, and **13** against the number of repeating units.

4. Electrochemical Properties

Methods. Cyclic voltammograms of **4a–d** were recorded on an ALS/chi-610E electrochemical analyzer with a glassy-carbon working electrode, a Pt-wire counter electrode, and an $Ag/AgNO_3$ reference electrode under an argon atmosphere. The measurements were carried out using a 1.0 mM solution of the sample in toluene/acetonitrile (5.4:1, v/v) containing 0.1 M [*n*-Bu_4N][PF_6] as the supporting electrolyte at a scan rate of 0.1 V s^{-1}. The potentials were calibrated against the Fc/Fc^+ redox couple. The results were summarized in Figure 8 and Table 1.

5. Quantum Chemical Calculations

Quantum chemical calculations. Density functional theory (DFT) calculations were performed using the Gaussian 16 Revision B.01 program.[S8] Geometry optimizations were carried out at the PBE0/6-31G* level of theory using the default thresholds and algorithms. The PBE0 functional was employed based on our previous benchmark calculations on dibenzopentafulvalene oligomers, in which PBE0 was selected as the most suitable functional among those examined.[S9] The HOMO and LUMO energy levels and orbital distributions were obtained from the optimized structures. Calculations were performed for oligoenes **14a–m** (n = 3–15), fulvalene oligomers **15a–e** (n = 1–5), 3,4-diphenylhexatriene **16**, and methoxy-substituted fulvalene oligomers **17a,b** (n = 1, 2). The HOMO and LUMO distributions of the phenyl-capped trimer shown in Figure 1d were obtained from our previously reported calculation at the PBE0/6-31+G* level of theory.[S9]

Investigation of substituent effects. To elucidate the substituent effects of alkoxy groups on the frontier molecular orbitals of fulvalene oligomers, a computational comparison was performed. To reduce the computational cost, a methoxy group was employed as a model alkoxy substituent, and the orbital shapes and energy levels of the fulvalene monomer and dimer were compared in the presence and absence of methoxy substitution (Figure S5). Methoxy substitution caused no appreciable change in the shapes of the frontier molecular orbitals of either the monomer or the dimer. Although both the HOMO and LUMO energy levels shifted slightly upward, the HOMO–LUMO gaps remained essentially unchanged. These results indicate that alkoxy substituents have a negligible effect on the frontier molecular orbitals of fulvalene oligomers.

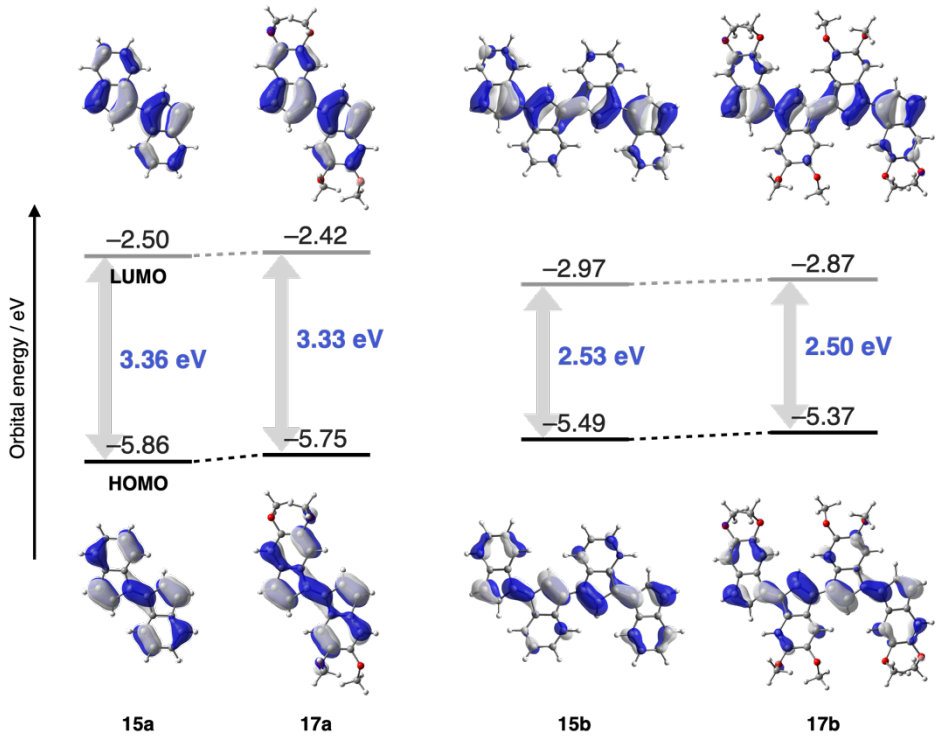


Figure S5. Orbital energy diagrams and selected Kohn–Sham molecular orbitals (isovalue = 0.03) for **15a,b** and **17a,b**, calculated at the PBE0/6-31G* level of theory.

6. MD Simulations

Computational details. Initial geometry optimizations of compounds **1d** and **1e** were performed using DFTB+ version 22.2 with the third-order density-functional tight-binding method (DFTB)[S10] and the 3ob parameter set. Grimme's D3 dispersion correction with Becke–Johnson damping was included.[S11] The Hubbard parameters were set to −0.1492, −0.1575, and −0.1857 for C, O, and H, respectively.[S12] Using the optimized structures as the initial geometries, molecular dynamics (MD) simulations were subsequently performed at the same level of theory. The DFTB-MD simulations were carried out for 1 ns with a time step of 0.2 fs at 300 K under the NVT ensemble using a Nosé-Hoover thermostat[S13,14]. Snapshots were collected at intervals of 1 ps, yielding 1000 structures for subsequent conformational analysis. Each of the 1000 snapshots was further optimized at the same level of theory to remove instantaneous thermal distortions and identify the corresponding local-minimum structures. The most stable conformations of **1d** and **1e** obtained from the DFTB optimizations were then further optimized at the B3LYP/cc-pVDZ level of theory using Gaussian 16 Rev. C.01.[S8]

For the conformational analysis, the 1000 conformations obtained for each of **1d** and **1e** from the DFTB optimizations were classified on the basis of the signs of the dihedral angles defined in Figure S6. The dihedral angles θ_2–θ_8 for **1d** and θ_2–θ_{10} for **1e** were considered, respectively. Two optimized structures showing the same sequence of positive and negative signs for the dihedral angles were regarded as the same conformation. The number of occurrences of each distinct conformation among the 1000 optimized snapshots was then counted. The MD trajectories of **1d** and **1e** were visualized as animated GIFs using the 1000 snapshots (Movies S1 and S2, respectively).

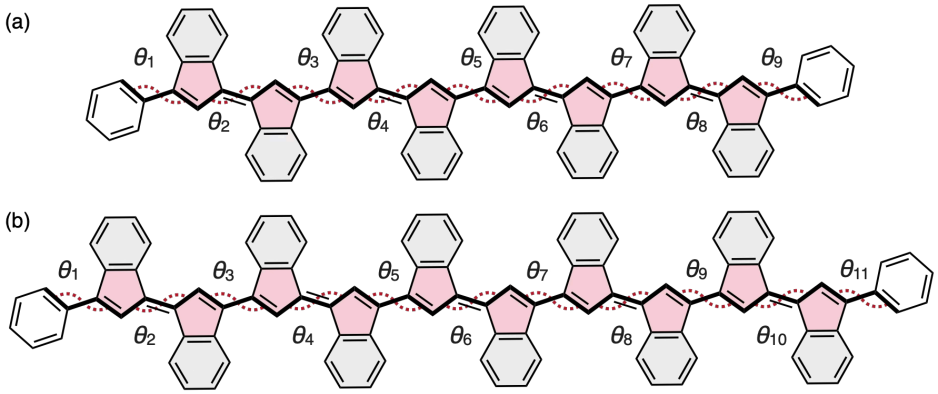


Figure S6. Definition of dihedral angles of (a) **1d** and (b) **1e**.

7. References

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Appendix 1. NMR Spectra of New Compounds

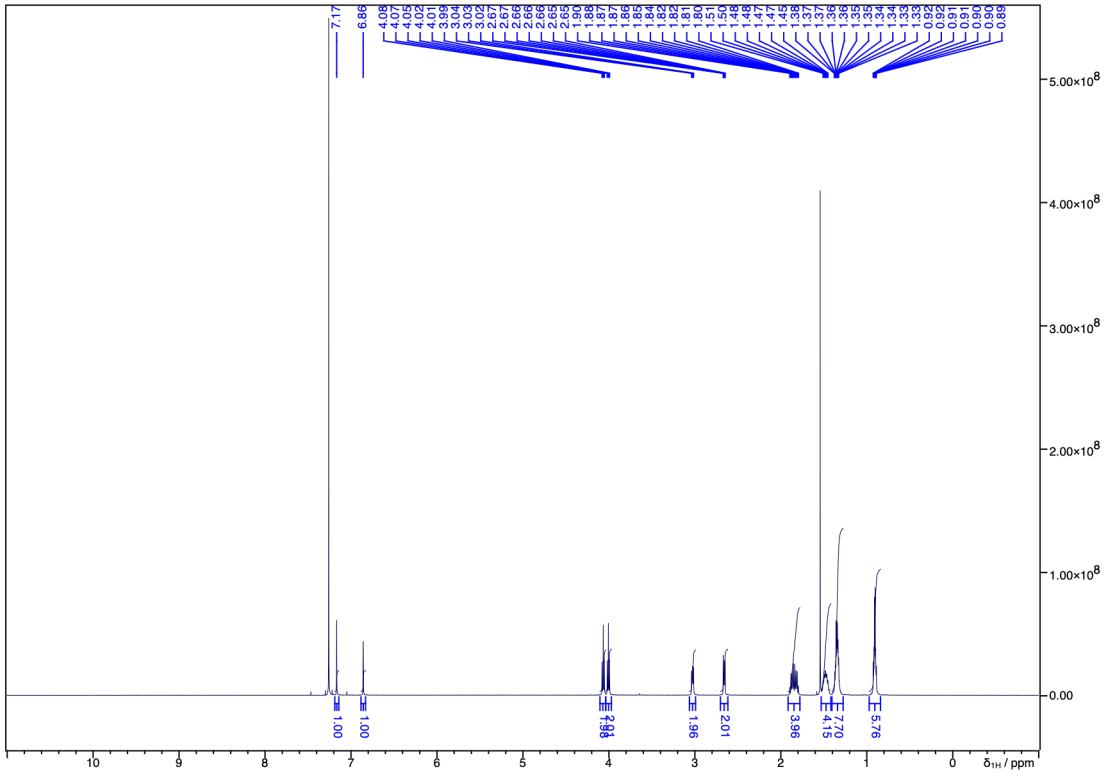


Figure S7. ^{1}H NMR spectrum of **S1** (500 MHz, $CDCl_3$)

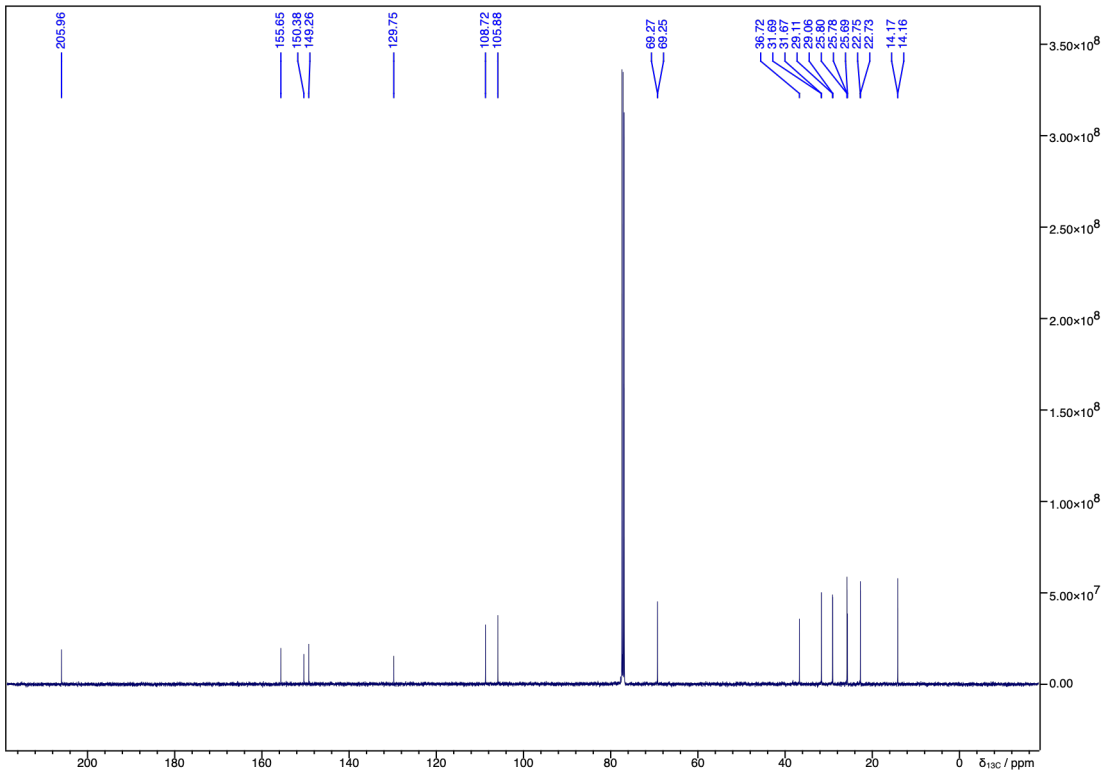


Figure S8. ^{13}C NMR spectrum of **S1** (126 MHz, $CDCl_3$)

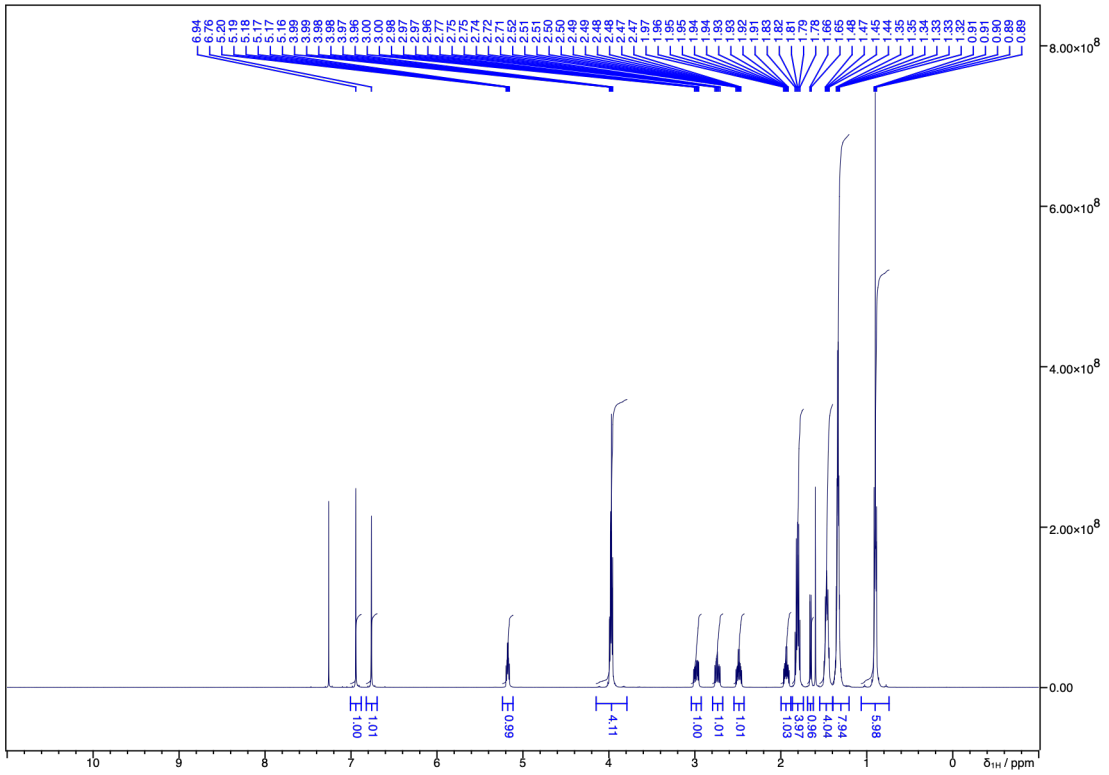


Figure S9. ^{1}H NMR spectrum of **S2** (500 MHz, $CDCl_3$)

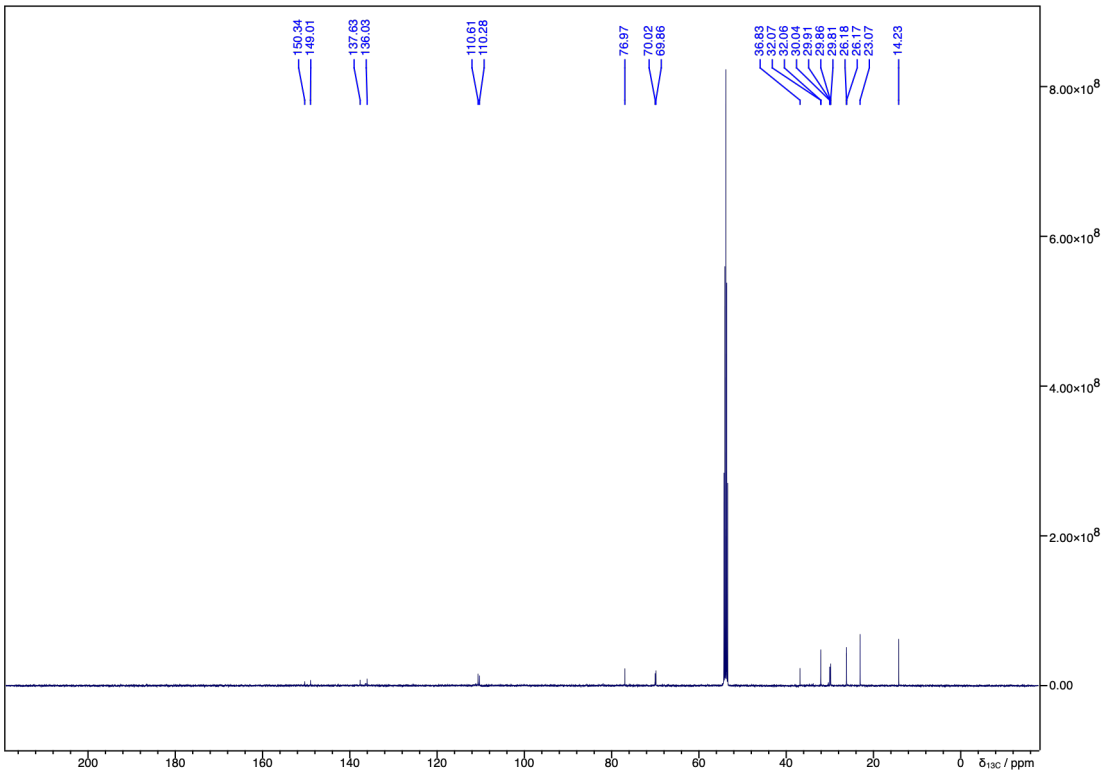


Figure S10. ^{13}C NMR spectrum of **S2** (126 MHz, CD_2Cl_2)

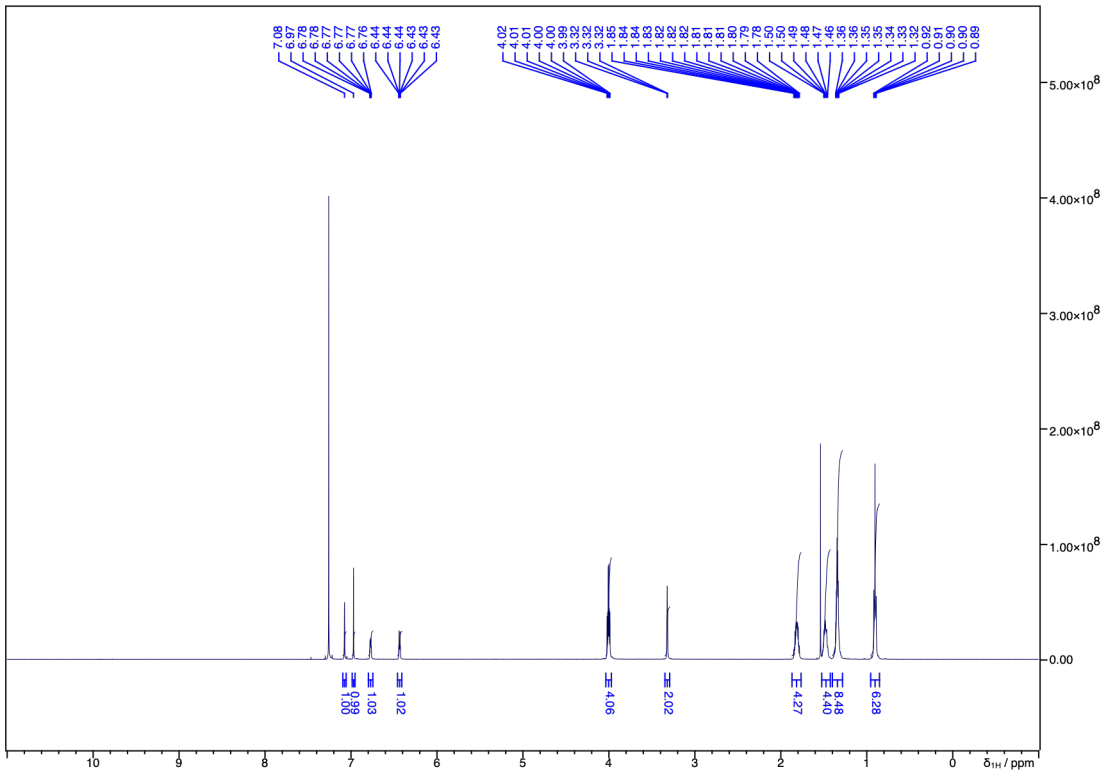


Figure S11. ^{1}H NMR spectrum of **S3** (500 MHz, $CDCl_3$)

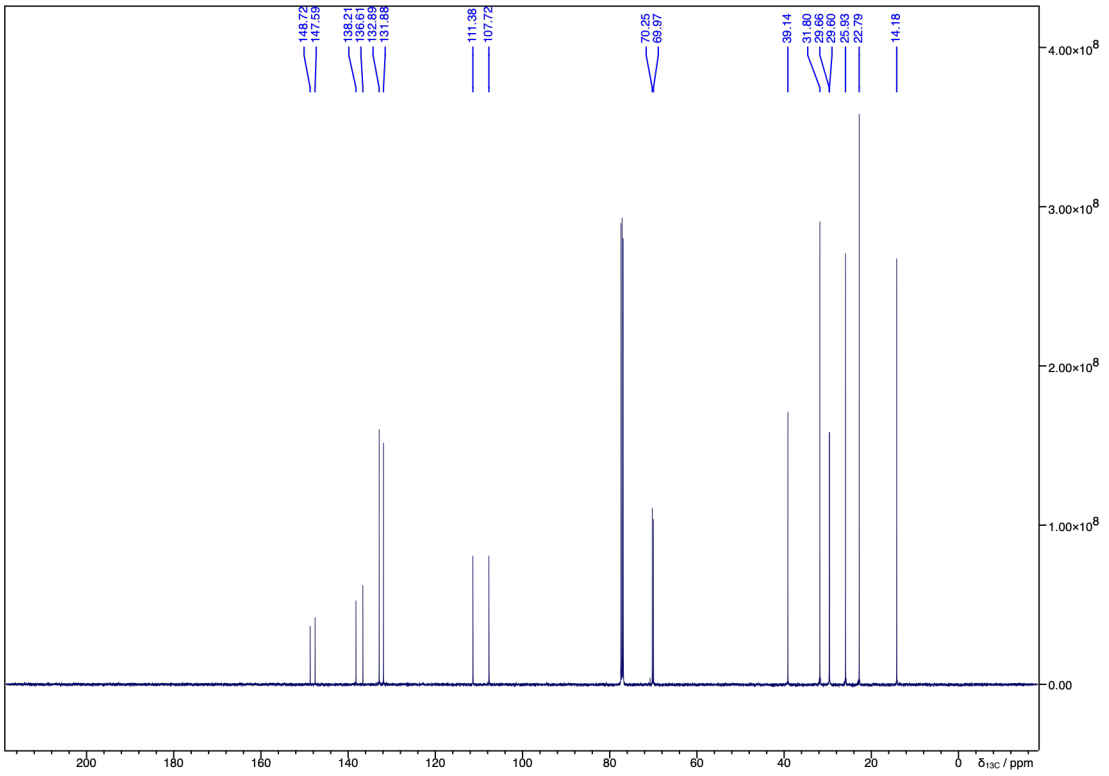


Figure S12. ^{13}C NMR spectrum of **S3** (126 MHz, $CDCl_3$)

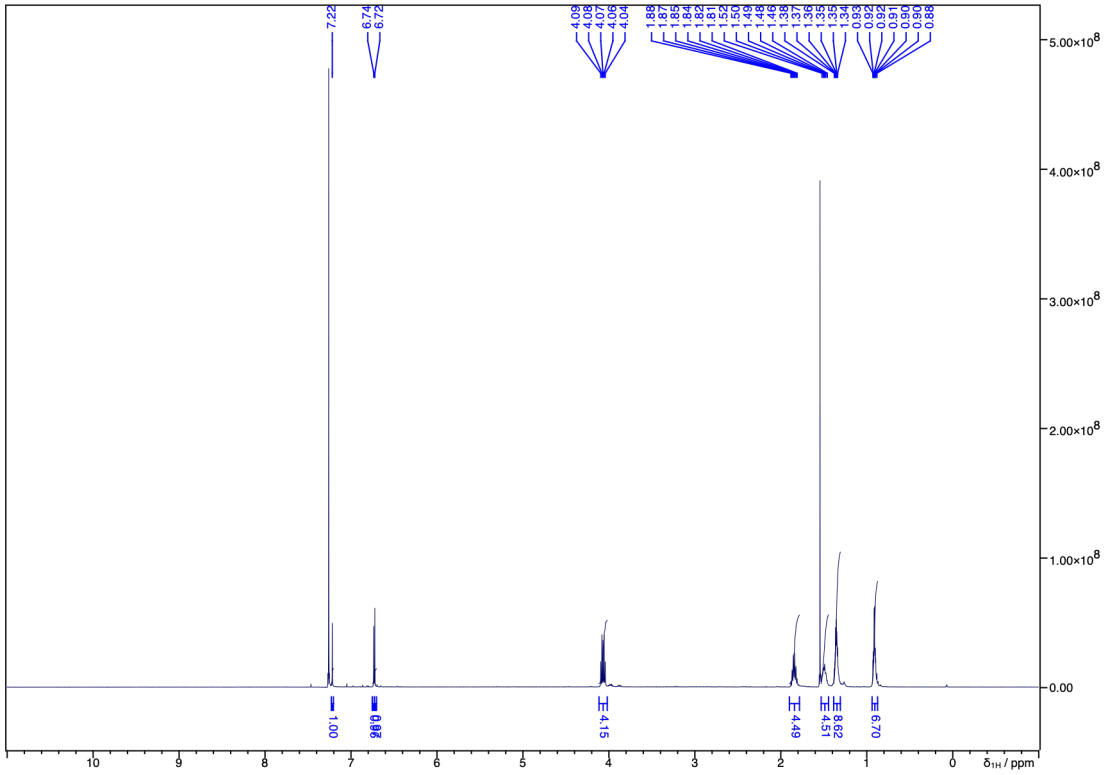


Figure S13. ^{1}H NMR spectrum of **S4** (500 MHz, $CDCl_3$)

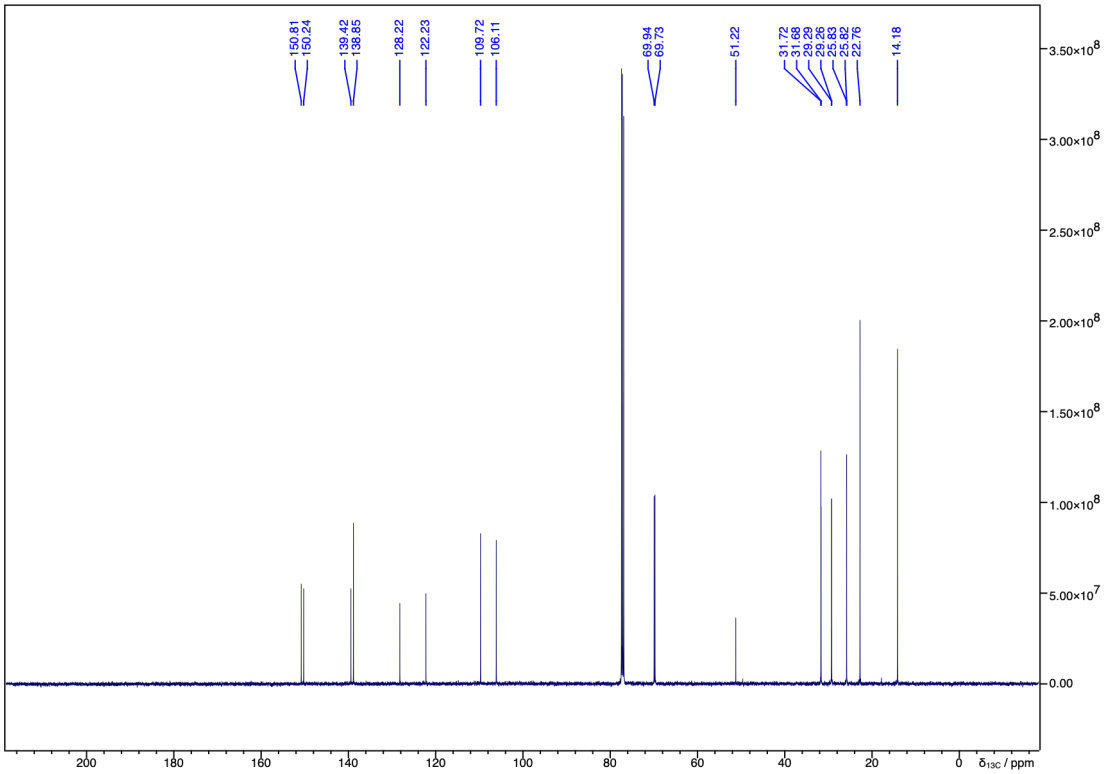


Figure S14. ^{13}C NMR spectrum of **S4** (126 MHz, $CDCl_3$)

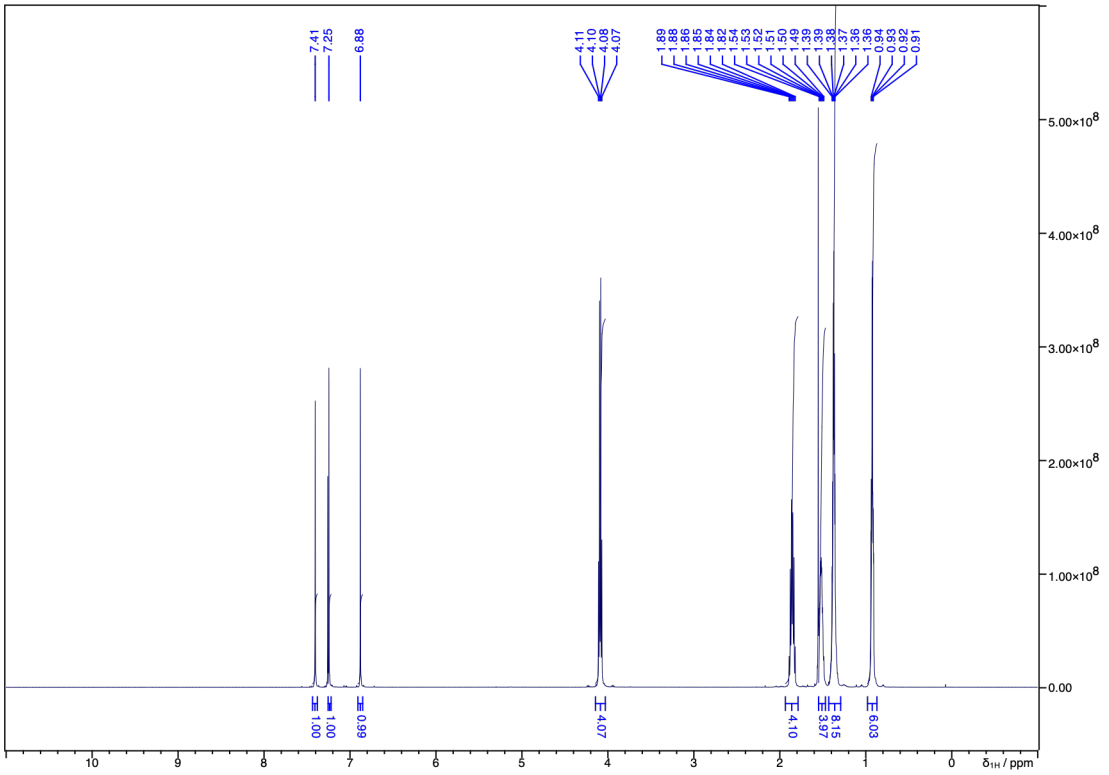


Figure S15. ^{1}H NMR spectrum of **5b** (500 MHz, $CDCl_3$)

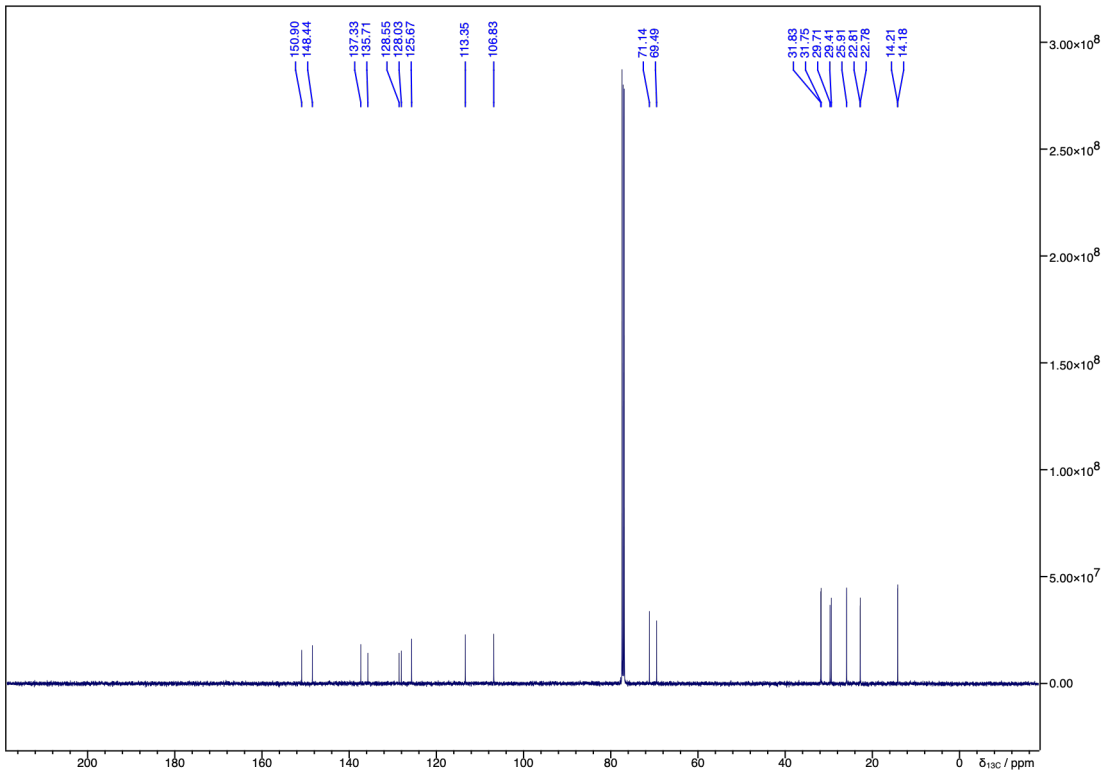


Figure S16. ^{13}C NMR spectrum of **5b** (126 MHz, $CDCl_3$)

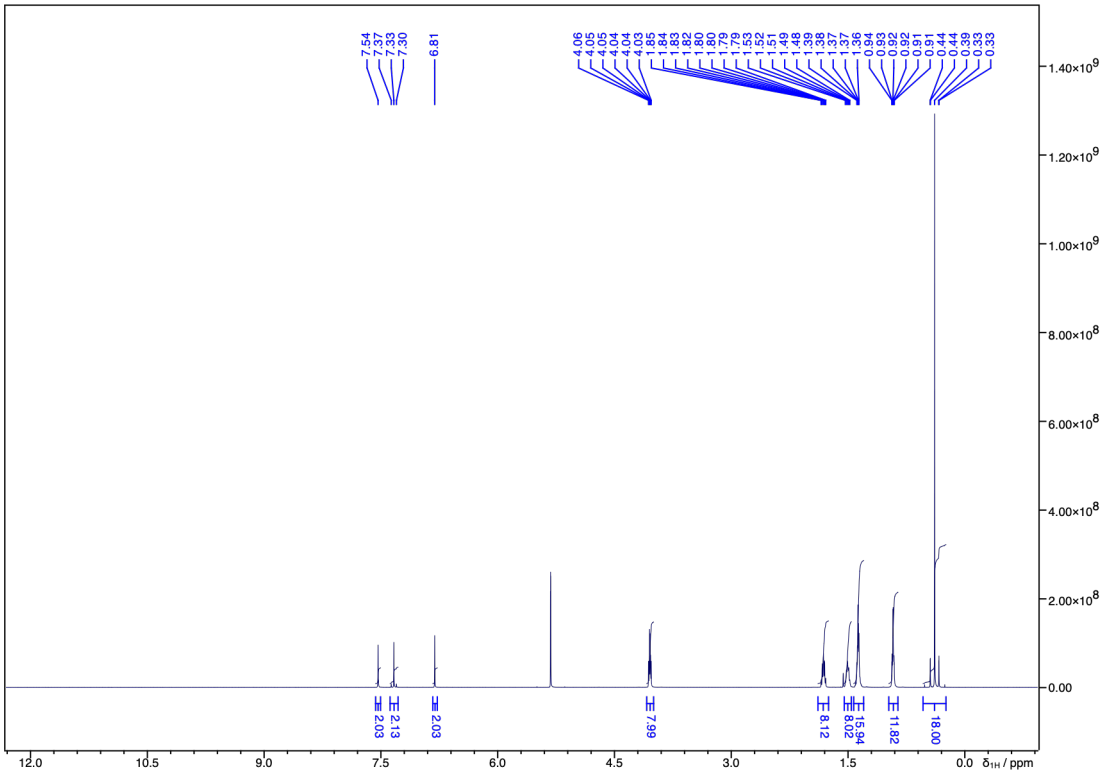


Figure S17. ^{1}H NMR spectrum of **7b** (500 MHz, CD_2Cl_2)

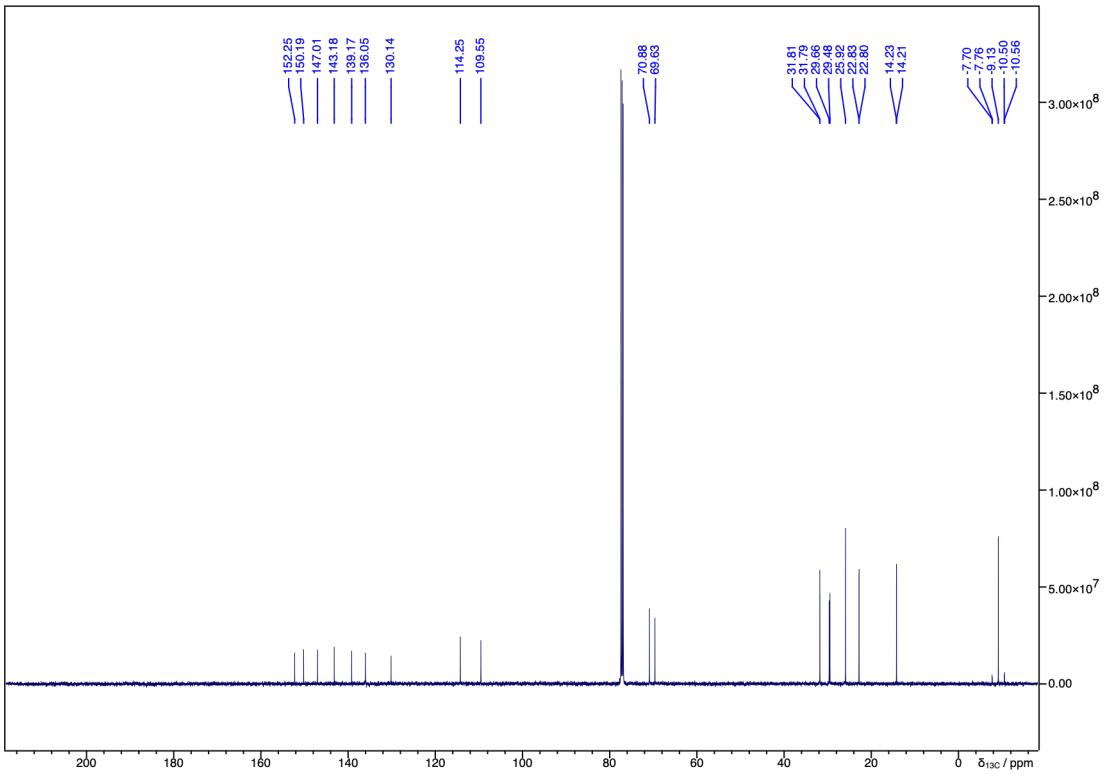


Figure S18. ^{13}C NMR spectrum of **7b** (126 MHz, $CDCl_3$)

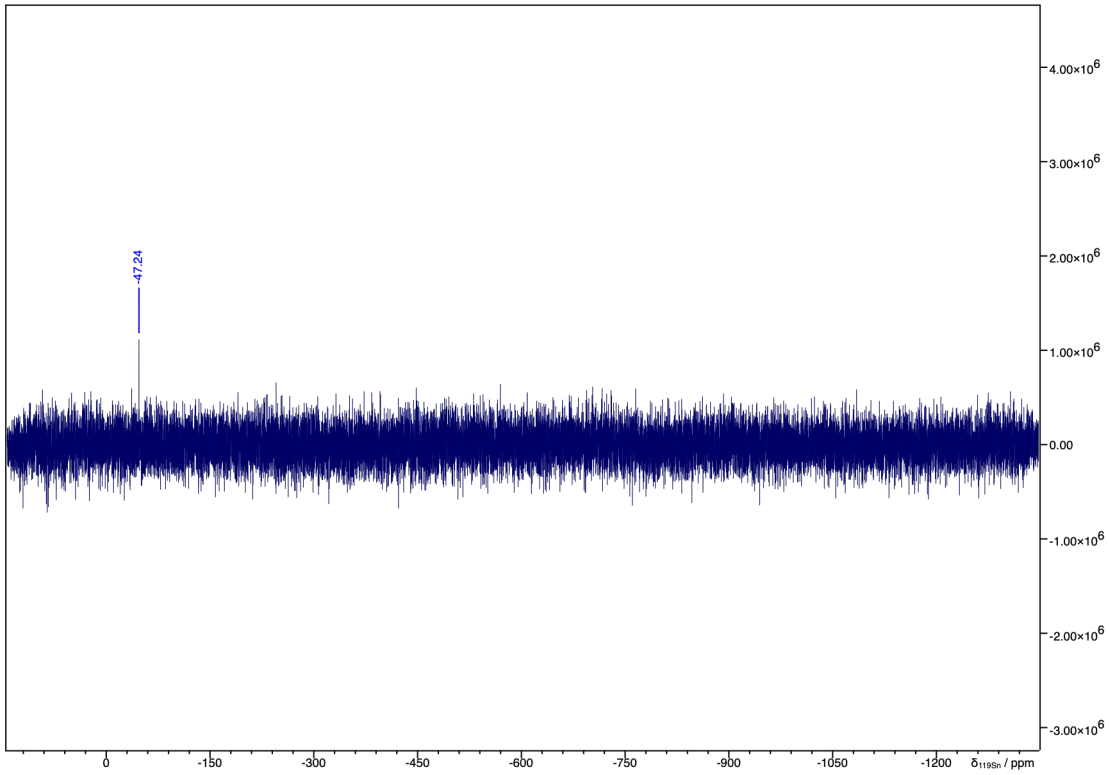


Figure S19. ^{119}Sn NMR spectrum of **7b** (187 MHz, CD_2Cl_2)

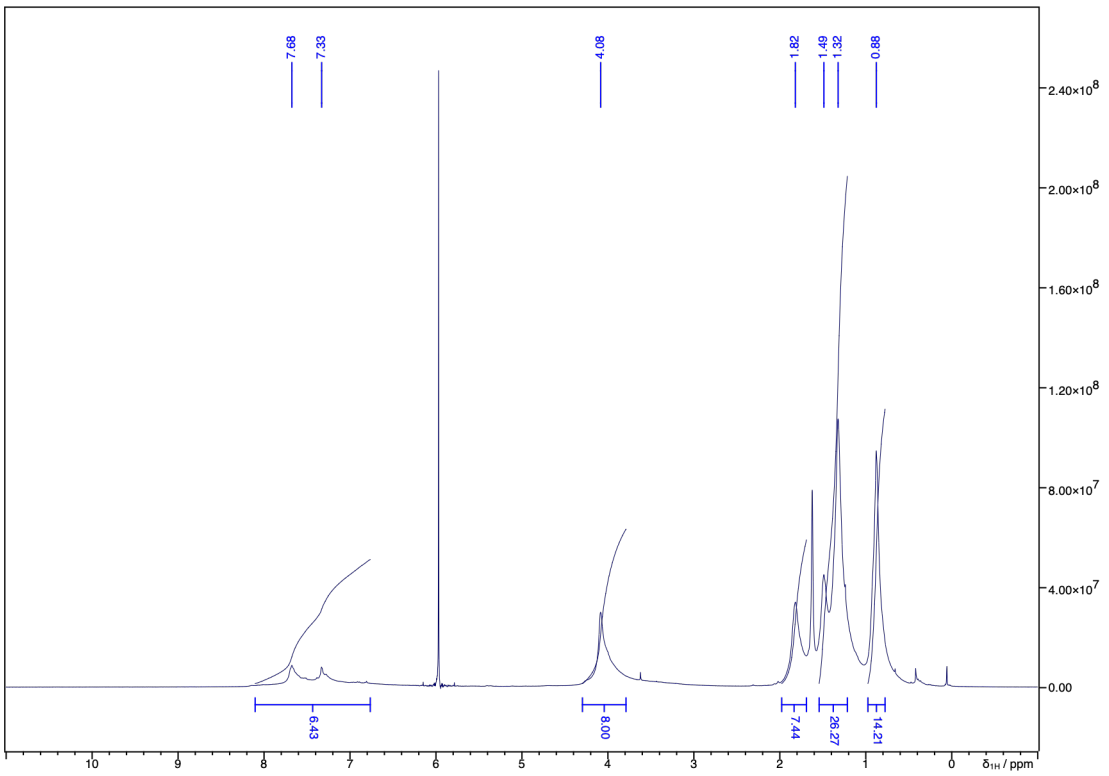


Figure S20. ^{1}H NMR spectrum of **3b** (500 MHz, TCE-d_2)

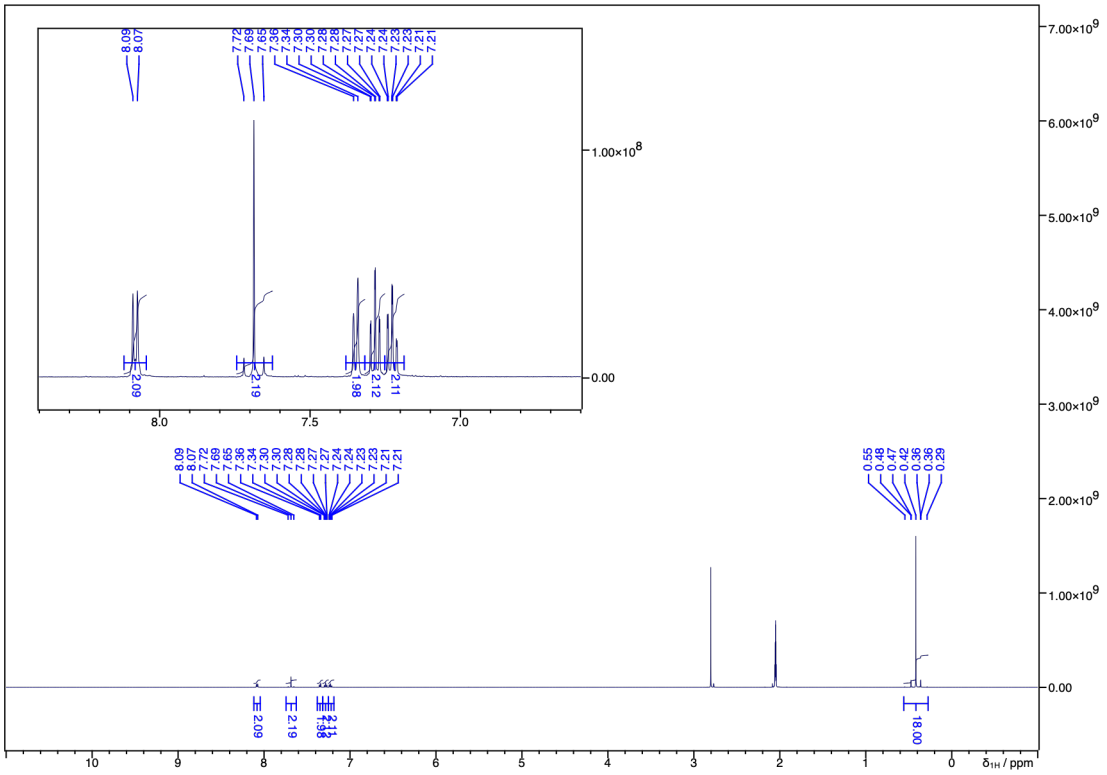


Figure S21. [1]H NMR spectrum of **7a** (500 MHz, acetone-d_6)

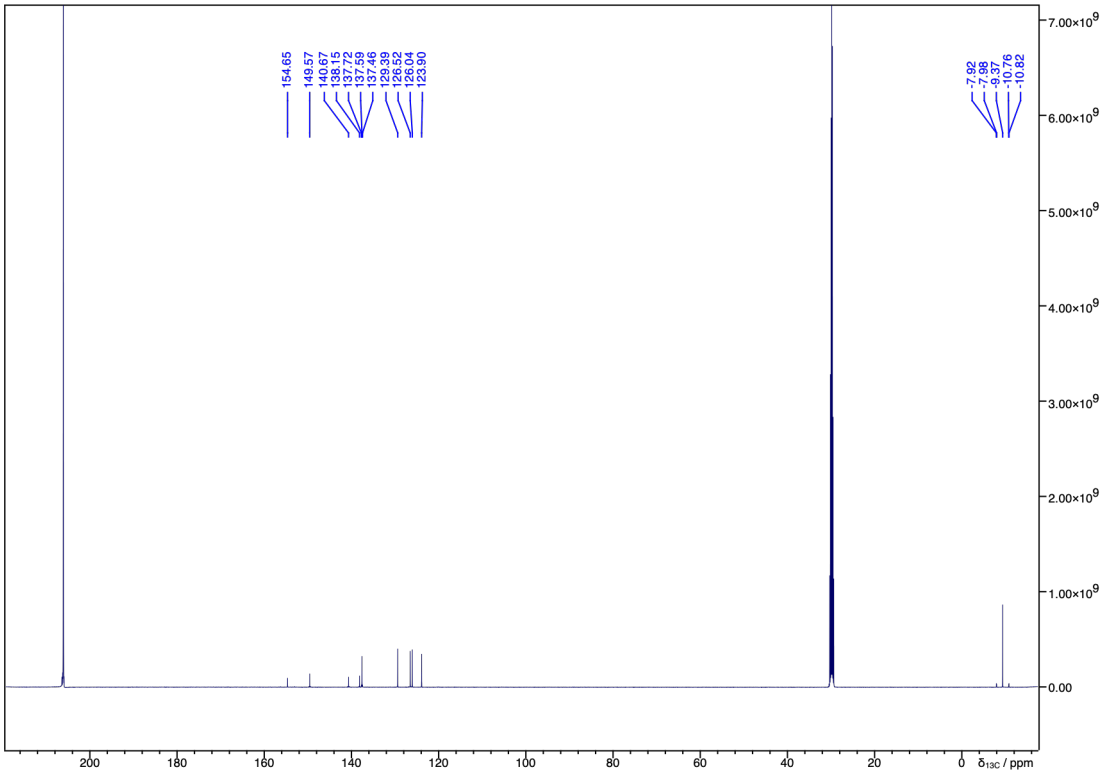


Figure S22. [13]C NMR spectrum of **7a** (126 MHz, acetone-d_6)

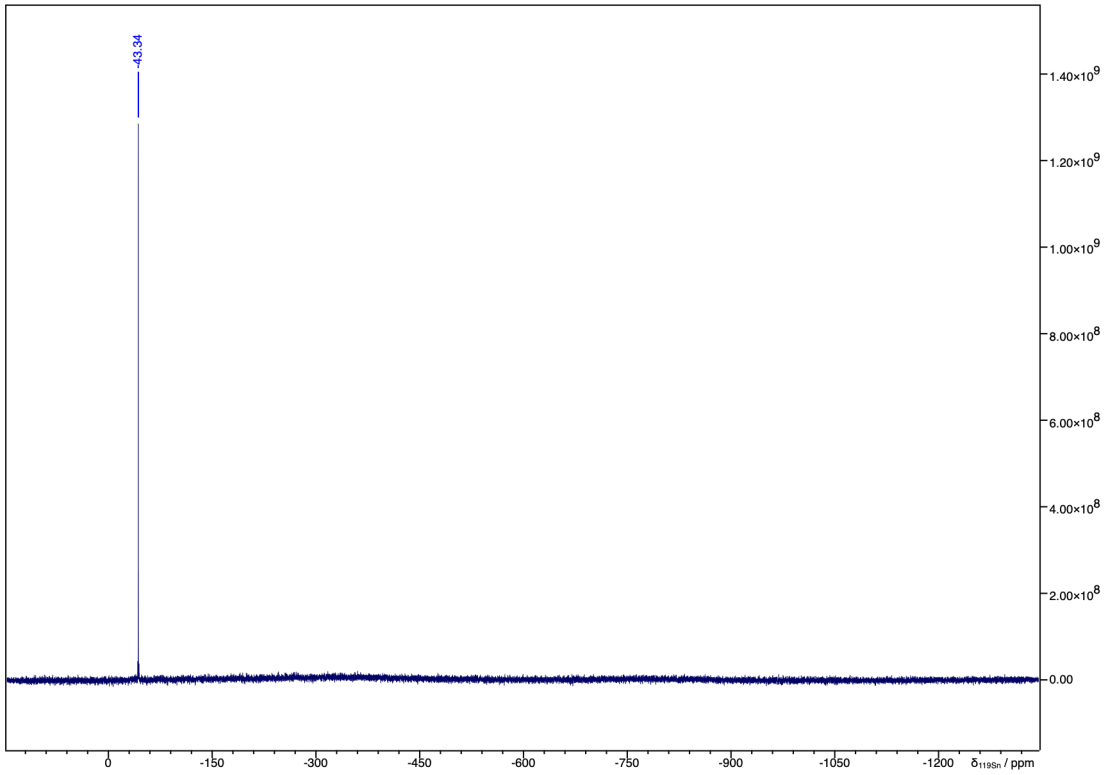


Figure S23. ^{119}Sn NMR spectrum of **7a** (187 MHz, acetone-d_6)

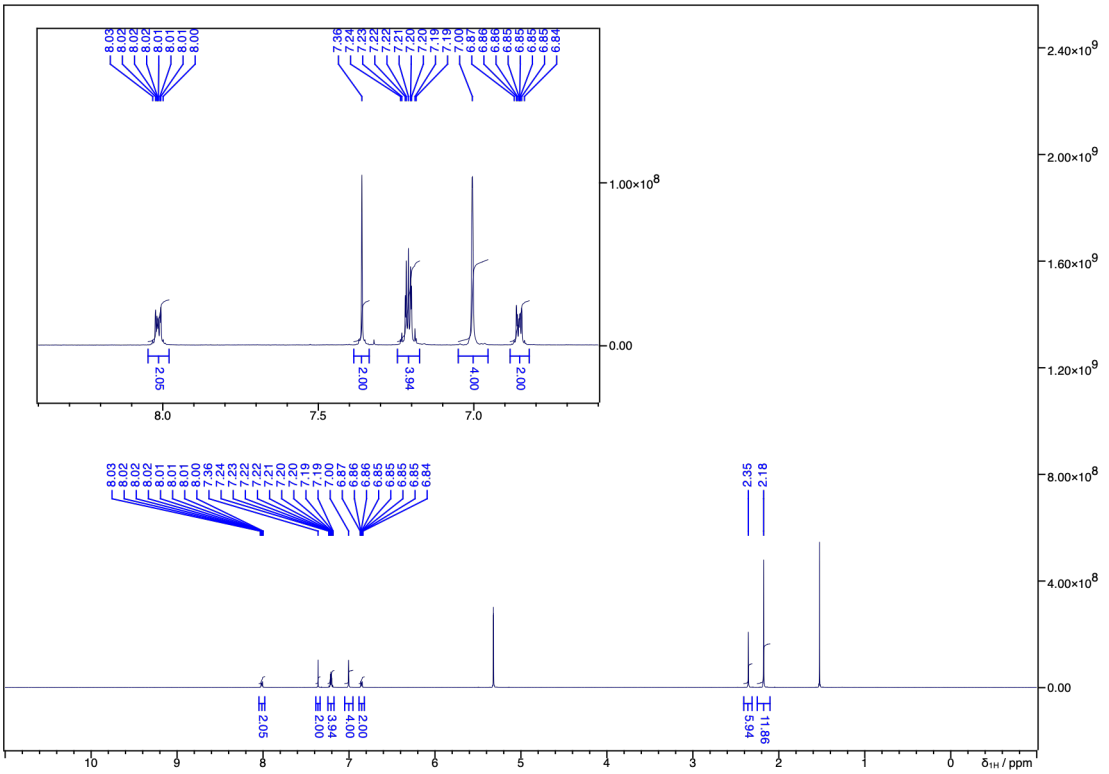


Figure S24. ^{1}H NMR spectrum of **4a** (500 MHz, CD_2Cl_2)

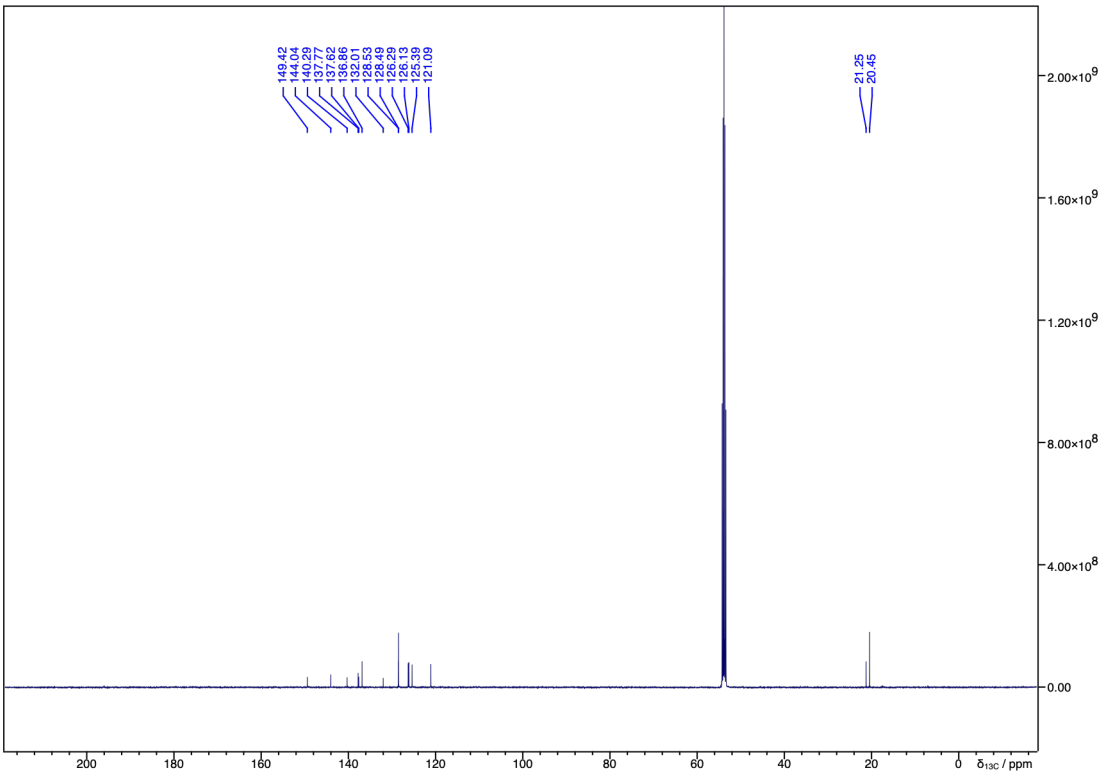


Figure S25. ^{13}C NMR spectrum of **4a** (126 MHz, CD_2Cl_2)

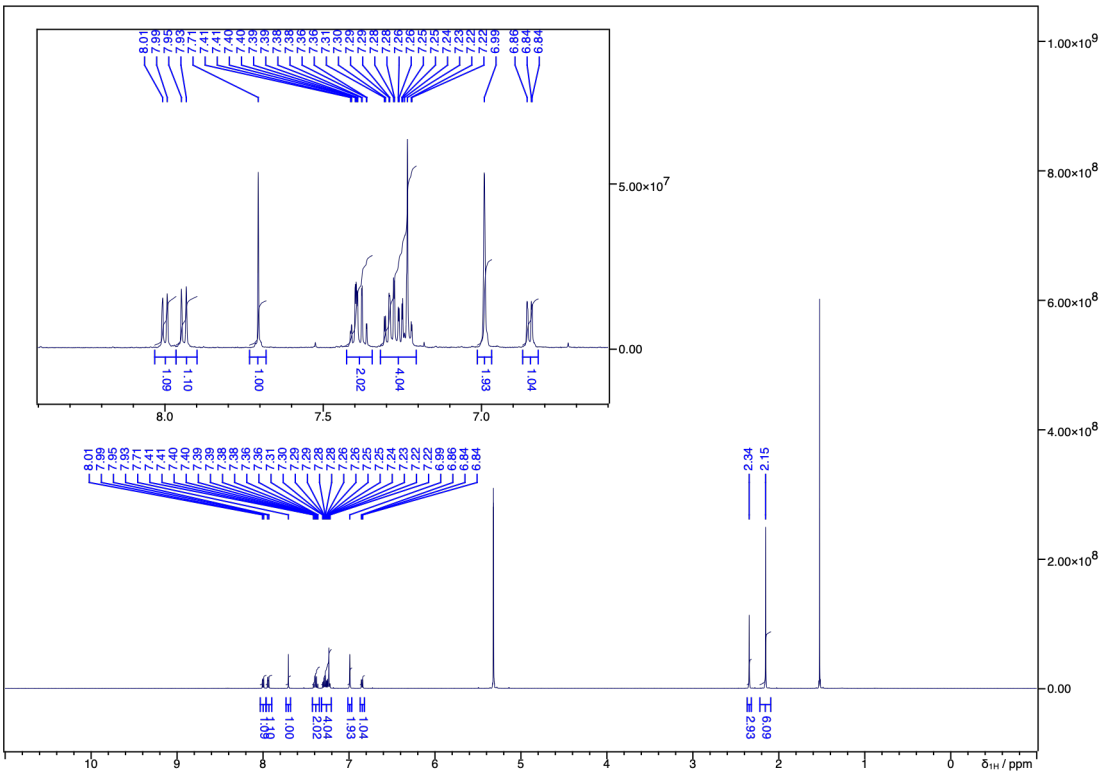


Figure S26. ^{1}H NMR spectrum of **9** (500 MHz, CD_2Cl_2)

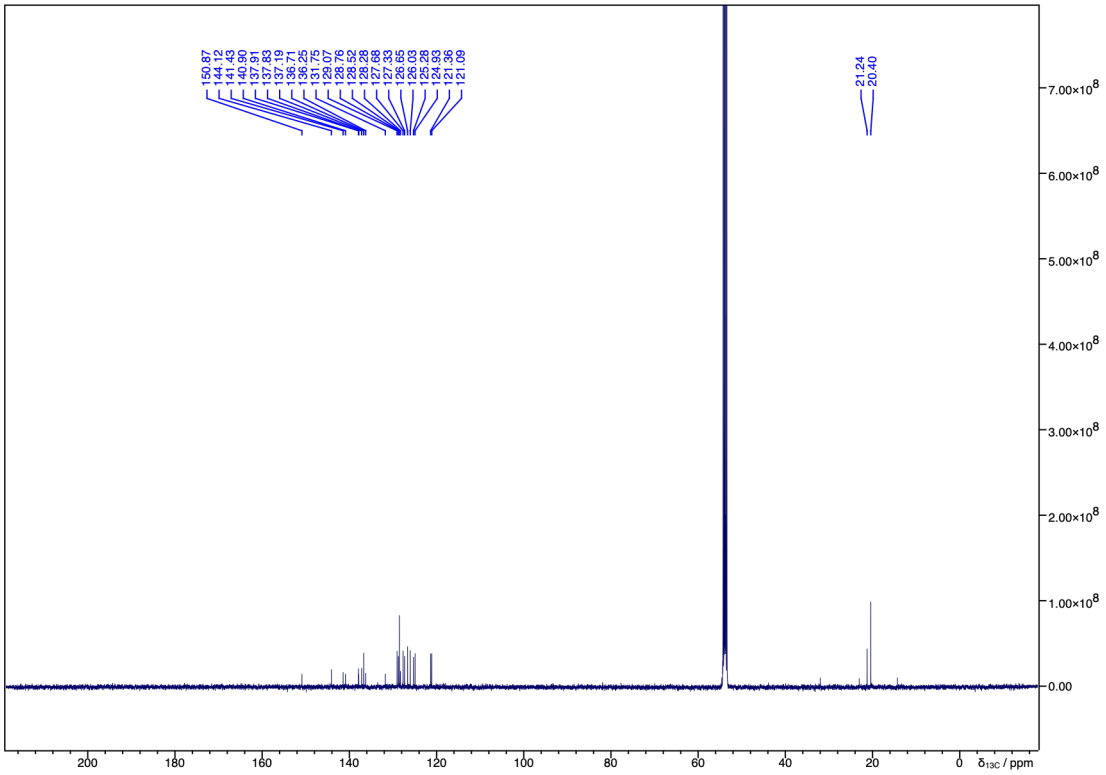


Figure S27. ^{13}C NMR spectrum of **9** (126 MHz, CD_2Cl_2)

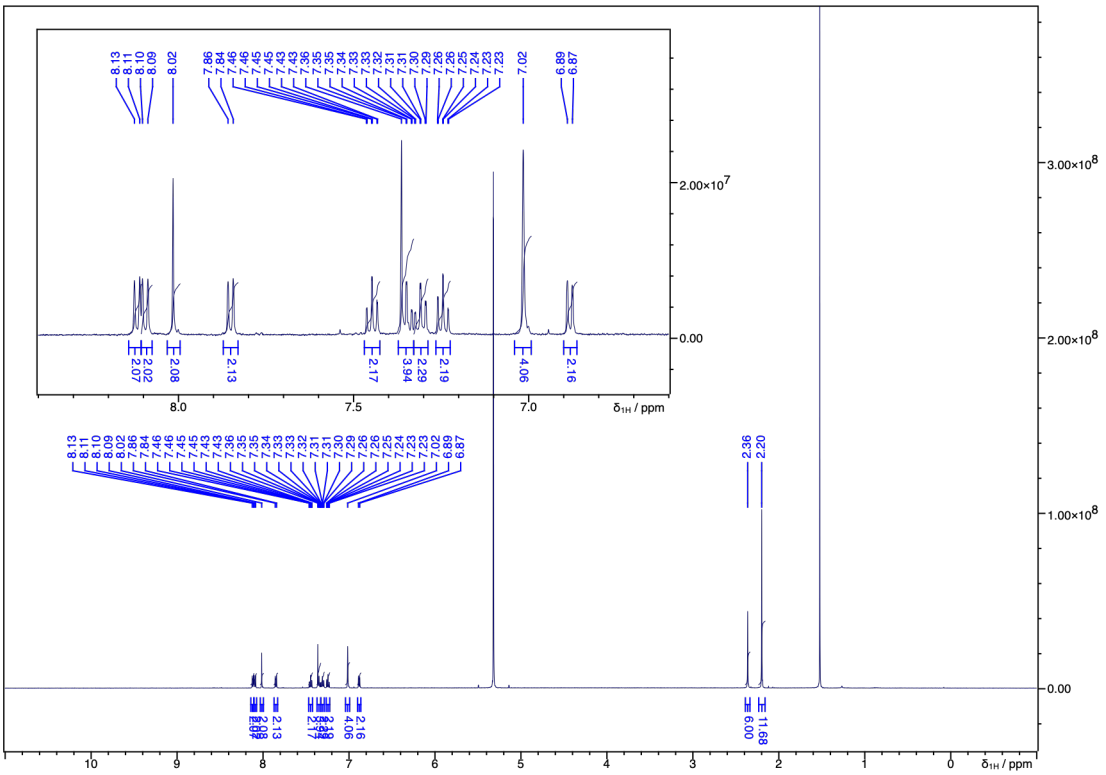


Figure S28. ^{1}H NMR spectrum of **4b** (500 MHz, CD_2Cl_2)

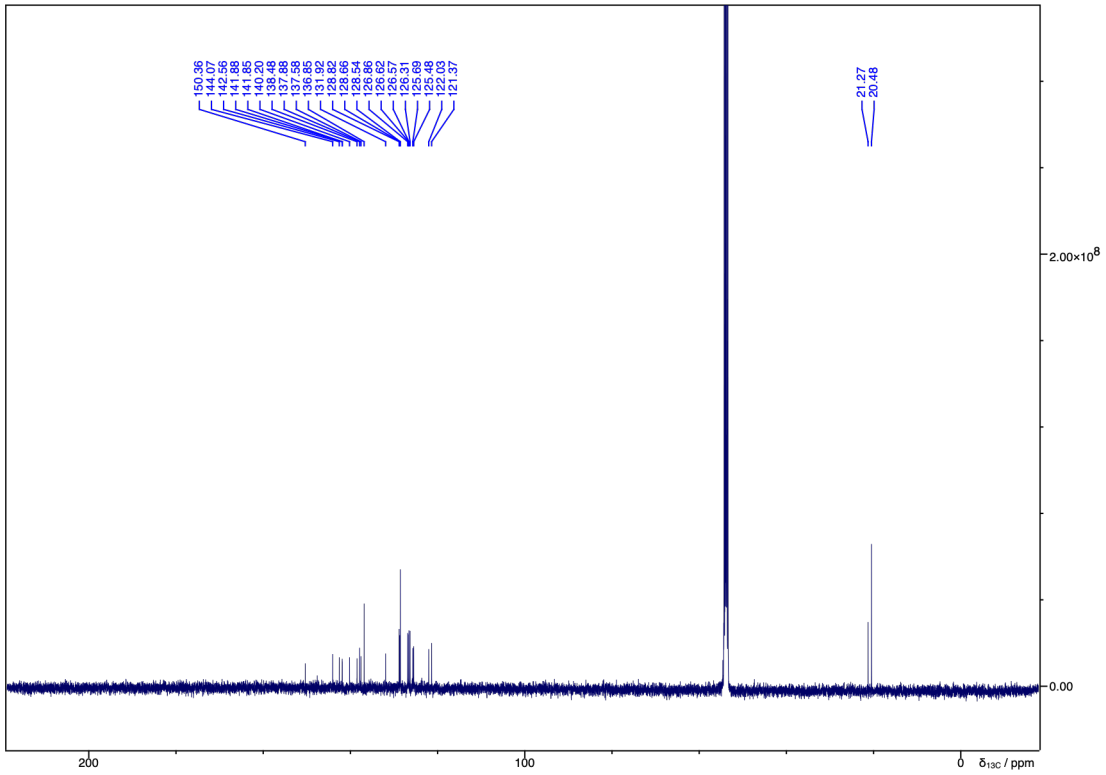


Figure S29. ^{13}C NMR spectrum of **4b** (126 MHz, CD_2Cl_2)

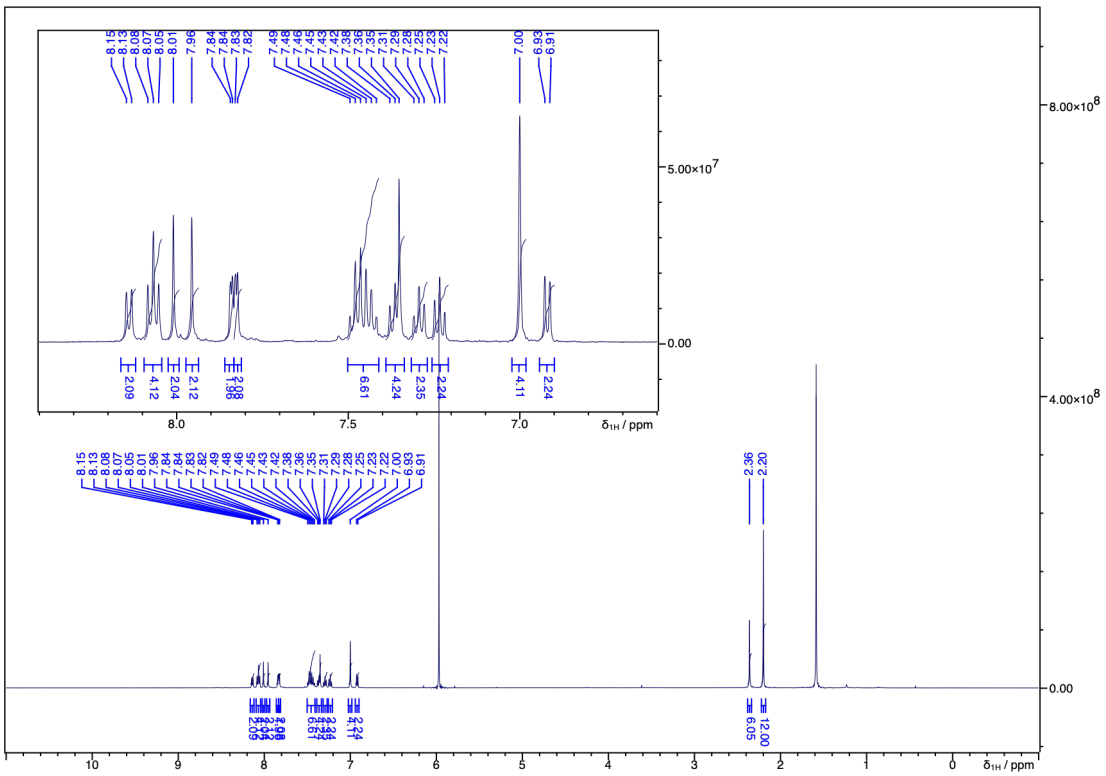


Figure S30. ^{1}H NMR spectrum of **4c** (500 MHz, TCE-d_2)

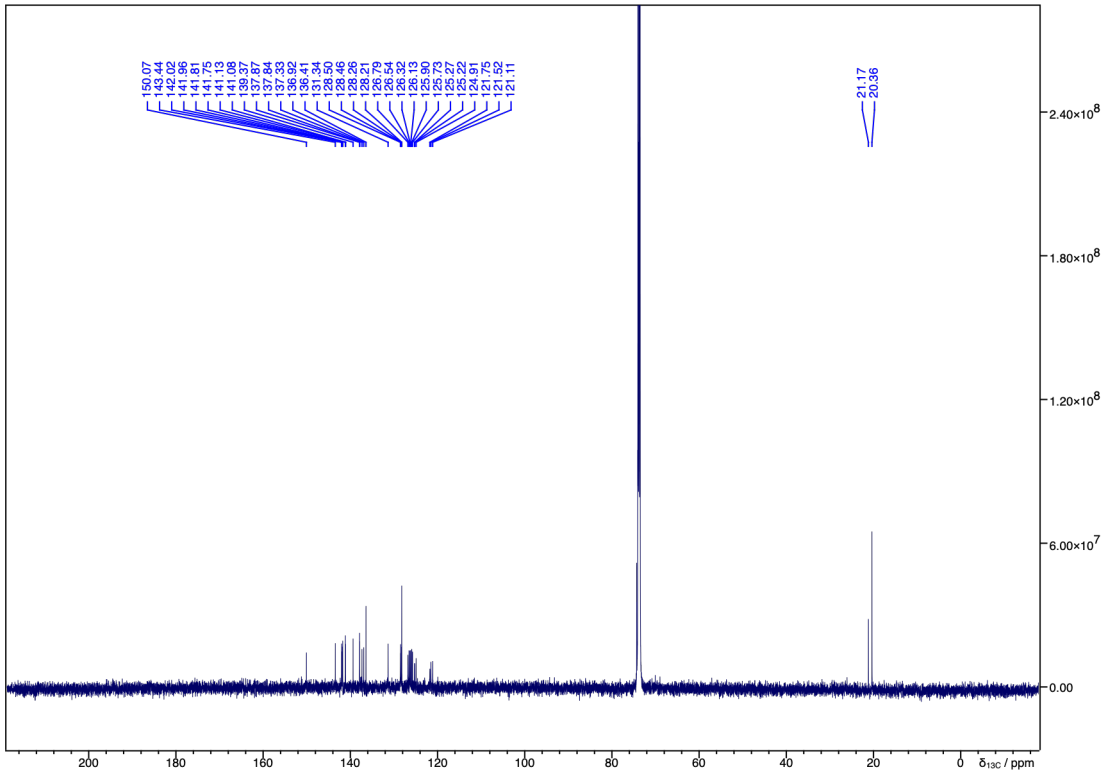


Figure S31. ^{13}C NMR spectrum of **4c** (126 MHz, TCE-d_2)

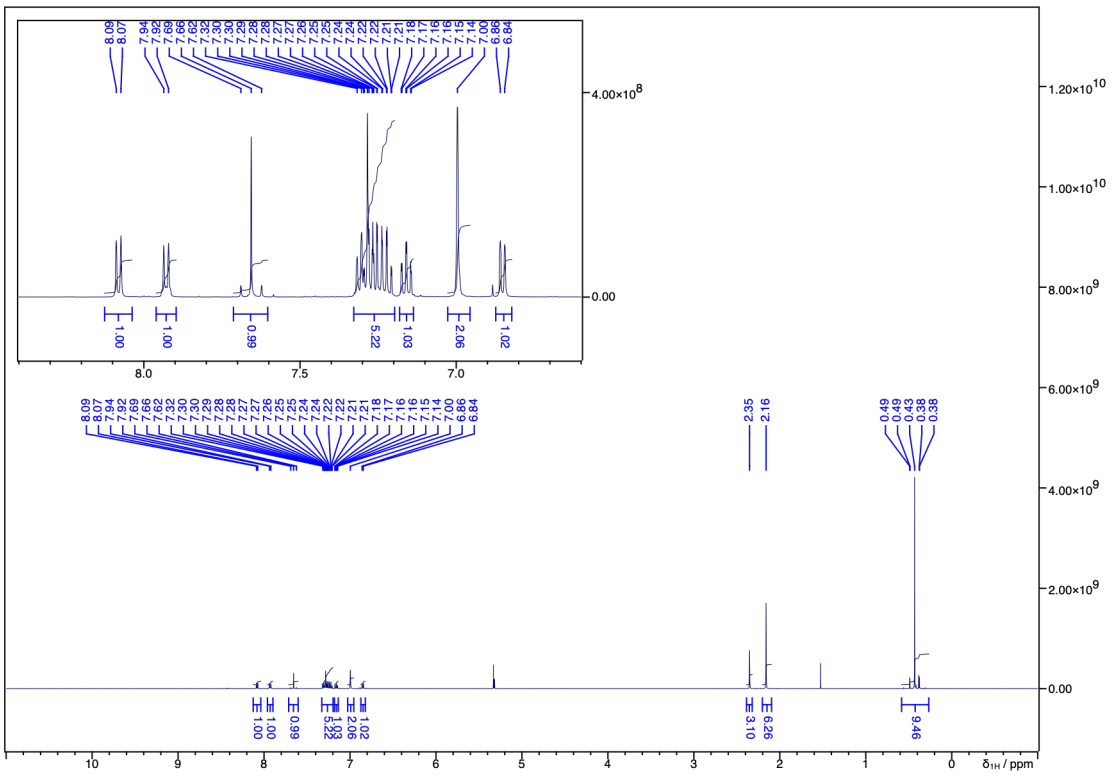


Figure S32. ^{1}H NMR spectrum of **10** (500 MHz, CD_2Cl_2)

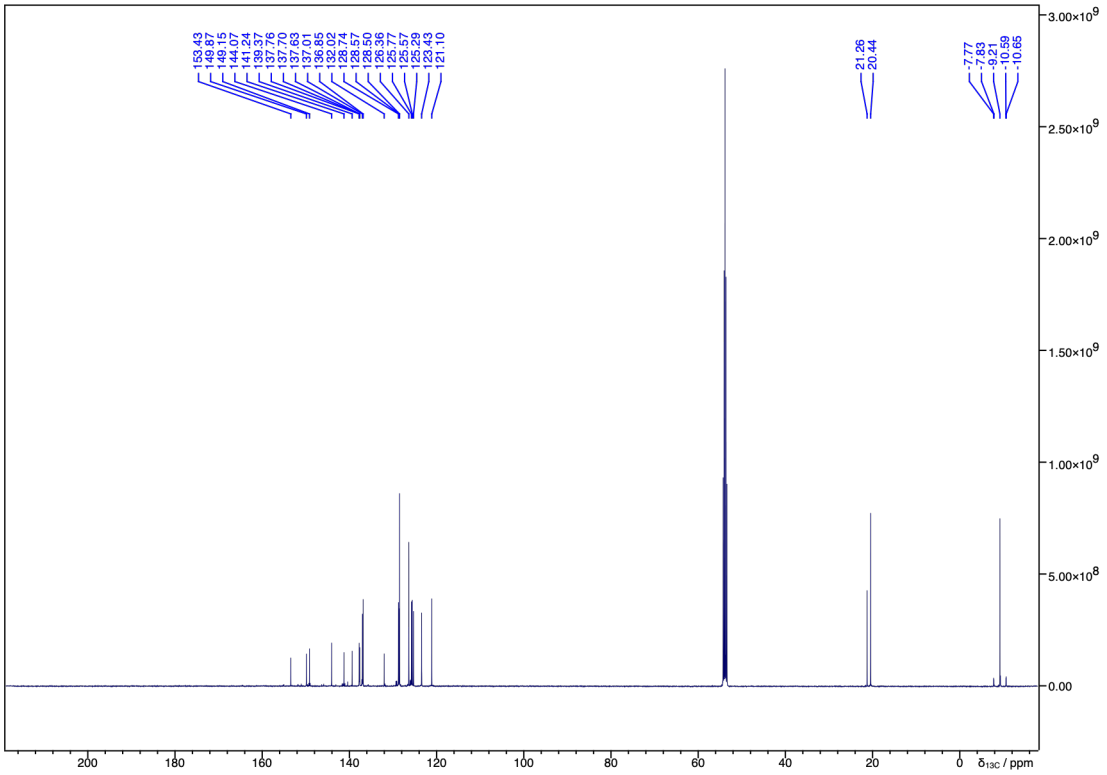


Figure S33. ^{13}C NMR spectrum of **10** (126 MHz, CD_2Cl_2)

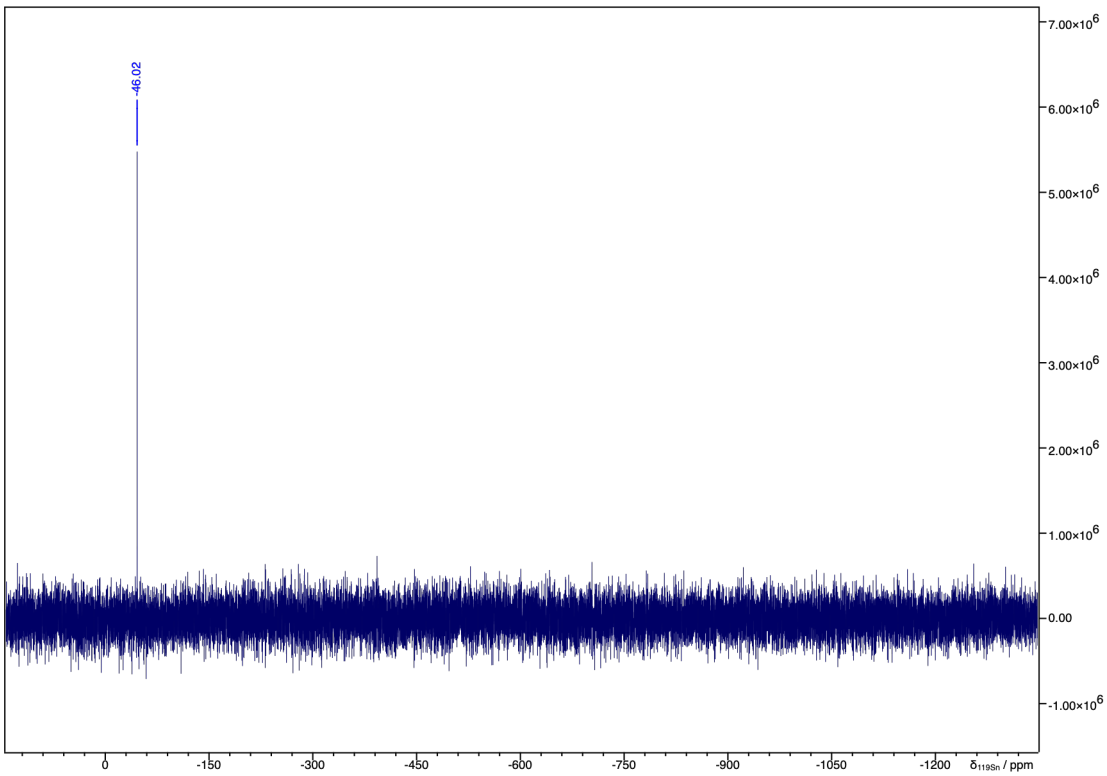


Figure S34. ^{119}Sn NMR spectrum of **10** (187 MHz, CD_2Cl_2)

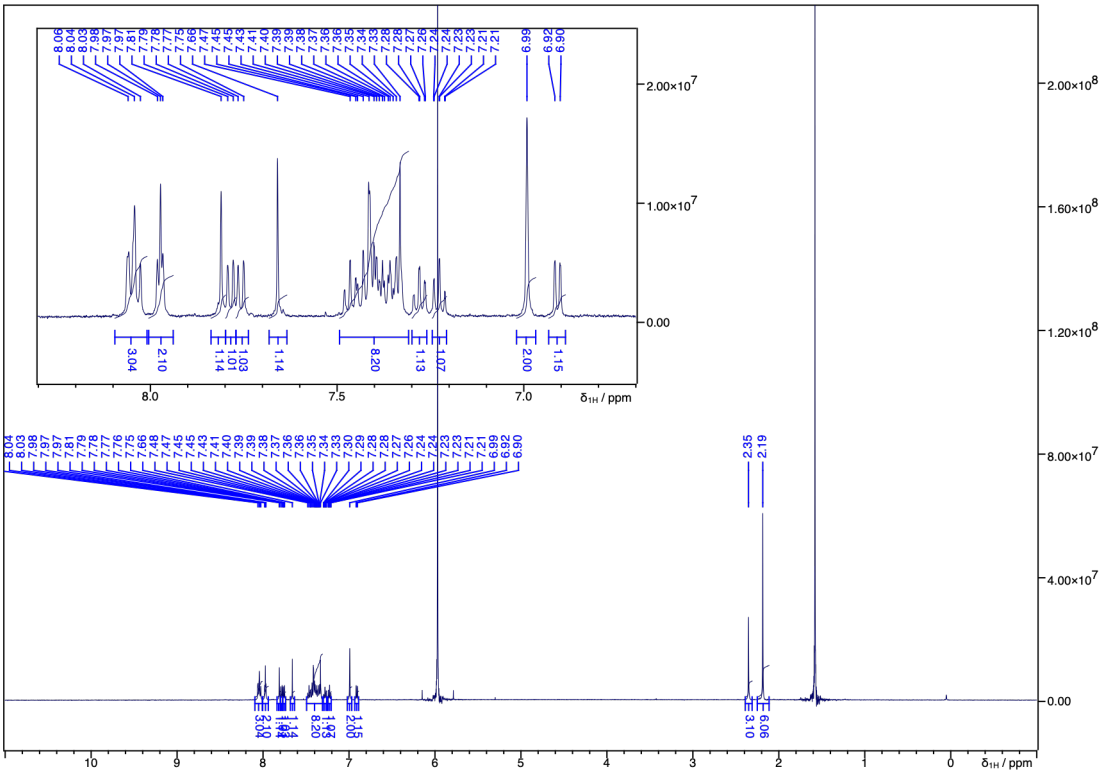


Figure S35. ^{1}H NMR spectrum of **11** (500 MHz, TCE-d_2)

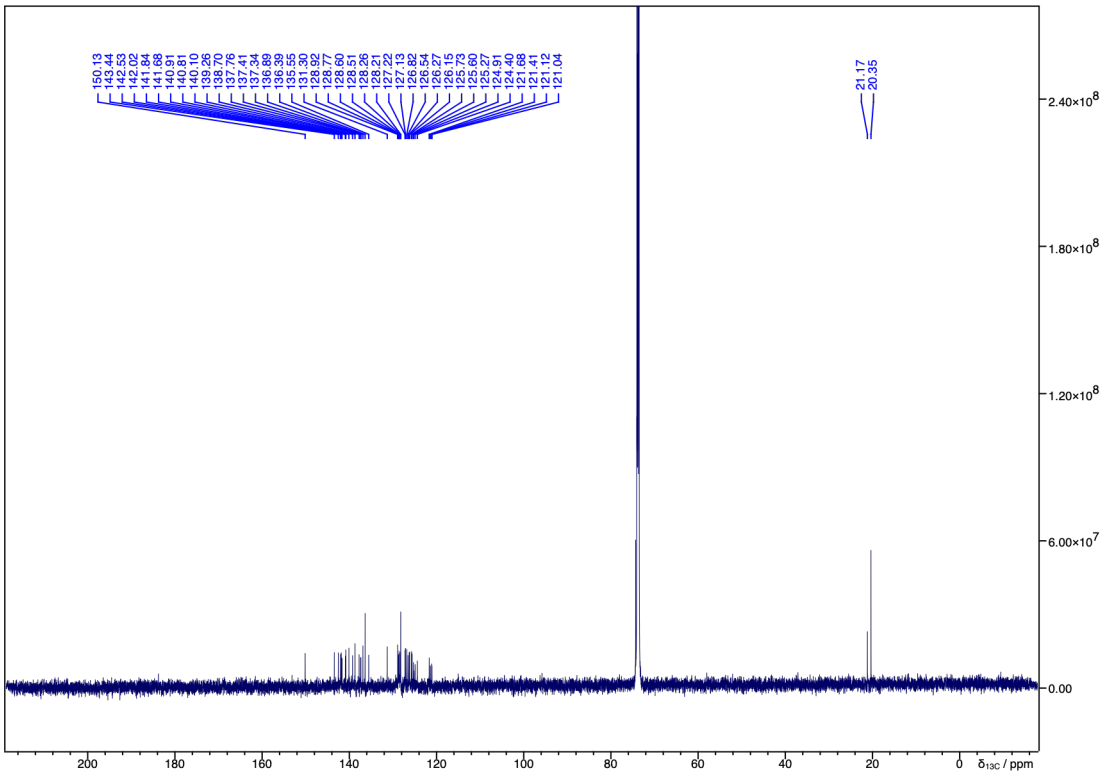


Figure S36. ^{13}C NMR spectrum of **11** (126 MHz, TCE-d_2)

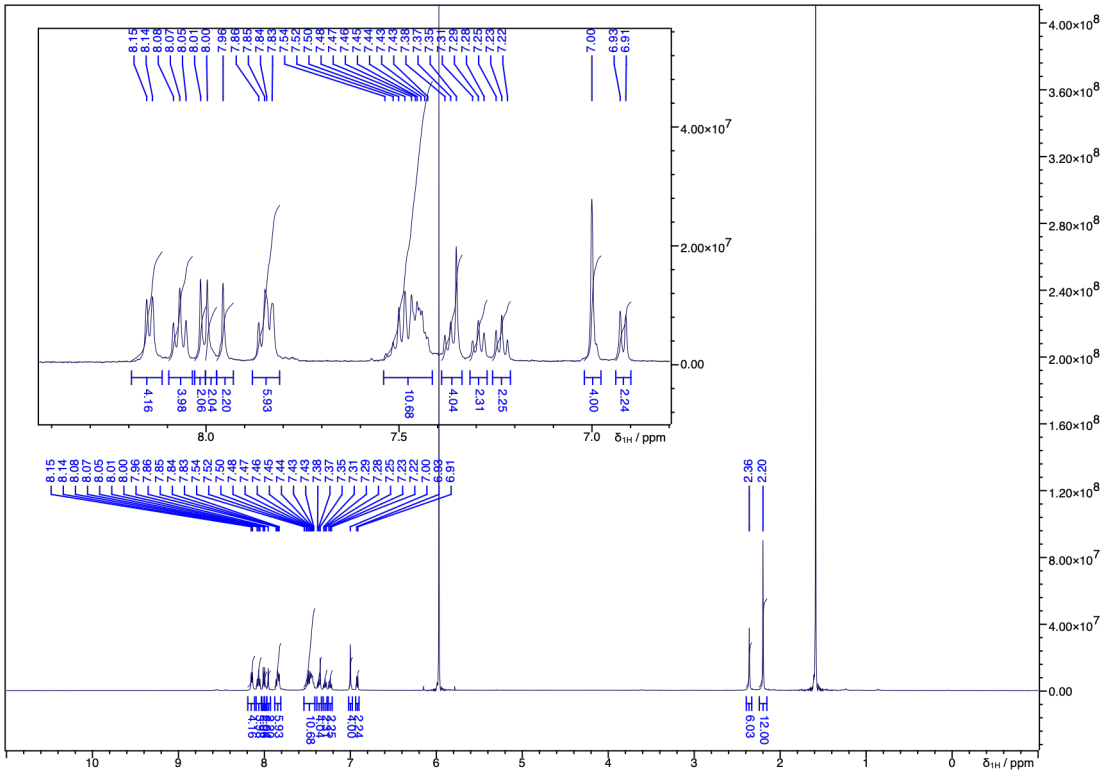


Figure S37. ^{1}H NMR spectrum of **4d** (500 MHz, TCE-d_2)

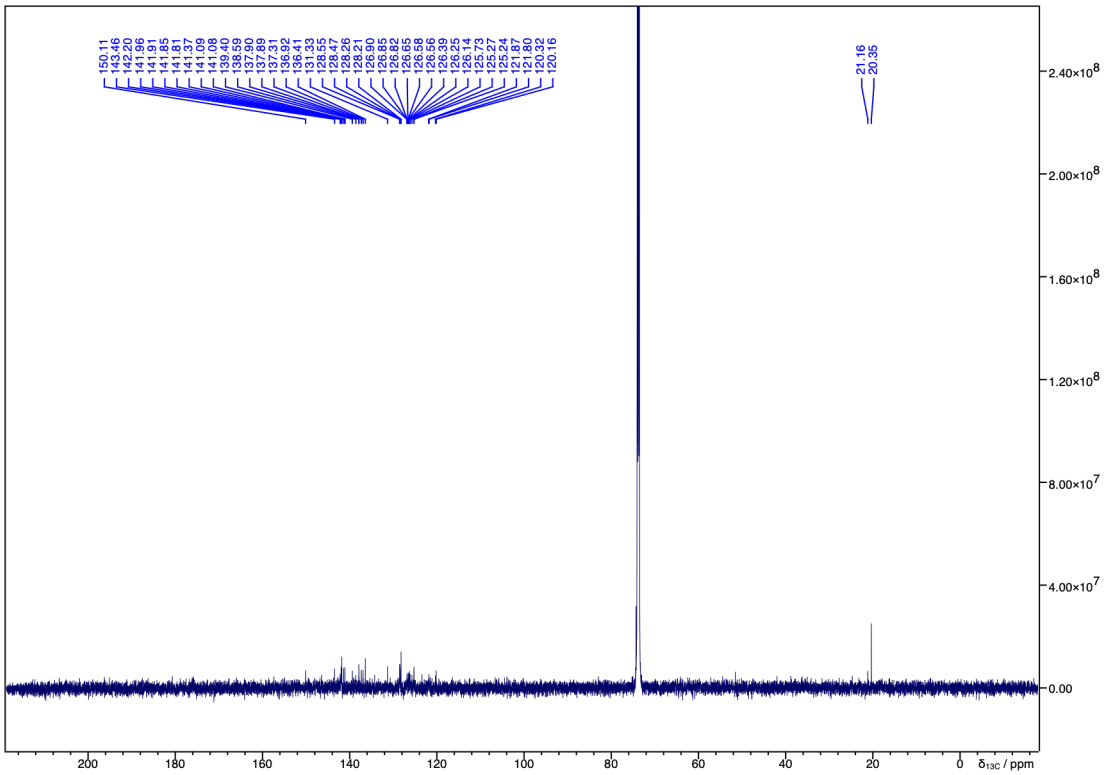


Figure S38. ^{13}C NMR spectrum of **4d** (126 MHz, TCE-d_2)

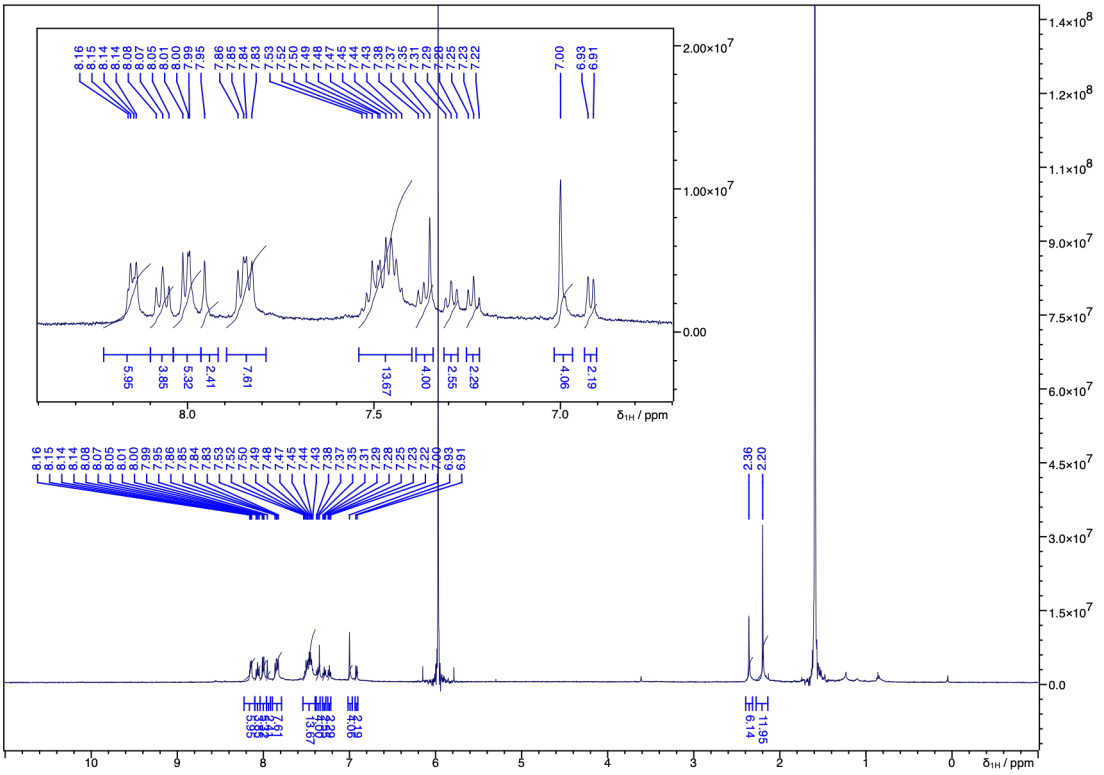


Figure S39. ^{1}H NMR spectrum of **4e** (500 MHz, TCE-d_2)

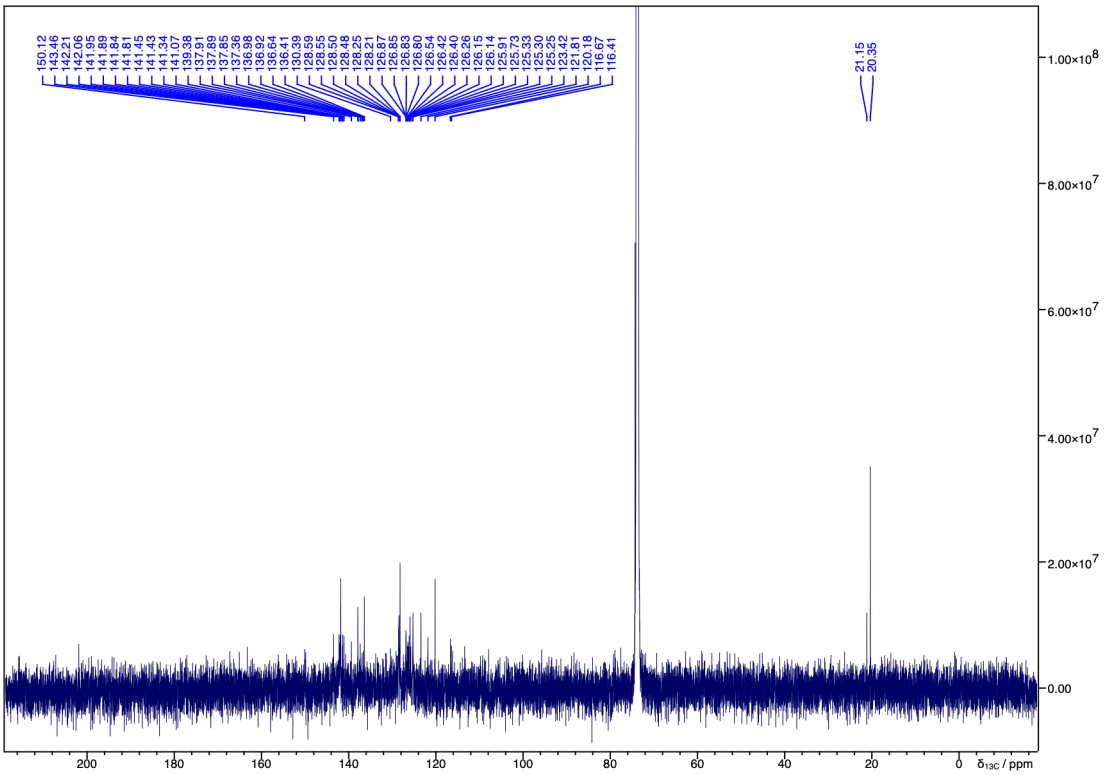


Figure S40. ^{13}C NMR spectrum of **4e** (126 MHz, TCE-d_2)

Appendix 2. Cartesian Coordinates of the Optimized Geometries

All Cartesian coordinates are given in Å.

Table S3. Cartesian Coordinates of the Optimized Geometries for **14a**

C	1.193420	2.829785	0.000000
C	1.193420	1.489573	0.000000
C	-0.001463	0.674548	0.000000
C	0.001463	-0.674548	0.000000
C	-1.193420	-1.489573	0.000000
C	-1.193420	-2.829785	0.000000
H	2.116535	3.400987	0.000000
H	0.265158	3.396939	0.000000
H	2.144549	0.956168	0.000000
H	-0.956680	1.201917	0.000000
H	0.956680	-1.201917	0.000000
H	-2.144549	-0.956168	0.000000
H	-0.265158	-3.396939	0.000000
H	-2.116535	-3.400987	0.000000

Table S4. Cartesian Coordinates of the Optimized Geometries for **14b**

C	-4.296948	-0.198144	0.000009
C	-3.102889	0.412536	0.000002
C	-0.632539	0.341701	-0.000007
C	-1.834396	-0.278178	-0.000005
C	0.632533	-0.341617	-0.000007
C	1.834406	0.278228	-0.000005
C	3.102866	-0.412550	0.000002
C	4.296966	0.198049	0.000009
H	-4.378628	-1.282829	0.000007
H	-3.061831	1.502301	0.000002
H	-0.604091	1.432496	-0.000006
H	-1.867271	-1.368695	-0.000009
H	0.604064	-1.432413	-0.000006
H	1.867322	1.368743	-0.000009
H	3.061736	-1.502313	0.000002
H	4.378725	1.282727	0.000007
H	5.225823	-0.363624	0.000016
H	-5.225847	0.363460	0.000016

Table S5. Cartesian Coordinates of the Optimized Geometries for **14c**

C	5.528570	0.215644	0.000000

C	4.338522	-0.403698	0.000000
C	3.065727	0.276994	0.000000
C	1.867840	-0.353554	0.000000
C	0.599344	0.317033	0.000000
C	-0.599343	-0.316990	0.000000
H	6.461283	-0.339612	0.000000
H	5.602995	1.300838	0.000000
H	4.306017	-1.493734	0.000000
H	3.089091	1.367665	0.000000
H	1.849971	-1.444609	0.000000
H	0.615785	1.407925	0.000000
H	-0.615780	-1.407882	0.000000
C	-1.867845	0.353586	0.000000
C	-3.065721	-0.276983	0.000000
C	-4.338532	0.403679	0.000000
C	-5.528562	-0.215698	0.000000
H	-1.849992	1.444641	0.000000
H	-3.089063	-1.367655	0.000000
H	-4.306057	1.493715	0.000000
H	-5.602955	-1.300894	0.000000
H	-6.461292	0.339531	0.000000

Table S6. Cartesian Coordinates of the Optimized Geometries for **14d**

C	5.573035	-0.396653	0.000002
C	6.761110	0.226943	0.000007
C	4.298143	0.279136	0.000000
C	3.102025	-0.356193	-0.000003
C	1.831928	0.308606	-0.000004
C	0.634529	-0.331281	-0.000005
C	-1.831926	-0.308576	-0.000004
C	-0.634531	0.331319	-0.000005
C	-3.102029	0.356210	-0.000003
C	-4.298138	-0.279137	0.000000
C	-5.573042	0.396628	0.000002
C	-6.761105	-0.226991	0.000007
H	6.831732	1.312384	0.000009
H	3.088842	-1.447342	-0.000004
H	5.544580	-1.486814	0.000001
H	4.317218	1.369881	0.000001
H	1.843199	1.399536	-0.000003
H	0.624176	-1.422315	-0.000005
H	-1.843188	-1.399506	-0.000003
H	-0.624181	1.422352	-0.000005

H	-3.088862	1.447359	-0.000004
H	-4.317195	-1.369882	0.000001
H	-5.544609	1.486791	0.000000
H	-6.831703	-1.312434	0.000009
H	7.695740	-0.325063	0.000010
H	-7.695747	0.324994	0.000010

Table S7. Cartesian Coordinates of the Optimized Geometries for **14e**

C	-6.806780	0.391377	-0.000004
C	-7.993873	-0.234332	-0.000004
C	-5.530865	-0.281970	0.000001
C	-4.335606	0.355760	0.000000
C	-3.064747	-0.306115	0.000004
C	-1.867982	0.336709	0.000002
C	0.598621	0.321898	0.000003
C	-0.598621	-0.322039	0.000004
C	1.867993	-0.336830	0.000002
C	3.064739	0.306031	0.000002
C	4.335626	-0.355790	-0.000001
C	5.530847	0.282006	-0.000001
C	6.806791	-0.391257	-0.000004
C	7.993862	0.234503	-0.000002
H	-8.062673	-1.319870	-0.000002
H	-4.324797	1.446957	-0.000005
H	6.780427	-1.481483	-0.000009
H	8.929467	-0.315839	-0.000005
H	-6.780334	1.481585	-0.000008
H	-5.547741	-1.372747	0.000005
H	-3.073383	-1.397065	0.000008
H	-1.860670	1.427801	-0.000001
H	0.605424	1.412939	0.000004
H	-0.605430	-1.413079	0.000005
H	1.860710	-1.427922	-0.000001
H	3.073340	1.396981	0.000006
H	4.324890	-1.446986	-0.000005
H	5.547678	1.372789	0.000003
H	8.062570	1.320066	0.000002
H	-8.929502	0.316163	-0.000009

Table S8. Cartesian Coordinates of the Optimized Geometries for **14f**

C	9.226561	0.238806	-0.000027
C	8.039702	-0.387539	-0.000016
C	5.568669	-0.354066	0.000001

C	6.763540	0.284924	-0.000009
C	4.297524	0.306344	0.000007
C	3.101101	-0.338226	0.000014
C	1.831633	0.318451	0.000017
C	0.634512	-0.327828	0.000019
C	-1.831631	-0.318406	0.000017
C	-0.634513	0.327878	0.000019
C	-3.101104	0.338260	0.000013
C	-4.297520	-0.306325	0.000007
C	-5.568674	0.354067	0.000001
C	-6.763534	-0.284944	-0.000009
C	-8.039709	0.387494	-0.000016
C	-9.226556	-0.238876	-0.000027
H	6.779463	1.375670	-0.000011
H	3.095533	-1.429301	0.000015
H	-8.014020	1.477714	-0.000014
H	-10.162347	0.311158	-0.000033
H	9.294891	1.324376	-0.000030
H	8.013990	-1.477759	-0.000014
H	5.559101	-1.445242	0.000003
H	4.304687	1.397257	0.000005
H	1.836422	1.409447	0.000016
H	0.630156	-1.418865	0.000020
H	-1.836414	-1.409403	0.000016
H	-0.630160	1.418914	0.000020
H	-3.095548	1.429335	0.000015
H	-4.304668	-1.397237	0.000005
H	-5.559125	1.445243	0.000003
H	-6.779437	-1.375690	-0.000011
H	-9.294862	-1.324448	-0.000030
H	10.162340	-0.311248	-0.000033

Table S9. Cartesian Coordinates of the Optimized Geometries for **14g**

C	7.996374	0.287022	-0.000022
C	9.272769	-0.384802	-0.000040
C	10.459396	0.242078	-0.000059
C	6.801697	-0.352605	-0.000005
C	4.334100	-0.338357	0.000022
C	5.530402	0.307017	0.000009
C	3.064615	0.317263	0.000031
C	1.867488	-0.330117	0.000038
C	0.598679	0.324041	0.000041
C	-0.598678	-0.323901	0.000040

C	-3.064609	-0.317157	0.000028
C	-1.867493	0.330246	0.000036
C	-4.334111	0.338429	0.000019
C	-5.530390	-0.306986	0.000006
C	-6.801712	0.352584	-0.000007
C	-7.996359	-0.287099	-0.000023
C	-9.272788	0.384662	-0.000038
C	-10.459382	-0.242282	-0.000056
H	5.536856	1.397930	0.000007
H	1.864310	-1.421171	0.000039
H	-9.247653	1.474897	-0.000036
H	-11.395425	0.307319	-0.000067
H	8.011730	1.377774	-0.000025
H	9.247577	-1.475036	-0.000037
H	10.527255	1.327676	-0.000062
H	6.792763	-1.443789	-0.000004
H	4.329359	-1.429442	0.000024
H	3.068444	1.408256	0.000030
H	0.601601	1.415070	0.000041
H	-0.601596	-1.414929	0.000039
H	-3.068413	-1.408150	0.000027
H	-1.864331	1.421299	0.000038
H	-4.329405	1.429514	0.000021
H	-5.536804	-1.397900	0.000004
H	-6.792826	1.443769	-0.000005
H	-8.011663	-1.377851	-0.000025
H	-10.527181	-1.327884	-0.000059
H	11.395409	-0.307574	-0.000072

Table S10. Cartesian Coordinates of the Optimized Geometries for **14h**

C	6.763273	0.307915	0.000006
C	8.034615	-0.351305	-0.000019
C	9.229215	0.288635	-0.000048
C	5.567003	-0.337874	0.000030
C	3.100380	-0.330781	0.000061
C	4.297552	0.317186	0.000047
C	1.831737	0.322546	0.000070
C	0.634207	-0.326282	0.000073
C	-0.634210	0.326542	0.000072
C	-1.831731	-0.322304	0.000064
C	-4.297537	-0.317028	0.000038
C	-3.100392	0.330989	0.000054
C	-5.567022	0.337966	0.000021

C	-6.763253	-0.307899	-0.000001
C	-8.034640	0.351230	-0.000023
C	-9.229191	-0.288805	-0.000048
C	-10.505739	0.382604	-0.000073
C	-11.692211	-0.244631	-0.000099
H	4.300868	1.408177	0.000045
H	0.632271	-1.417326	0.000072
H	-10.480889	1.472846	-0.000070
H	-12.628379	0.304754	-0.000118
C	10.505707	-0.382882	-0.000078
C	11.692235	0.244246	-0.000110
H	6.769362	1.398828	0.000003
H	8.025999	-1.442495	-0.000017
H	9.244294	1.379389	-0.000051
H	5.562696	-1.428965	0.000031
H	3.097832	-1.421843	0.000062
H	1.833893	1.413569	0.000069
H	-0.632284	1.417584	0.000073
H	-1.833866	-1.413328	0.000062
H	-4.300804	-1.408019	0.000035
H	-3.097883	1.422050	0.000056
H	-5.562781	1.429057	0.000023
H	-6.769268	-1.398812	-0.000004
H	-8.026109	1.442420	-0.000021
H	-9.244180	-1.379560	-0.000051
H	-11.759757	-1.330248	-0.000103
H	10.480760	-1.473122	-0.000075
H	11.759879	1.329857	-0.000115
H	12.628353	-0.305223	-0.000133

Table S11. Cartesian Coordinates of the Optimized Geometries for **14i**

C	5.530453	0.317600	0.000072
C	6.799841	-0.337187	0.000040
C	7.996134	0.308790	0.000003
C	4.333215	-0.330663	0.000095
C	1.867042	-0.327075	0.000119
C	3.064711	0.322227	0.000112
C	0.598895	0.325035	0.000120
C	-0.598893	-0.324587	0.000113
C	-1.867052	0.327497	0.000102
C	-3.064697	-0.321852	0.000083
C	-5.530427	-0.317391	0.000036
C	-4.333238	0.330963	0.000062

C	-6.799874	0.337280	0.000008
C	-7.996099	-0.308824	-0.000022
C	-9.267501	0.350076	-0.000054
C	-10.462017	-0.290136	-0.000086
C	-11.738611	0.381101	-0.000120
C	-12.925032	-0.246268	-0.000152
H	3.066461	1.413247	0.000112
H	-0.600146	-1.415624	0.000110
H	-11.713897	1.471347	-0.000118
H	-13.861257	0.303019	-0.000177
C	9.267460	-0.350258	-0.000042
C	10.462056	0.289802	-0.000089
C	11.738561	-0.381608	-0.000143
C	12.925070	0.245594	-0.000196
H	5.533514	1.408589	0.000071
H	6.795741	-1.428281	0.000041
H	8.002060	1.399702	0.000001
H	4.330993	-1.421729	0.000095
H	1.865633	-1.418125	0.000117
H	0.600156	1.416072	0.000122
H	-1.865680	1.418547	0.000104
H	-3.066394	-1.412873	0.000079
H	-5.533397	-1.408381	0.000032
H	-4.331093	1.422029	0.000064
H	-6.795884	1.428374	0.000010
H	-8.001904	-1.399737	-0.000024
H	-9.259152	1.441268	-0.000053
H	-10.476848	-1.380893	-0.000087
H	-12.992465	-1.331892	-0.000155
H	9.258974	-1.441450	-0.000042
H	10.477031	1.380556	-0.000091
H	11.713694	-1.471851	-0.000140
H	12.992658	1.331208	-0.000200
H	13.861217	-0.303827	-0.000236

Table S12. Cartesian Coordinates of the Optimized Geometries for **14j**

C	4.297631	0.322479	0.000161
C	5.566013	-0.330201	0.000134
C	6.763329	0.318186	0.000105
C	3.099841	-0.327057	0.000170
C	0.633912	-0.325359	0.000171
C	1.831893	0.324675	0.000177
C	-0.633919	0.326070	0.000161

C	-1.831883	-0.323995	0.000141
C	-3.099864	0.327673	0.000118
C	-4.297605	-0.321955	0.000091
C	-6.763288	-0.317935	0.000024
C	-5.566051	0.330598	0.000058
C	-8.032692	0.336566	-0.000015
C	-9.228933	-0.309658	-0.000051
C	-10.500327	0.349126	-0.000095
C	-11.694837	-0.291171	-0.000133
C	-12.971447	0.379985	-0.000179
C	-14.157849	-0.247447	-0.000218
H	1.832801	1.415708	0.000181
H	-1.832752	-1.415029	0.000139
H	-12.946798	1.470232	-0.000182
H	-15.094101	0.301795	-0.000253
C	8.032641	-0.336497	0.000055
C	9.228986	0.309531	0.000005
C	10.500263	-0.349484	-0.000069
C	11.694896	0.290578	-0.000141
C	12.971367	-0.380843	-0.000241
C	14.157904	0.246331	-0.000334
H	4.299179	1.413496	0.000166
H	5.563942	-1.421268	0.000129
H	6.766283	1.409173	0.000110
H	3.098706	-1.418111	0.000166
H	0.633133	-1.416403	0.000167
H	-0.633157	1.417113	0.000163
H	-3.098801	1.418726	0.000120
H	-4.299059	-1.412973	0.000090
H	-6.766099	-1.408923	0.000025
H	-5.564105	1.421666	0.000057
H	-8.028835	1.427660	-0.000017
H	-9.234630	-1.400570	-0.000049
H	-10.492069	1.440319	-0.000098
H	-11.709593	-1.381927	-0.000131
H	-14.225234	-1.333073	-0.000217
H	8.028611	-1.427592	0.000048
H	9.234868	1.400442	0.000011
H	10.491793	-1.440676	-0.000077
H	11.709875	1.381331	-0.000134
H	12.946496	-1.471085	-0.000239
H	14.225610	1.331937	-0.000335
H	15.094048	-0.303144	-0.000408

Table S13. Cartesian Coordinates of the Optimized Geometries for **14k**

C	3.064806	0.324708	0.000047
C	4.332619	-0.326772	0.000041
C	5.530485	0.322940	0.000037
C	1.866697	-0.325605	0.000048
C	-0.599088	-0.325082	0.000046
C	0.599089	0.325421	0.000048
C	-1.866703	0.325930	0.000041
C	-3.064798	-0.324409	0.000037
C	-4.332631	0.327032	0.000029
C	-5.530471	-0.322730	0.000023
C	-7.996128	-0.318819	0.000005
C	-6.798802	0.329737	0.000012
C	-9.265430	0.335711	-0.000008
C	-10.461771	-0.310433	-0.000016
C	-11.733054	0.348492	-0.000030
C	-12.927693	-0.291624	-0.000039
C	-14.204165	0.379783	-0.000054
C	-15.390727	-0.247377	-0.000062
H	0.599481	1.416460	0.000048
H	-3.065448	-1.415439	0.000040
H	-14.179274	1.470025	-0.000058
H	-16.326838	0.302102	-0.000074
C	6.798784	-0.329591	0.000026
C	7.996148	0.318894	0.000019
C	9.265406	-0.335722	0.000003
C	10.461794	0.310334	-0.000008
C	11.733025	-0.348693	-0.000029
C	12.927718	0.291320	-0.000042
C	14.204131	-0.380199	-0.000066
C	15.390752	0.246846	-0.000084
H	3.065483	1.415738	0.000049
H	4.331667	-1.417828	0.000040
H	5.531897	1.413955	0.000042
H	1.866218	-1.416652	0.000048
H	-0.599475	-1.416122	0.000047
H	-1.866242	1.416976	0.000040
H	-4.331721	1.418087	0.000026
H	-5.531834	-1.413745	0.000026
H	-7.998953	-1.409805	0.000009
H	-6.796892	1.420805	0.000009
H	-9.261506	1.426806	-0.000012

H	-10.467574	-1.401341	-0.000012
H	-11.724638	1.439683	-0.000034
H	-12.942649	-1.382374	-0.000035
H	-15.458356	-1.332986	-0.000059
H	6.796812	-1.420660	0.000023
H	7.999041	1.409879	0.000025
H	9.261404	-1.426817	-0.000001
H	10.467681	1.401241	0.000000
H	11.724517	-1.439883	-0.000033
H	12.942771	1.382069	-0.000035
H	14.179147	-1.470440	-0.000070
H	15.458514	1.332446	-0.000080
H	16.326812	-0.302726	-0.000104

Table S14. Cartesian Coordinates of the Optimized Geometries for **14l**

C	1.832023	0.325553	0.000046
C	3.099459	-0.325289	0.000043
C	4.297687	0.325124	0.000041
C	0.633680	-0.325157	0.000047
C	-1.832019	-0.325214	0.000043
C	-0.633682	0.325507	0.000045
C	-3.099467	0.325603	0.000037
C	-4.297676	-0.324844	0.000034
C	-5.565391	0.326506	0.000024
C	-6.763317	-0.323294	0.000020
C	-9.228955	-0.319397	0.000002
C	-8.031560	0.329150	0.000008
C	-10.498188	0.335155	-0.000012
C	-11.694587	-0.310949	-0.000018
C	-12.965814	0.348030	-0.000033
C	-14.160504	-0.292022	-0.000040
C	-15.436925	0.379461	-0.000055
C	-16.623536	-0.247619	-0.000063
H	-0.633547	1.416550	0.000044
H	-4.298237	-1.415873	0.000038
H	-15.411964	1.469702	-0.000062
H	-17.559608	0.301927	-0.000075
C	5.565377	-0.326273	0.000033
C	6.763332	0.323471	0.000029
C	8.031541	-0.329040	0.000016
C	9.228975	0.319433	0.000010
C	10.498164	-0.335204	-0.000006
C	11.694610	0.310812	-0.000014

C	12.965785	-0.348265	-0.000034
C	14.160528	0.291689	-0.000043
C	15.436893	-0.379902	-0.000064
C	16.623560	0.247070	-0.000075
H	1.832241	1.416590	0.000048
H	3.099105	-1.416338	0.000043
H	4.298283	1.416153	0.000045
H	0.633539	-1.416201	0.000048
H	-1.832222	-1.416252	0.000046
H	-3.099140	1.416652	0.000035
H	-5.564537	1.417563	0.000020
H	-6.764652	-1.414308	0.000025
H	-9.231793	-1.410382	0.000008
H	-8.029656	1.420219	0.000003
H	-10.494236	1.426251	-0.000018
H	-11.700432	-1.401856	-0.000012
H	-12.957345	1.439221	-0.000039
H	-14.175522	-1.382771	-0.000033
H	-16.691240	-1.333223	-0.000057
H	5.564476	-1.417330	0.000030
H	6.764722	1.414485	0.000035
H	8.029572	-1.420110	0.000012
H	9.231883	1.410418	0.000018
H	10.494134	-1.426300	-0.000012
H	11.700537	1.401719	-0.000005
H	12.957228	-1.439456	-0.000040
H	14.175639	1.382436	-0.000034
H	15.411845	-1.470141	-0.000071
H	16.691389	1.332665	-0.000069
H	17.559582	-0.302564	-0.000092

Table S15. Cartesian Coordinates of the Optimized Geometries for **14m**

C	0.599259	0.325685	0.000057
C	1.866433	-0.324805	0.000054
C	3.064910	0.325992	0.000052
C	-0.599257	-0.325233	0.000057
C	-3.064901	-0.325579	0.000051
C	-1.866439	0.325244	0.000054
C	-4.332218	0.325151	0.000044
C	-5.530523	-0.325332	0.000039
C	-6.798140	0.325993	0.000028
C	-7.996141	-0.323801	0.000021
C	-10.461765	-0.319860	-0.000001

C	-9.264310	0.328656	0.000007
C	-11.730938	0.334729	-0.000017
C	-12.927387	-0.311330	-0.000026
C	-14.198566	0.347698	-0.000043
C	-15.393296	-0.292303	-0.000052
C	-16.669678	0.379237	-0.000069
C	-17.856322	-0.247788	-0.000079
H	-1.866434	1.416290	0.000054
H	-5.531054	-1.416360	0.000044
H	-16.644669	1.469477	-0.000075
H	-18.792367	0.301804	-0.000092
C	4.332206	-0.324779	0.000044
C	5.530538	0.325653	0.000039
C	6.798122	-0.325738	0.000027
C	7.996162	0.323982	0.000020
C	9.264286	-0.328563	0.000005
C	10.461790	0.319861	-0.000003
C	11.730910	-0.334835	-0.000019
C	12.927416	0.311116	-0.000028
C	14.198532	-0.348033	-0.000043
C	15.393326	0.291848	-0.000052
H	0.599226	1.416726	0.000059
H	1.866410	-1.415851	0.000055
H	3.065054	1.417027	0.000055
H	-0.599220	-1.416274	0.000059
H	-3.065017	-1.416615	0.000055
H	-4.331949	1.416201	0.000042
H	-6.797299	1.417051	0.000024
H	-7.997481	-1.414815	0.000026
H	-10.464629	-1.410845	0.000004
H	-9.262389	1.419726	0.000002
H	-11.726954	1.425825	-0.000022
H	-12.933270	-1.402237	-0.000020
H	-14.190056	1.438889	-0.000048
H	-15.408359	-1.383051	-0.000046
H	-17.924080	-1.333388	-0.000074
H	4.331895	-1.415830	0.000043
H	5.531120	1.416680	0.000044
H	6.797217	-1.416796	0.000024
H	7.997573	1.414996	0.000025
H	9.262284	-1.419633	0.000001
H	10.464742	1.410845	0.000002
H	11.726829	-1.425931	-0.000024

H	12.933400	1.402023	-0.000022
H	14.189914	-1.439223	-0.000049
H	15.408500	1.382595	-0.000047
C	16.669639	-0.379823	-0.000068
C	17.856349	0.247074	-0.000076
H	16.644514	-1.470061	-0.000073
H	17.924225	1.332667	-0.000071
H	18.792335	-0.302618	-0.000088

Table S16. Cartesian Coordinates of the Optimized Geometries for **15a**

C	-1.771563	0.777523	-0.266874
C	-2.065270	2.096670	-0.339230
C	-0.846509	2.871924	-0.144570
C	0.227513	1.962787	0.019459
C	-0.333825	0.597424	-0.077932
C	-0.644294	4.242632	-0.077577
C	0.644294	4.720623	0.173089
C	1.697932	3.833551	0.374219
C	1.495196	2.451072	0.311406
C	0.333825	-0.597424	-0.077932
C	1.771563	-0.777523	-0.266874
C	2.065270	-2.096670	-0.339230
C	0.846509	-2.871924	-0.144570
C	-0.227513	-1.962787	0.019459
C	-1.495196	-2.451072	0.311406
C	-1.697932	-3.833551	0.374219
C	-0.644294	-4.720623	0.173089
C	0.644294	-4.242632	-0.077577
H	-2.470413	-0.029268	-0.435026
H	-3.043972	2.524624	-0.527524
H	-1.474234	4.933030	-0.205873
H	0.821683	5.791284	0.228065
H	2.691349	4.216197	0.591325
H	2.327171	1.786632	0.517087
H	2.470413	0.029268	-0.435026
H	-2.327171	-1.786632	0.517087
H	-2.691349	-4.216197	0.591325
H	-0.821683	-5.791284	0.228065
H	1.474234	-4.933030	-0.205873
H	3.043972	-2.524624	-0.527524

Table S17. Cartesian Coordinates of the Optimized Geometries for **15b**

C	-4.974391	1.979155	-0.718608

C	-6.146464	1.503759	-1.204287
C	-6.158774	0.051447	-1.090545
C	-4.917614	-0.352958	-0.538386
C	-4.122562	0.869947	-0.304583
C	-2.825047	0.969632	0.133257
C	-1.856364	-0.104596	0.171704
C	-0.616659	0.374376	0.504855
C	-0.751695	1.811904	0.792680
C	-2.098150	2.180891	0.567277
C	0.616605	-0.374062	0.504966
C	1.856357	0.104881	0.171919
C	2.824946	-0.969426	0.133418
C	2.098023	-2.180617	0.567509
C	0.751591	-1.811593	0.792903
C	4.122374	-0.869968	-0.304699
C	4.973713	-1.979339	-0.719292
C	6.145795	-1.504151	-1.205179
C	6.158612	-0.051875	-1.090928
C	4.917726	0.352733	-0.538314
C	-4.704428	-1.695434	-0.250056
C	-5.696919	-2.629014	-0.566120
C	-6.899045	-2.227150	-1.141260
C	-7.143384	-0.875576	-1.398638
C	-2.517820	3.470378	0.872047
C	-1.585865	4.398567	1.343830
C	-0.258455	4.033566	1.547555
C	0.164667	2.727643	1.289617
C	-0.164690	-2.727235	1.290162
C	0.258529	-4.033067	1.548435
C	1.585956	-4.398064	1.344772
C	2.517839	-3.469957	0.872684
C	4.704972	1.695155	-0.249430
C	5.697636	2.628569	-0.565422
C	6.899493	2.226547	-1.141031
C	7.143394	0.874995	-1.398948
H	-4.664129	3.014299	-0.733821
H	-6.949638	2.095686	-1.629587
H	-2.047079	-1.106829	-0.184206
H	2.047146	1.107097	-0.184038
H	4.663072	-3.014373	-0.734811
H	6.948613	-2.096200	-1.630977
H	-3.795591	-2.035884	0.234275
H	-5.526154	-3.680387	-0.352076

H	-7.658521	-2.967401	-1.377850
H	-8.092371	-0.557195	-1.822859
H	-3.558107	3.762928	0.782115
H	-1.909021	5.410764	1.570639
H	0.449541	4.762236	1.932917
H	1.185952	2.428767	1.507266
H	-1.185984	-2.428368	1.507781
H	-0.449411	-4.761650	1.934066
H	1.909187	-5.410167	1.571895
H	3.558163	-3.762407	0.782809
H	3.796355	2.035623	0.235322
H	5.527246	3.679917	-0.350956
H	7.659122	2.966662	-1.377557
H	8.092190	0.556508	-1.823520

Table S18. Cartesian Coordinates of the Optimized Geometries for **15c**

C	8.277526	-1.461653	1.784432
C	9.138295	-0.847292	2.631649
C	8.903186	0.589914	2.595075
C	7.827885	0.832902	1.704216
C	7.389626	-0.476080	1.178762
C	9.553649	1.633643	3.236304
C	9.140272	2.943482	2.978737
C	8.108887	3.192914	2.077978
C	7.454681	2.142199	1.426359
C	6.324318	-0.748279	0.355692
C	5.221608	0.142853	0.071358
C	4.251590	-0.503038	-0.651235
C	4.742478	-1.858382	-0.951604
C	6.007813	-2.013296	-0.339504
C	6.742434	-3.172026	-0.561768
C	6.191049	-4.191046	-1.342812
C	4.938009	-4.039274	-1.928977
C	4.211054	-2.859854	-1.751621
H	8.182541	-2.530380	1.654542
H	9.877620	-1.335425	3.257306
H	10.376934	1.438826	3.919047
H	9.637402	3.773332	3.473647
H	7.809534	4.216488	1.870576
H	6.682511	2.371521	0.699944
H	5.099957	1.119243	0.517848
H	7.746385	-3.289191	-0.169087
H	6.757944	-5.103121	-1.507561

H	4.531634	-4.833464	-2.549331
H	3.261852	-2.719180	-2.260408
C	2.953675	0.020135	-0.994527
C	1.787276	-0.697377	-1.091646
C	0.664160	0.184155	-1.298891
C	1.231014	1.541897	-1.413311
C	2.629697	1.436446	-1.228914
C	3.450735	2.546691	-1.363821
C	2.871835	3.782485	-1.663075
C	1.498156	3.890146	-1.858123
C	0.672054	2.767866	-1.754216
C	-0.664110	-0.183728	-1.298887
C	-1.787235	0.697756	-1.091448
C	-2.953622	-0.019785	-0.994453
C	-2.629622	-1.436059	-1.229025
C	-1.230942	-1.541460	-1.413460
C	-0.671957	-2.767380	-1.754495
C	-1.498040	-3.889664	-1.858523
C	-2.871721	-3.782050	-1.663453
C	-3.450642	-2.546302	-1.364043
H	1.699384	-1.751220	-0.868803
H	4.529009	2.454663	-1.271480
H	3.502887	4.661008	-1.766152
H	1.061956	4.852418	-2.111303
H	-0.386472	2.866620	-1.969060
H	-1.699348	1.751560	-0.868423
H	0.386575	-2.866094	-1.969331
H	-1.061823	-4.851901	-2.111805
H	-3.502758	-4.660573	-1.766624
H	-4.528917	-2.454302	-1.271664
C	-4.251560	0.503273	-0.651069
C	-4.742552	1.858622	-0.951238
C	-6.007910	2.013348	-0.339127
C	-6.324348	0.748173	0.355826
C	-5.221535	-0.142807	0.071401
C	-7.389683	0.475729	1.178780
C	-8.277819	1.461120	1.784407
C	-9.138655	0.846564	2.631408
C	-8.903348	-0.590607	2.594727
C	-7.827827	-0.833368	1.704062
C	-4.211141	2.860255	-1.751056
C	-4.938139	4.039678	-1.928227
C	-6.191201	4.191275	-1.342070

C	-6.742557	3.172095	-0.561208
C	-7.454351	-2.142602	1.426277
C	-8.108531	-3.193443	2.077723
C	-9.140145	-2.944220	2.978274
C	-9.553775	-1.634452	3.235797
H	-5.099813	-1.119265	0.517721
H	-8.182952	2.529868	1.654620
H	-3.261916	2.719696	-2.259830
H	-4.531785	4.834000	-2.548424
H	-6.758153	5.103340	-1.506673
H	-7.746538	3.289167	-0.168572
H	-6.681982	-2.371829	0.700044
H	-7.808955	-4.216955	1.870339
H	-9.637243	-3.774168	3.473049
H	-9.878189	1.334520	3.256954
H	-10.377216	-1.439766	3.918389

Table S19. Cartesian Coordinates of the Optimized Geometries for **15d**

C	10.683149	3.371837	1.332452
C	11.197993	4.493221	0.772380
C	10.861389	4.519309	-0.644729
C	10.081469	3.370296	-0.928384
C	9.933012	2.618603	0.334308
C	11.200201	5.420846	-1.642741
C	10.770367	5.171432	-2.948888
C	10.034652	4.026943	-3.242443
C	9.695965	3.113764	-2.238449
C	9.186685	1.489555	0.568753
C	8.144946	0.967720	-0.287199
C	7.491637	-0.071666	0.324141
C	8.164721	-0.332705	1.607526
C	9.200651	0.617505	1.761643
C	10.065543	0.522959	2.845467
C	9.860445	-0.478978	3.797555
C	8.828870	-1.401215	3.648887
C	7.982767	-1.344991	2.538989
H	10.723326	3.121487	2.383062
H	11.757963	5.266165	1.287465
H	11.797841	6.300428	-1.416810
H	11.025080	5.868021	-3.742992
H	9.723106	3.833431	-4.265094
H	9.156914	2.211462	-2.506384
H	7.812308	1.444591	-1.197917

H	10.914460	1.189278	2.951380
H	10.527397	-0.547193	4.652490
H	8.694661	-2.186018	4.388336
H	7.218258	-2.102624	2.393598
C	6.312974	-0.741070	-0.163445
C	5.294327	-1.255635	0.600561
C	4.209338	-1.704954	-0.236374
C	4.663006	-1.511562	-1.626946
C	5.949321	-0.924730	-1.577635
C	6.672663	-0.698491	-2.739750
C	6.100709	-1.038317	-3.968359
C	4.841109	-1.627469	-4.022312
C	4.123122	-1.885988	-2.851625
C	2.975579	-2.139011	0.200694
C	1.768484	-2.186958	-0.585354
C	0.690248	-2.512105	0.202309
C	1.192970	-2.782018	1.559213
C	2.588945	-2.552722	1.563323
C	3.330360	-2.834600	2.704544
C	2.673676	-3.292742	3.849725
C	1.298374	-3.505327	3.843481
C	0.549363	-3.267723	2.688365
H	5.237245	-1.172717	1.676643
H	7.680005	-0.294558	-2.697440
H	6.654298	-0.861242	-4.886415
H	4.415693	-1.906761	-4.982179
H	3.170082	-2.399891	-2.915863
H	1.694943	-1.840797	-1.606497
H	4.410281	-2.733991	2.716123
H	3.248751	-3.501201	4.747594
H	0.805435	-3.880337	4.736246
H	-0.513082	-3.492229	2.669045
C	-0.690365	-2.512109	-0.202384
C	-1.193078	-2.781951	-1.559306
C	-2.589058	-2.552691	-1.563404
C	-2.975705	-2.139070	-0.200751
C	-1.768611	-2.187044	0.585298
C	-4.209473	-1.705064	0.236342
C	-5.294465	-1.255699	-0.600563
C	-6.313117	-0.741187	0.163473
C	-5.949465	-0.924945	1.577654
C	-4.663143	-1.511767	1.626925
C	-7.491768	-0.071728	-0.324067

C	-8.165037	-0.332854	-1.607341
C	-9.200862	0.617471	-1.761464
C	-9.186666	1.489657	-0.568678
C	-8.144894	0.967805	0.287221
C	-9.932822	2.618829	-0.334298
C	-10.683010	3.372003	-1.332446
C	-11.197661	4.493522	-0.772467
C	-10.860845	4.519784	0.644594
C	-10.081023	3.370718	0.928307
C	-0.549459	-3.267573	-2.688487
C	-1.298464	-3.505128	-3.843617
C	-2.673771	-3.292578	-3.849849
C	-3.330466	-2.834523	-2.704641
C	-4.123250	-1.886275	2.851575
C	-4.841238	-1.627852	4.022283
C	-6.100848	-1.038720	3.968373
C	-6.672809	-0.698811	2.739789
C	-7.983327	-1.345292	-2.538688
C	-8.829560	-1.401545	-3.648487
C	-9.861024	-0.479189	-3.797163
C	-10.065884	0.522900	-2.845182
C	-9.695355	3.114349	2.238356
C	-10.033781	4.027729	3.242257
C	-10.769401	5.172261	2.948635
C	-11.199399	5.421518	1.642513
H	-1.695083	-1.840948	1.606464
H	-5.237367	-1.172686	-1.676636
H	-7.812092	1.444759	1.197834
H	-10.723363	3.121495	-2.383013
H	0.512992	-3.492056	-2.669180
H	-0.805514	-3.880074	-4.736403
H	-3.248840	-3.500999	-4.747731
H	-4.410391	-2.733943	-2.716211
H	-3.170201	-2.400166	2.915772
H	-4.415819	-1.907210	4.982129
H	-6.654442	-0.861728	4.886441
H	-7.680161	-0.294903	2.697515
H	-7.218912	-2.103014	-2.393277
H	-8.695541	-2.186463	-4.387850
H	-10.528085	-0.547428	-4.652010
H	-10.914724	1.189316	-2.951090
H	-9.156381	2.212026	2.506367
H	-9.722099	3.834339	4.264890

H	-11.023903	5.869008	3.742669
H	-11.796973	6.301131	1.416528
H	-11.757618	5.266447	-1.287583

Table S20. Cartesian Coordinates of the Optimized Geometries for **15e**

C	-12.333010	-5.168210	1.511412
C	-12.478335	-6.450102	1.096854
C	-12.038599	-6.558960	-0.287683
C	-11.581200	-5.282714	-0.701116
C	-11.741225	-4.366835	0.446541
C	-12.040999	-7.637507	-1.159631
C	-11.595387	-7.444227	-2.469997
C	-11.178304	-6.185948	-2.894324
C	-11.179672	-5.095585	-2.018107
C	-11.351070	-3.053693	0.547185
C	-10.427967	-2.372613	-0.332772
C	-10.131733	-1.123508	0.149752
C	-10.947520	-0.902569	1.355526
C	-11.694241	-2.077798	1.602485
C	-12.633353	-2.093713	2.626971
C	-12.783110	-0.962838	3.433892
C	-12.033383	0.185248	3.196006
C	-11.120738	0.229085	2.139587
H	-12.521046	-4.808964	2.513234
H	-12.844999	-7.277752	1.694242
H	-12.392428	-8.613537	-0.834445
H	-11.588644	-8.278487	-3.166186
H	-10.852841	-6.043300	-3.920991
H	-10.887286	-4.119654	-2.390386
H	-9.908866	-2.847681	-1.152726
H	-13.274351	-2.952393	2.793238
H	-13.507529	-0.977223	4.243461
H	-12.175636	1.062856	3.820728
H	-10.582680	1.146095	1.917776
C	-9.146801	-0.216295	-0.381224
C	-8.369116	0.654561	0.341906
C	-7.389494	1.285700	-0.507375
C	-7.665647	0.801863	-1.873713
C	-8.741153	-0.113666	-1.792138
C	-9.284159	-0.676840	-2.937913
C	-8.735966	-0.344060	-4.179223
C	-7.686305	0.565375	-4.266815
C	-7.158024	1.158902	-3.117235

C	-6.360839	2.103133	-0.088326
C	-5.159513	2.397819	-0.827704
C	-4.276890	3.112273	-0.053305
C	-4.934925	3.390042	1.233786
C	-6.208071	2.773454	1.217491
C	-7.081915	2.968789	2.280445
C	-6.666406	3.731597	3.374975
C	-5.408192	4.325991	3.390686
C	-4.538432	4.173452	2.308130
H	-8.373450	0.724463	1.420370
H	-10.141995	-1.340305	-2.878007
H	-9.148201	-0.782418	-5.083955
H	-7.282258	0.832439	-5.239308
H	-6.380784	1.908894	-3.216533
H	-4.915937	1.965461	-1.787722
H	-8.088913	2.566099	2.265732
H	-7.342396	3.874861	4.213335
H	-5.107410	4.932192	4.240802
H	-3.582230	4.688335	2.297456
C	-2.926197	3.461162	-0.403889
C	-2.421305	3.699560	-1.765825
C	-1.021296	3.887419	-1.687664
C	-0.635587	3.770746	-0.268232
C	-1.860976	3.560082	0.459447
C	0.635521	3.770736	0.268330
C	1.860901	3.560083	-0.459364
C	2.926123	3.461104	0.403963
C	2.421256	3.699539	1.765905
C	1.021248	3.887424	1.687757
C	4.276805	3.112201	0.053350
C	4.934797	3.389944	-1.233766
C	6.207956	2.773384	-1.217490
C	6.360740	2.103042	0.088310
C	5.159443	2.397740	0.827727
C	7.389406	1.285617	0.507334
C	8.369085	0.654564	-0.341943
C	9.146772	-0.216300	0.381174
C	8.741050	-0.113776	1.792071
C	7.665539	0.801742	1.873658
C	10.131764	-1.123451	-0.149802
C	10.947589	-0.902442	-1.355537
C	11.694379	-2.077631	-1.602492
C	11.351216	-3.053563	-0.547218

C	10.428075	-2.372532	0.332734
C	11.741412	-4.366692	-0.446579
C	12.333214	-5.168065	-1.511443
C	12.478619	-6.449939	-1.096856
C	12.038840	-6.558804	0.287668
C	11.581364	-5.282575	0.701069
C	-3.090355	3.837564	-2.973466
C	-2.355669	4.141820	-4.122200
C	-0.980108	4.339468	-4.049148
C	-0.307572	4.233496	-2.828853
C	0.307549	4.233547	2.828950
C	0.980104	4.339533	4.049233
C	2.355661	4.141856	4.122273
C	3.090325	3.837562	2.973535
C	4.538260	4.173323	-2.308115
C	5.407994	4.325863	-3.390692
C	6.666220	3.731493	-3.375002
C	7.081771	2.968712	-2.280468
C	7.157846	1.158696	3.117176
C	7.686079	0.565096	4.266740
C	8.735757	-0.344320	4.179137
C	9.284009	-0.677021	2.937831
C	11.120788	0.229239	-2.139564
C	12.033478	0.185472	-3.195945
C	12.783275	-0.962569	-3.433823
C	12.633540	-2.093471	-2.626934
C	11.179760	-5.095452	2.018036
C	11.178409	-6.185805	2.894266
C	11.595575	-7.444068	2.469972
C	12.041255	-7.637340	1.159628
H	-1.909058	3.333314	1.515014
H	1.908969	3.333349	-1.514940
H	4.915898	1.965393	1.787757
H	8.373465	0.724529	-1.420401
H	9.908996	-2.847624	1.152689
H	12.521231	-4.808833	-2.513273
H	12.845334	-7.277579	-1.694226
H	-4.171233	3.744142	-3.023857
H	-2.866886	4.246787	-5.075237
H	-0.422622	4.596587	-4.945487
H	0.754220	4.451043	-2.787110
H	-0.754235	4.451133	2.787218
H	0.422637	4.596691	4.945572

H	2.866893	4.246837	5.075300
H	4.171201	3.744121	3.023912
H	3.582046	4.688185	-2.297423
H	5.107177	4.932038	-4.240815
H	7.342181	3.874743	-4.213386
H	8.088765	2.566015	-2.265785
H	6.380563	1.908641	3.216478
H	7.281967	0.832075	5.239229
H	9.147944	-0.782745	5.083859
H	10.141849	-1.340480	2.877915
H	10.582674	1.146217	-1.917754
H	12.175715	1.063102	-3.820641
H	13.507736	-0.976895	-4.243356
H	13.274600	-2.952107	-2.793186
H	10.887307	-4.119530	2.390288
H	10.852892	-6.043161	3.920917
H	11.588844	-8.278320	3.166170
H	12.392744	-8.613358	0.834469

Table S21. Cartesian Coordinates of the Optimized Geometries for **16**

C	0.934742	-2.946949	0.009892
C	0.100324	-1.907262	-0.156762
C	0.472861	-0.495581	-0.086531
C	-0.472866	0.495577	-0.086547
C	-0.100319	1.907255	-0.156785
C	-0.934723	2.946951	0.009880
C	1.926168	-0.175015	-0.025788
C	2.773763	-0.492880	-1.095022
C	4.133119	-0.196415	-1.037861
C	4.671593	0.409426	0.096988
C	3.840534	0.719994	1.171448
C	2.478676	0.431290	1.108686
C	-1.926171	0.175013	-0.025785
C	-2.773766	0.492862	-1.095032
C	-4.133117	0.196401	-1.037871
C	-4.671597	-0.409425	0.096990
C	-3.840545	-0.719974	1.171454
C	-2.478682	-0.431271	1.108693
H	0.569479	-3.966806	-0.073228
H	1.989679	-2.816495	0.236512
H	-0.947628	-2.113128	-0.365019
H	0.947634	2.113111	-0.365052
H	-1.989659	2.816510	0.236512

H	-0.569449	3.966804	-0.073244
H	2.356879	-0.969646	-1.978937
H	4.774070	-0.439861	-1.881805
H	5.733667	0.636916	0.143304
H	4.251840	1.188876	2.062098
H	1.830175	0.675220	1.946955
H	-2.356870	0.969613	-1.978949
H	-4.774071	0.439831	-1.881818
H	-5.733673	-0.636908	0.143301
H	-4.251847	-1.188838	2.062115
H	-1.830193	-0.675187	1.946975

Table S22. Cartesian Coordinates of the Optimized Geometries for **17a**

C	-4.725315	-0.666243	-0.025320
C	-4.152126	0.596921	0.196928
C	-3.904883	-1.772138	-0.257343
C	-2.530245	-1.598124	-0.308418
C	-1.949044	-0.317137	-0.154803
C	-2.767292	0.768159	0.132899
C	-1.452594	-2.561955	-0.490847
C	-0.263755	-1.918603	-0.421780
C	-0.483557	-0.484245	-0.244145
C	0.483557	0.484236	-0.244174
C	0.263759	1.918581	-0.421914
C	1.452600	2.561926	-0.491013
C	2.530247	1.598107	-0.308505
C	1.949042	0.317133	-0.154803
C	2.767284	-0.768145	0.132988
C	4.152118	-0.596905	0.197018
C	4.725312	0.666242	-0.025312
C	3.904886	1.772122	-0.257427
O	-6.077919	-0.858776	0.038753
O	-4.929822	1.695392	0.456133
H	-4.370505	-2.745104	-0.387075
H	-2.384293	1.760261	0.343117
H	-1.597384	-3.622011	-0.668408
H	0.703633	-2.373029	-0.580422
H	-0.703626	2.372998	-0.580601
H	1.597394	3.621970	-0.668649
H	2.384282	-1.760232	0.343270
O	4.929814	-1.695358	0.456300
O	6.077915	0.858773	0.038772
H	4.370512	2.745078	-0.387222

C	-6.816184	-0.211405	-0.989732
C	-5.578422	1.664632	1.720763
C	5.578422	-1.664501	1.720924
C	6.816195	0.211298	-0.989635
H	-7.865549	-0.455710	-0.810868
H	-6.518601	-0.591323	-1.975319
H	-6.677159	0.873984	-0.960360
H	-6.149485	2.592881	1.794327
H	-4.841131	1.627110	2.532832
H	-6.254065	0.806743	1.803093
H	6.149473	-2.592751	1.794562
H	4.841136	-1.626899	2.532994
H	6.254077	-0.806614	1.803177
H	7.865558	0.455617	-0.810777
H	6.518631	0.591120	-1.975266
H	6.677165	-0.874087	-0.960158

Table S23. Cartesian Coordinates of the Optimized Geometries for **17b**

C	7.225261	0.551917	-1.303616
C	6.255157	-0.274257	-0.712239
C	6.940451	1.897694	-1.546194
C	5.682276	2.391620	-1.237316
C	4.677651	1.549201	-0.703227
C	4.985994	0.225970	-0.412226
C	5.129210	3.734656	-1.347817
C	3.858145	3.732064	-0.878047
C	3.480166	2.380928	-0.477378
C	2.236640	1.980212	-0.055552
C	1.755597	0.615920	-0.012825
C	0.424976	0.582262	0.312535
C	-0.005090	1.963158	0.584390
C	1.094637	2.820136	0.357165
C	0.979302	4.173547	0.650822
C	-0.237233	4.683589	1.107416
C	-1.338284	3.831811	1.293372
C	-1.209998	2.462518	1.055608
C	-0.424976	-0.582251	0.312554
C	0.005091	-1.963138	0.584452
C	-1.094636	-2.820123	0.357256
C	-2.236640	-1.980213	-0.055486
C	-1.755598	-0.615920	-0.012803
C	-3.480166	-2.380943	-0.477298
C	-4.677653	-1.549224	-0.703172

C	-3.858145	-3.732093	-0.877923
C	-5.129210	-3.734700	-1.347690
C	-5.682278	-2.391661	-1.237232
C	-4.985996	-0.225984	-0.412213
C	-6.255160	0.274232	-0.712240
C	-7.225265	-0.551962	-1.303588
C	-6.940454	-1.897747	-1.546123
C	1.210000	-2.462482	1.055684
C	1.338286	-3.831767	1.293492
C	0.237236	-4.683552	1.107563
C	-0.979300	-4.173525	0.650955
O	8.476871	0.086180	-1.598834
O	6.514864	-1.591926	-0.437152
H	7.720597	2.524265	-1.969402
H	4.291358	-0.453056	0.069475
H	5.657980	4.587686	-1.758718
H	3.183793	4.576512	-0.892119
H	2.320418	-0.237199	-0.359524
H	1.808641	4.867098	0.571719
O	-0.330572	6.028504	1.347340
O	-2.530157	4.306429	1.767820
H	-2.057904	1.820732	1.273386
H	-2.320419	0.237189	-0.359529
H	-3.183792	-4.576540	-0.891969
H	-5.657981	-4.587744	-1.758563
H	-4.291360	0.453058	0.069464
O	-6.514868	1.591909	-0.437196
O	-8.476876	-0.086236	-1.598819
H	-7.720599	-2.524332	-1.969309
H	2.057905	-1.820688	1.273442
O	2.530160	-4.306372	1.767954
O	0.330579	-6.028460	1.347527
H	-1.808638	-4.867079	0.571874
C	8.514915	-0.894491	-2.627882
C	7.452213	-1.792364	0.612896
C	3.237352	-5.130673	0.849611
C	0.477756	-6.370149	2.720053
C	-0.477750	6.370235	2.719856
C	-3.237346	5.130713	0.849459
C	-7.452215	1.792381	0.612848
C	-8.514923	0.894400	-2.627900
H	9.565949	-1.163985	-2.752556
H	8.132900	-0.480145	-3.569482

H	7.933351	-1.781582	-2.357528
H	7.557466	-2.873647	0.727022
H	7.079113	-1.363581	1.551612
H	8.424122	-1.349514	0.370951
H	4.144958	-5.452268	1.365064
H	3.513137	-4.561032	-0.046809
H	2.645395	-6.005734	0.561805
H	0.536104	-7.460153	2.758537
H	-0.392410	-6.033127	3.297673
H	1.388110	-5.934856	3.144687
H	-0.536100	7.460240	2.758307
H	0.392417	6.033231	3.297485
H	-1.388104	5.934953	3.144503
H	-4.144950	5.452324	1.364905
H	-3.513132	4.561053	-0.046949
H	-2.645385	6.005766	0.561634
H	-7.557469	2.873668	0.726939
H	-7.079113	1.363629	1.551577
H	-8.424124	1.349522	0.370919
H	-9.565957	1.163890	-2.752581
H	-8.132909	0.480023	-3.569486
H	-7.933359	1.781501	-2.357577

Table S24. Cartesian Coordinates of the Optimized Geometries for **1d**

C	-14.604293	5.963554	-1.919275
C	-13.737815	6.394296	-0.910638
C	-14.606090	4.617395	-2.299347
H	-15.276019	6.672951	-2.406784
C	-12.884056	5.488187	-0.281259
H	-13.722054	7.445517	-0.615275
C	-13.751440	3.711078	-1.674542
H	-15.283973	4.271015	-3.082333
C	-12.875065	4.129473	-0.653557
H	-12.189626	5.839686	0.482339
H	-13.772470	2.656689	-1.956078
C	-11.960744	3.171866	-0.025129
C	-11.549293	3.158369	1.393372
C	-11.318049	2.138996	-0.662603
C	-10.603552	2.108180	1.566578
C	-11.985777	3.909885	2.481699
C	-10.421428	1.450507	0.249695
H	-11.368732	1.953564	-1.730626
C	-10.119823	1.818528	2.841479

C	-11.462786	3.631395	3.753836
H	-12.739519	4.687929	2.357430
C	-9.539813	0.439340	-0.077638
C	-10.541715	2.597127	3.930662
H	-9.440778	0.985762	3.015468
H	-11.793350	4.217748	4.613502
C	-9.494957	-0.367656	-1.321228
C	-8.412453	0.001848	0.719143
H	-10.154041	2.378792	4.927545
C	-8.351043	-1.209755	-1.246075
C	-10.371291	-0.486616	-2.399619
C	-7.666710	-0.938748	0.037034
H	-8.109909	0.462760	1.652051
C	-8.088034	-2.143680	-2.245318
C	-10.086326	-1.408957	-3.417379
H	-11.285287	0.102254	-2.453796
C	-6.413038	-1.505303	0.463476
C	-8.957058	-2.227803	-3.342768
H	-7.239330	-2.822979	-2.165220
H	-10.765259	-1.496123	-4.267844
C	-5.963185	-1.664388	1.863497
C	-5.389552	-1.938965	-0.357167
H	-8.758166	-2.952669	-4.134690
C	-4.629225	-2.158759	1.843265
C	-6.637947	-1.486658	3.068457
C	-4.233495	-2.318899	0.423093
H	-5.396989	-1.859718	-1.437884
C	-3.993186	-2.479733	3.042118
C	-5.974195	-1.779188	4.269225
H	-7.676323	-1.155473	3.083262
C	-2.998486	-2.691399	-0.078956
C	-4.667119	-2.270744	4.254863
H	-2.994293	-2.911945	3.056553
H	-6.491441	-1.638125	5.220344
C	-2.655720	-3.063971	-1.472792
C	-1.756528	-2.726660	0.658945
H	-4.166348	-2.510720	5.194748
C	-1.248733	-3.267701	-1.528818
C	-3.435908	-3.316007	-2.601000
C	-0.696955	-3.020863	-0.178765
H	-1.651209	-2.413658	1.691013
C	-0.639703	-3.706594	-2.701721
C	-2.813403	-3.729264	-3.788381

H	-4.520034	-3.222716	-2.573499
C	0.696950	-3.020866	0.178747
C	-1.431103	-3.922108	-3.839593
H	0.430441	-3.911343	-2.732508
H	-3.421420	-3.916047	-4.675613
C	1.248735	-3.267735	1.528792
C	1.756521	-2.726652	-0.658963
H	-0.963491	-4.260684	-4.766293
C	2.655721	-3.063994	1.472766
C	0.639710	-3.706661	2.701685
C	2.998480	-2.691393	0.078935
H	1.651197	-2.413639	-1.691026
C	3.435911	-3.316049	2.600968
C	1.431113	-3.922194	3.839552
H	-0.430432	-3.911417	2.732467
C	4.233486	-2.318883	-0.423114
C	2.813410	-3.729337	3.788341
H	4.520036	-3.222750	2.573472
H	0.963505	-4.260796	4.766244
C	4.629211	-2.158720	-1.843286
C	5.389544	-1.938952	0.357147
H	3.421432	-3.916133	4.675567
C	5.963172	-1.664350	-1.863515
C	3.993166	-2.479666	-3.042142
C	6.413036	-1.505299	-0.463495
H	5.396987	-1.859714	1.437864
C	6.637930	-1.486599	-3.068474
C	4.667096	-2.270656	-4.254887
H	2.994273	-2.911875	-3.056584
C	7.666713	-0.938755	-0.037051
C	5.974173	-1.779104	-4.269245
H	7.676309	-1.155422	-3.083274
H	4.166319	-2.510613	-5.194774
C	8.351053	-1.209777	1.246052
C	8.412451	0.001851	-0.719151
H	6.491417	-1.638027	-5.220363
C	9.494970	-0.367681	1.321206
C	8.088053	-2.143717	2.245282
C	9.539814	0.439336	0.077630
H	8.109904	0.462769	-1.652054
C	10.371307	-0.486650	2.399593
C	8.957083	-2.227854	3.342727
H	7.239352	-2.823019	2.165175

C	10.421427	1.450509	-0.249691
C	10.086347	-1.409004	3.417344
H	11.285300	0.102224	2.453779
H	8.758197	-2.952731	4.134640
C	10.603546	2.108210	-1.566563
C	11.318057	2.138975	0.662615
H	10.765283	-1.496174	4.267806
C	11.549286	3.158396	-1.393337
C	10.119814	1.818589	-2.841469
C	11.960747	3.171859	0.025161
H	11.368756	1.953516	1.730634
C	11.985763	3.909943	-2.481646
C	10.541699	2.597218	-3.930635
H	9.440773	0.985825	-3.015483
C	12.875074	4.129451	0.653606
C	11.462767	3.631484	-3.753788
H	12.739504	4.687984	-2.357358
H	10.154022	2.378905	-4.927521
C	12.884050	5.488176	0.281351
C	13.751465	3.711028	1.674567
H	11.793325	4.217860	-4.613441
C	13.737812	6.394271	0.910748
H	12.189608	5.839695	-0.482226
C	14.606117	4.617331	2.299389
H	13.772505	2.656631	1.956068
C	14.604307	5.963502	1.919359
H	13.722042	7.445501	0.615419
H	15.284014	4.270931	3.082354
H	15.276034	6.672888	2.406882

Table S25. Cartesian Coordinates of the Optimized Geometries of **1e**

C	-15.653471	8.138475	-2.444450
C	-16.005647	6.801020	-2.653725
C	-14.659906	8.452456	-1.512822
H	-16.152125	8.931996	-3.004622
C	-15.371862	5.788335	-1.936092
H	-16.784985	6.547102	-3.375338
C	-14.026752	7.440744	-0.790616
H	-14.371086	9.493117	-1.351098
C	-14.371175	6.089420	-0.990848
H	-15.665866	4.747557	-2.083696
H	-13.231056	7.693162	-0.089181
C	-13.691831	5.013016	-0.264412

C	-13.228053	5.061928	1.137098
C	-13.355035	3.785236	-0.779630
C	-12.562767	3.835943	1.422766
C	-13.412734	6.019936	2.130998
C	-12.612100	3.005299	0.194983
H	-13.501458	3.495569	-1.815137
C	-12.104214	3.584407	2.714921
C	-12.913487	5.769036	3.418345
H	-13.956664	6.942096	1.924384
C	-12.022537	1.775205	-0.019905
C	-12.268142	4.564856	3.706087
H	-11.643685	2.634108	2.979017
H	-13.047422	6.515219	4.204077
C	-12.237316	0.843765	-1.153876
C	-10.999073	1.166493	0.804121
H	-11.898460	4.373692	4.715307
C	-11.331799	-0.241798	-0.995870
C	-13.167856	0.822310	-2.192481
C	-10.540151	-0.002603	0.230564
H	-10.547661	1.642795	1.666479
C	-11.354340	-1.321059	-1.875987
C	-13.167282	-0.253488	-3.092683
H	-13.911576	1.609655	-2.302426
C	-9.443391	-0.808944	0.700764
C	-12.270210	-1.313051	-2.937955
H	-10.694618	-2.176092	-1.728134
H	-13.887737	-0.267092	-3.912802
C	-8.977212	-0.909770	2.100890
C	-8.598253	-1.575812	-0.078334
H	-12.293586	-2.151319	-3.637157
C	-7.806535	-1.718129	2.117394
C	-9.527840	-0.430750	3.286681
C	-7.532379	-2.136002	0.721195
H	-8.639386	-1.624083	-1.160122
C	-7.209280	-2.044328	3.334605
C	-8.896648	-0.736116	4.501756
H	-10.452237	0.146818	3.279747
C	-6.450851	-2.857836	0.246081
C	-7.751068	-1.534468	4.524232
H	-6.346362	-2.705962	3.383965
H	-9.316176	-0.361003	5.437356
C	-6.277983	-3.463208	-1.096506
C	-5.219431	-3.110989	0.958091

H	-7.277938	-1.779443	5.477054
C	-4.966340	-4.011334	-1.154459
C	-7.151290	-3.645618	-2.168499
C	-4.304879	-3.752036	0.142934
H	-4.990494	-2.715041	1.940449
C	-4.540554	-4.720006	-2.275081
C	-6.706951	-4.334604	-3.306898
H	-8.178112	-3.286360	-2.131182
C	-2.936389	-4.051649	0.470205
C	-5.416328	-4.865635	-3.361021
H	-3.554390	-5.183519	-2.301339
H	-7.385541	-4.468480	-4.151492
C	-2.393521	-4.264407	1.829710
C	-1.879848	-4.127421	-0.418781
H	-5.091064	-5.414301	-4.247151
C	-0.983476	-4.419444	1.720557
C	-3.030549	-4.399162	3.060617
C	-0.631321	-4.309091	0.284406
H	-1.957833	-3.922564	-1.480021
C	-0.231159	-4.720212	2.855646
C	-2.258049	-4.665998	4.200724
H	-4.115576	-4.330553	3.138524
C	0.631337	-4.309074	-0.284388
C	-0.874386	-4.825236	4.098055
H	0.840721	-4.899662	2.794599
H	-2.746270	-4.767771	5.171977
C	0.983449	-4.419294	-1.720555
C	1.879884	-4.127440	0.418778
H	-0.285491	-5.049317	4.989530
C	2.393494	-4.264283	-1.829737
C	0.231083	-4.719932	-2.855644
C	2.936404	-4.051637	-0.470231
H	1.957901	-3.922653	1.480027
C	3.030480	-4.398962	-3.060674
C	0.874268	-4.824877	-4.098081
H	-0.840803	-4.899344	-2.794572
C	4.304906	-3.752056	-0.142977
C	2.257935	-4.665683	-4.200779
H	4.115507	-4.330380	-3.138609
H	0.285338	-5.048862	-4.989556
C	4.966402	-4.011449	1.154377
C	5.219433	-3.110951	-0.958114
H	2.746122	-4.767393	-5.172055

C	6.278053	-3.463338	1.096422
C	4.540632	-4.720199	2.274954
C	6.450873	-2.857845	-0.246121
H	4.990473	-2.714953	-1.940446
C	7.151386	-3.645858	2.168377
C	5.416433	-4.865932	3.360858
H	3.554459	-5.183695	2.301196
C	7.532370	-2.135970	-0.721242
C	6.707064	-4.334926	3.306735
H	8.178216	-3.286628	2.131071
H	5.091185	-5.414664	4.246954
C	7.806396	-1.717959	-2.117423
C	8.598304	-1.575829	0.078248
H	7.385680	-4.468882	4.151296
C	8.977088	-0.909620	-2.100955
C	7.209018	-2.044021	-3.334608
C	9.443386	-0.808915	-0.700860
H	8.639544	-1.624203	1.160025
C	9.527627	-0.430510	-3.286750
C	7.750714	-1.534067	-4.524239
H	6.346074	-2.705623	-3.383961
C	10.540178	-0.002605	-0.230672
C	8.896321	-0.735753	-4.501798
H	10.452039	0.147034	-3.279837
H	7.277487	-1.778943	-5.477037
C	11.331899	-0.241900	0.995691
C	10.999024	1.166568	-0.804133
H	9.315775	-0.360572	-5.437403
C	12.237403	0.843666	1.153752
C	11.354513	-1.321253	1.875694
C	12.022546	1.775208	0.019881
H	10.547539	1.642949	-1.666408
C	13.167994	0.822131	2.192309
C	12.270443	-1.313332	2.937612
H	10.694800	-2.176284	1.727786
C	12.612095	3.005322	-0.194935
C	13.167494	-0.253761	3.092400
H	13.911696	1.609485	2.302301
H	12.293883	-2.151673	3.636725
C	12.562680	3.836081	-1.422638
C	13.355099	3.785167	0.779700
H	13.887992	-0.267430	3.912480
C	13.227992	5.062037	-1.136901

C	12.104040	3.584674	-2.714788
C	13.691872	5.012989	0.264571
H	13.501595	3.495396	1.815167
C	13.412612	6.020139	-2.130720
C	12.267908	4.565218	-3.705871
H	11.643488	2.634405	-2.978949
C	14.371283	6.089321	0.991051
C	12.913278	5.769367	-3.418058
H	13.956561	6.942276	-1.924049
H	11.898155	4.374151	-4.715082
C	14.026861	7.440666	0.790961
C	15.372037	5.788140	1.936192
H	13.047165	6.515623	-4.203728
C	14.660083	8.452306	1.513208
H	13.231119	7.693155	0.089604
C	16.005890	6.800755	2.653866
H	15.666037	4.747344	2.083684
C	15.653713	8.138232	2.444734
H	14.371263	9.492986	1.351598
H	16.785278	6.546764	3.375398
H	16.152421	8.931698	3.004937